



(12) **United States Patent**
Kitazawa

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(45) **Date of Patent:** **Aug. 26, 2025**

(54) **CONNECTOR**

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(73) Assignee: **Molex, LLC**, Lisle, IL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 612 days.

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(30) **Foreign Application Priority Data**

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(51) **Int. Cl.**

H01R 12/71 (2011.01)

H01R 12/91 (2011.01)

H01R 13/426 (2006.01)

H01R 13/631 (2006.01)

H01R 13/6474 (2011.01)

(52) **U.S. Cl.**

CPC **H01R 12/716** (2013.01); **H01R 12/91** (2013.01); **H01R 13/426** (2013.01); **H01R 13/6315** (2013.01); **H01R 13/6474** (2013.01)

(58) **Field of Classification Search**

CPC H01R 12/716; H01R 12/91; H01R 13/426; H01R 13/6315; H01R 13/6474; H01R 12/7088

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2020/0412038 A1 12/2020 Yoshida

FOREIGN PATENT DOCUMENTS

CN	206076547 U	4/2017
EP	3696921 A1	8/2020
JP	2015035283 A	2/2015
JP	2017079214 A	4/2017
JP	2020035539 A	3/2020
JP	2020087837 A	6/2020
JP	2021093244 A	6/2021
JP	2021093249 A	6/2021

OTHER PUBLICATIONS

Extended European Search Report and Opinion received for EP Application No. 22183008.6, mailed on Dec. 6, 2022, 11 pages.

Primary Examiner — Brigitte R. Hammond

(57) **ABSTRACT**

The fixed housing includes a long wall portion extending in the longitudinal direction of the connector and a short wall portion extending in the width direction of the connector and connected to the long wall portion. The movable housing includes a long wall portion extending in the longitudinal direction of the connector and a short wall portion extending in the width direction of the connector and connected to the long wall portion. The power supply terminal includes a fixed-side retained portion retained on the short wall portion of the fixed housing, a movable-side retained portion retained on the movable housing, and a bent portion connected to the fixed-side retained portion and the movable-side retained portion. The movable-side retained portion includes a first retained portion retained on the short wall portion and a second retained portion retained on the long wall portion.

20 Claims, 37 Drawing Sheets

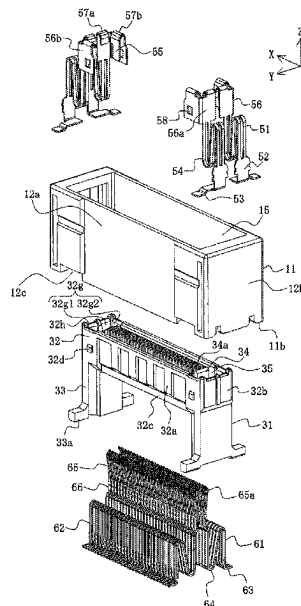


FIG. 1A

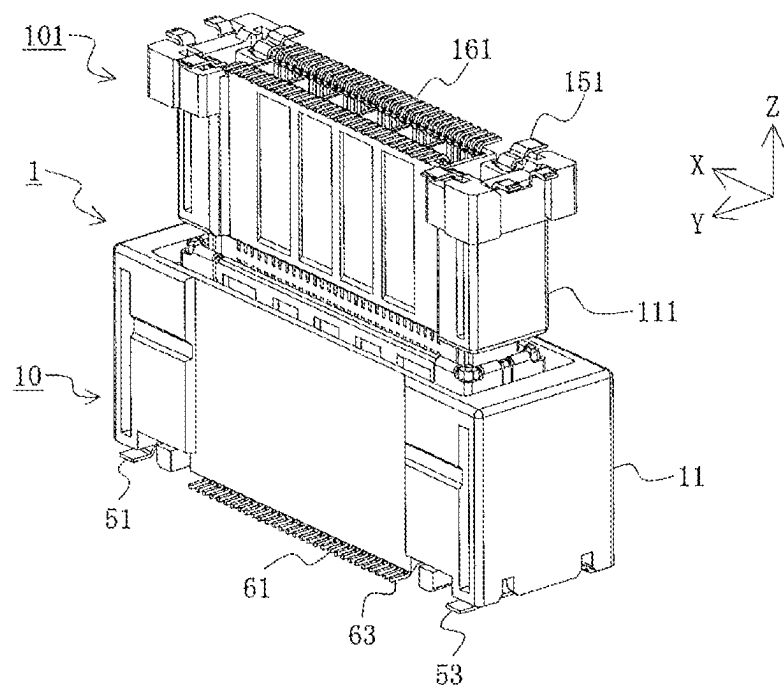


FIG. 1B

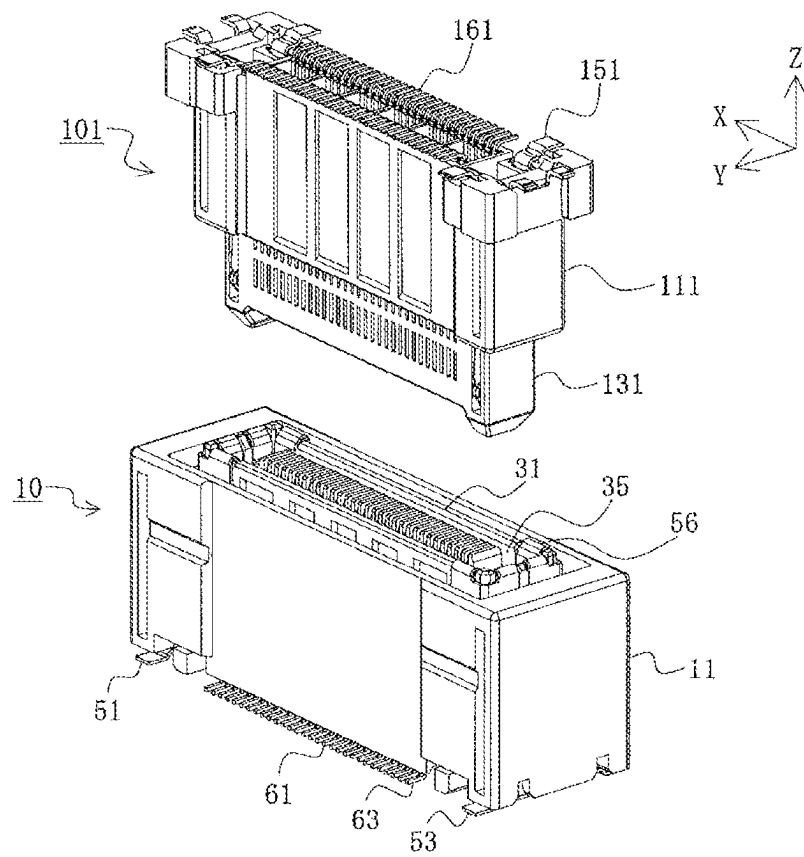
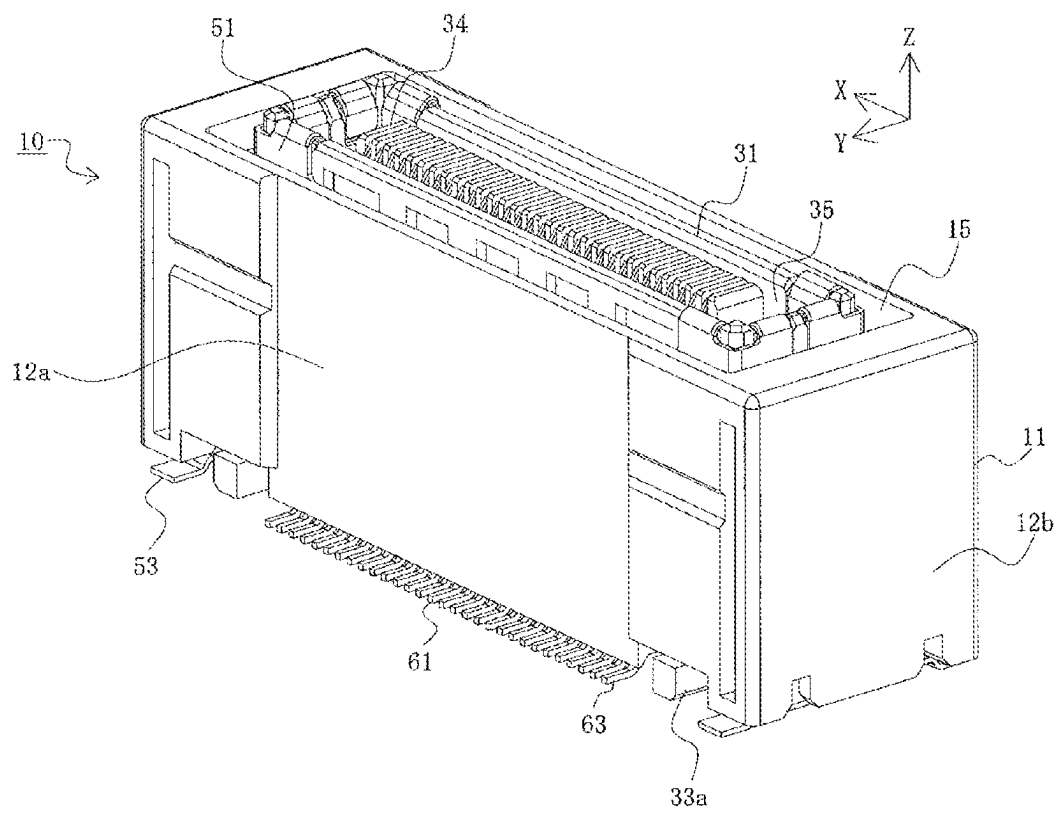


FIG. 2



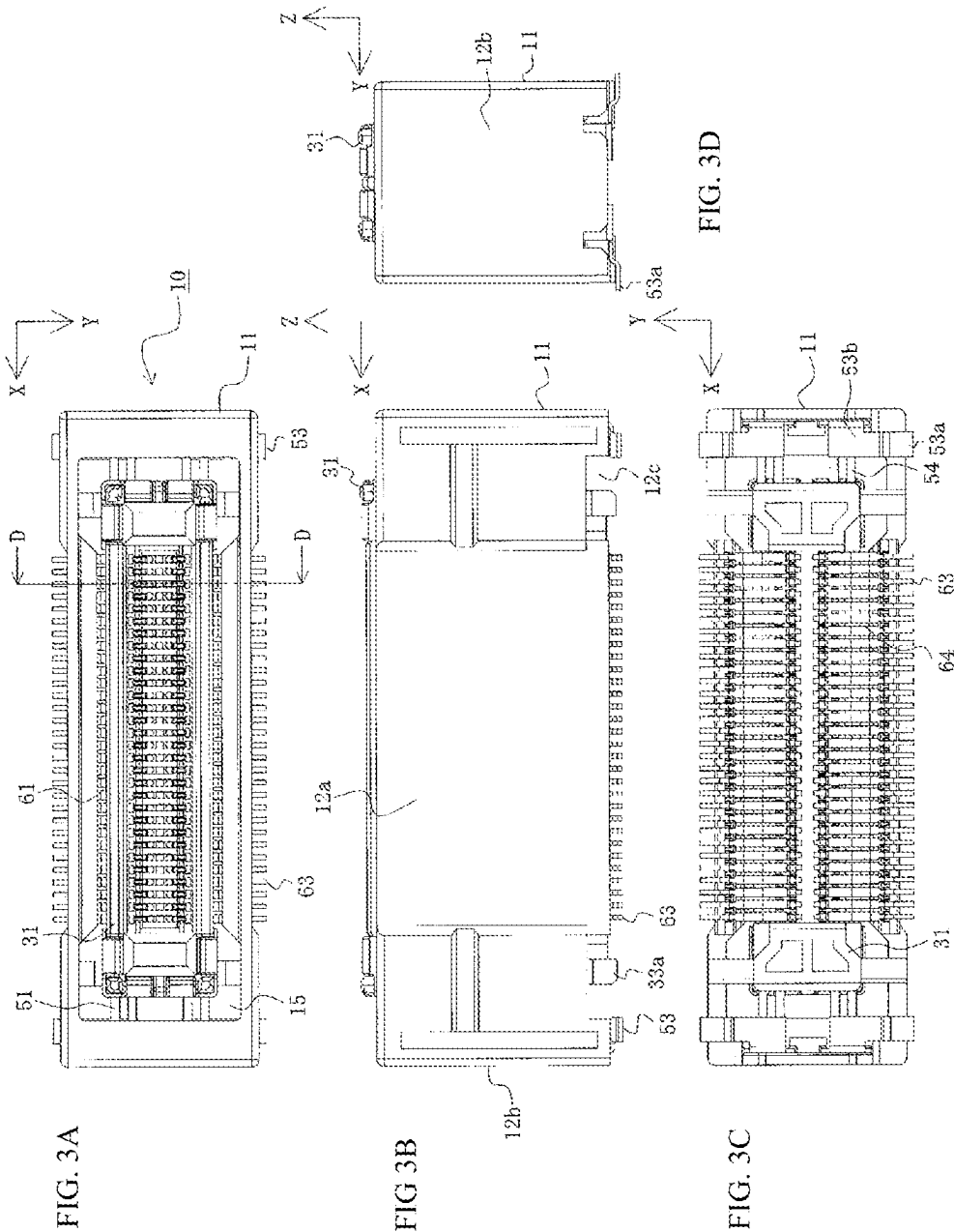


FIG. 4

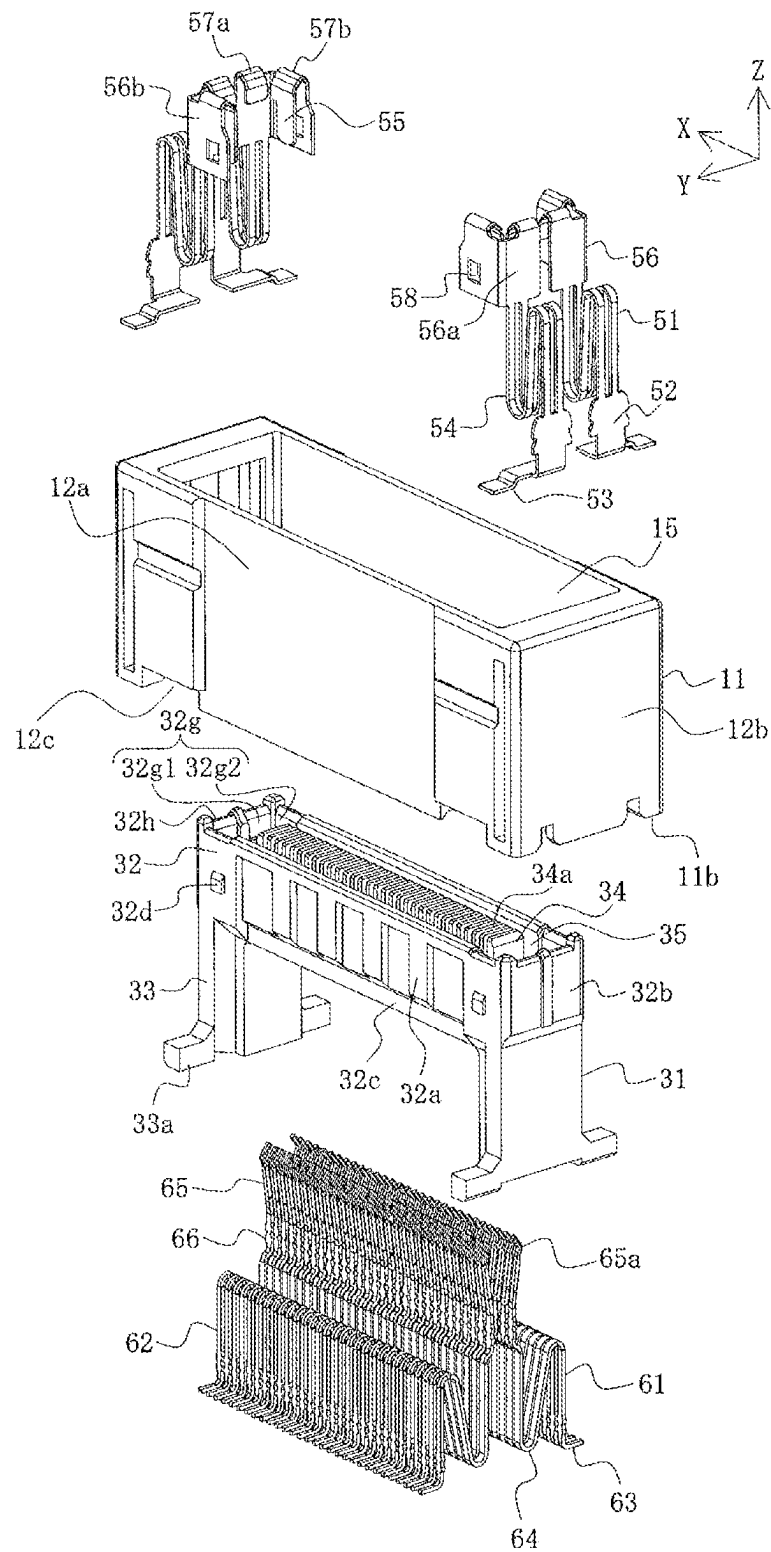


FIG. 5A

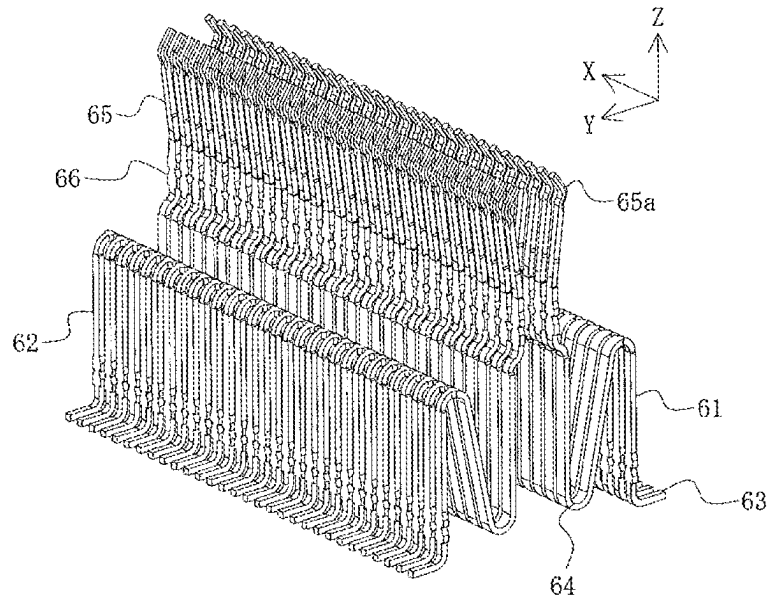


FIG. 5B

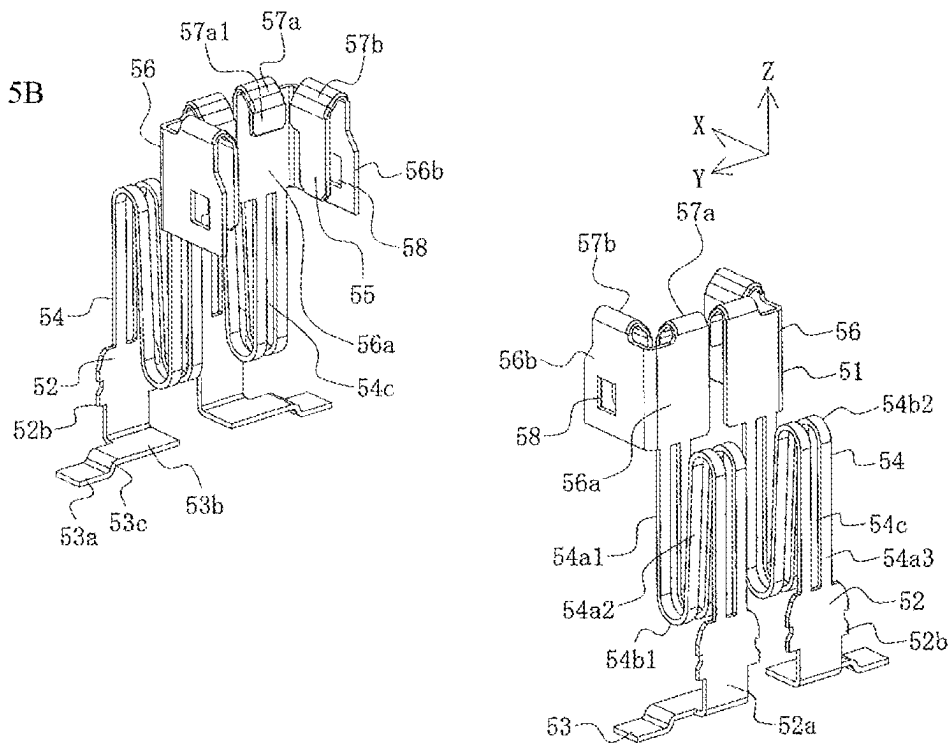


FIG. 6A

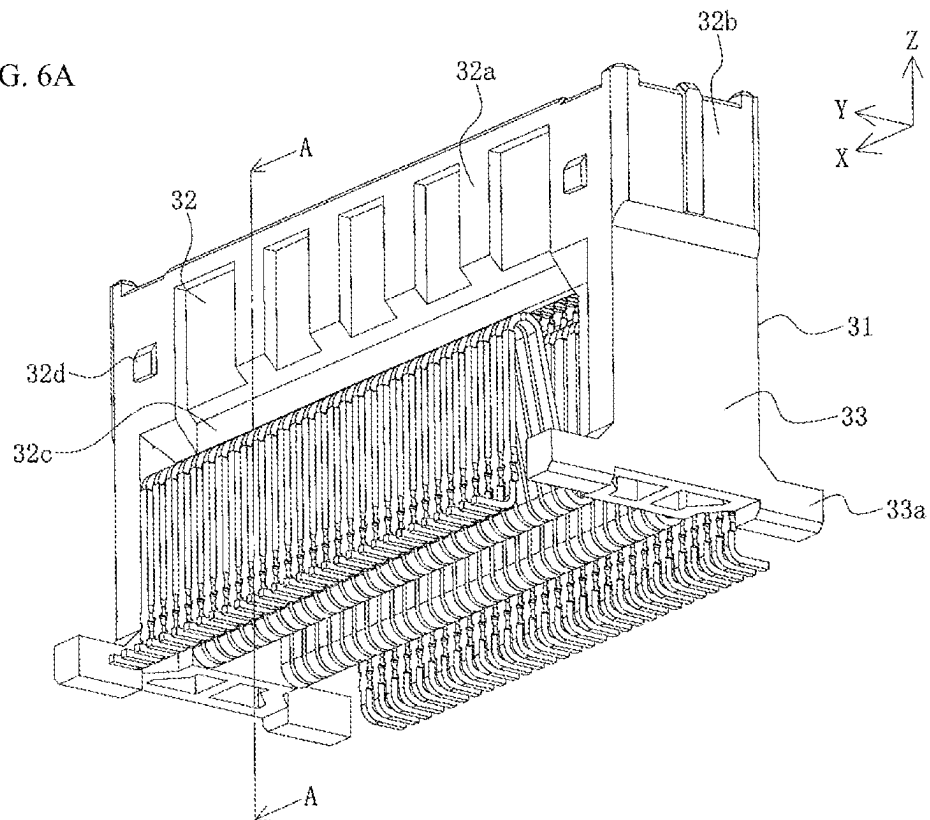


FIG. 6B

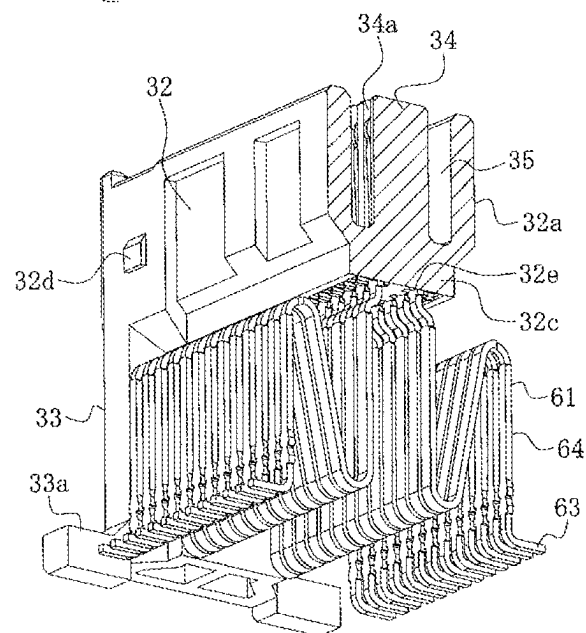
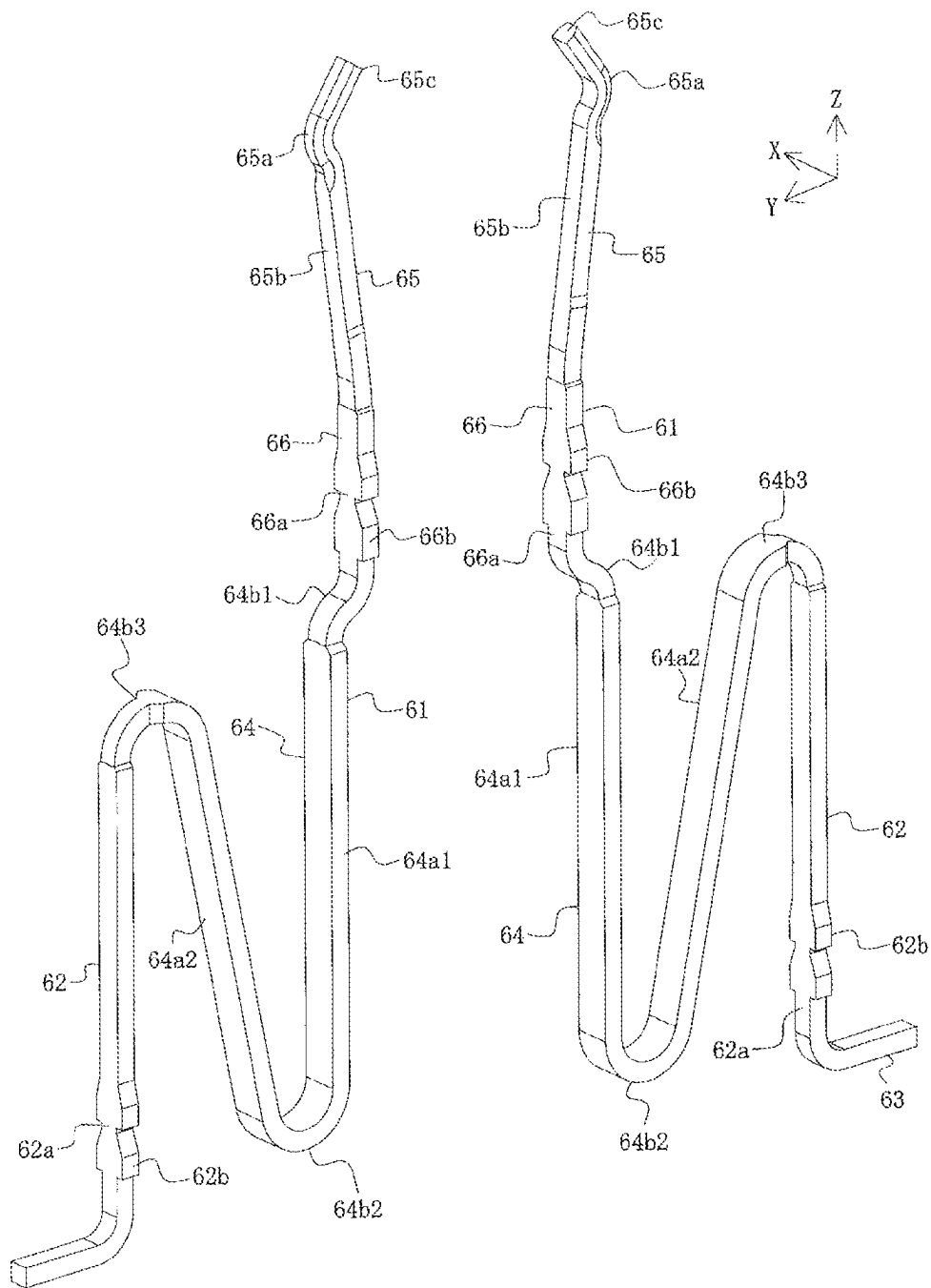
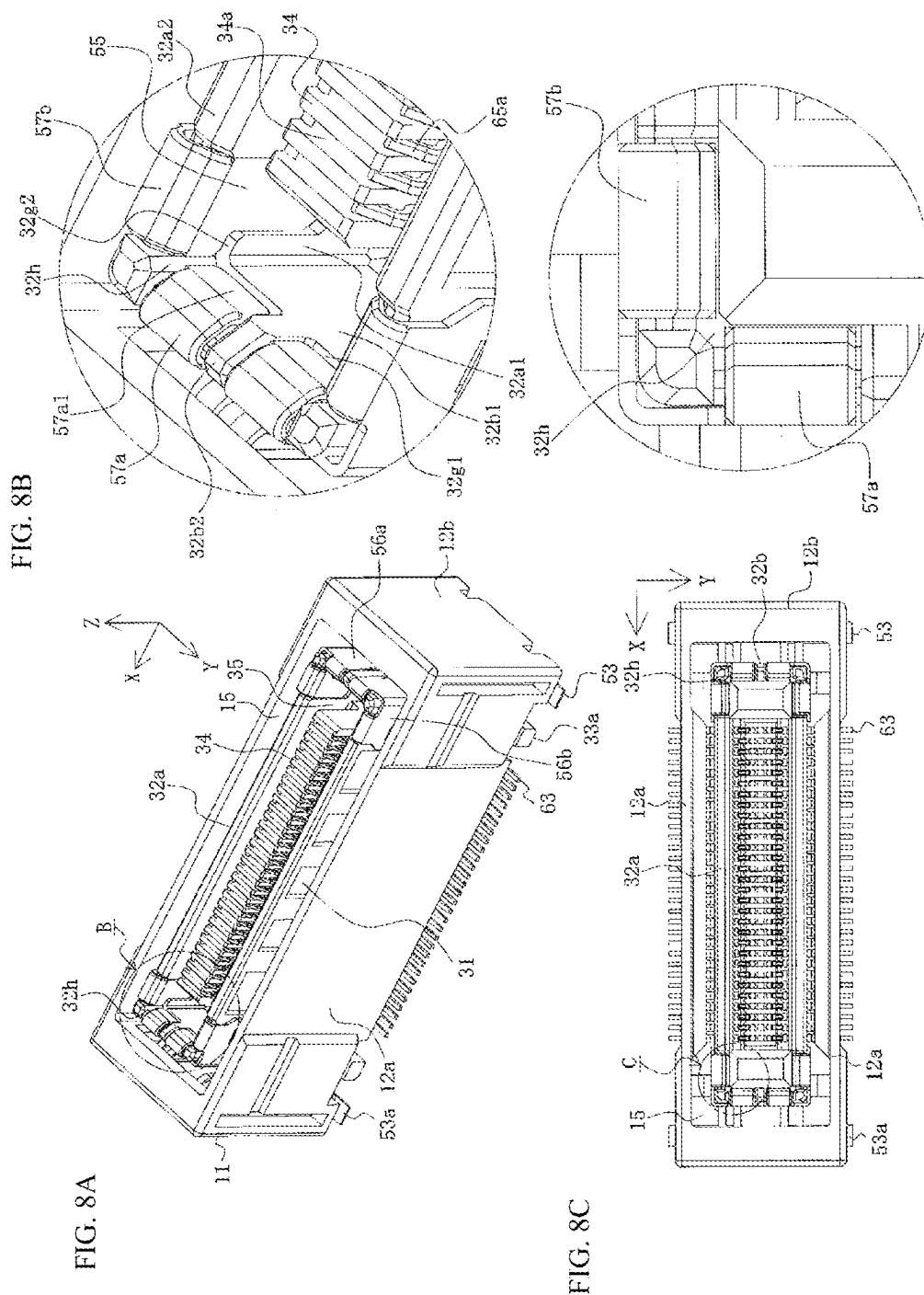


FIG. 7





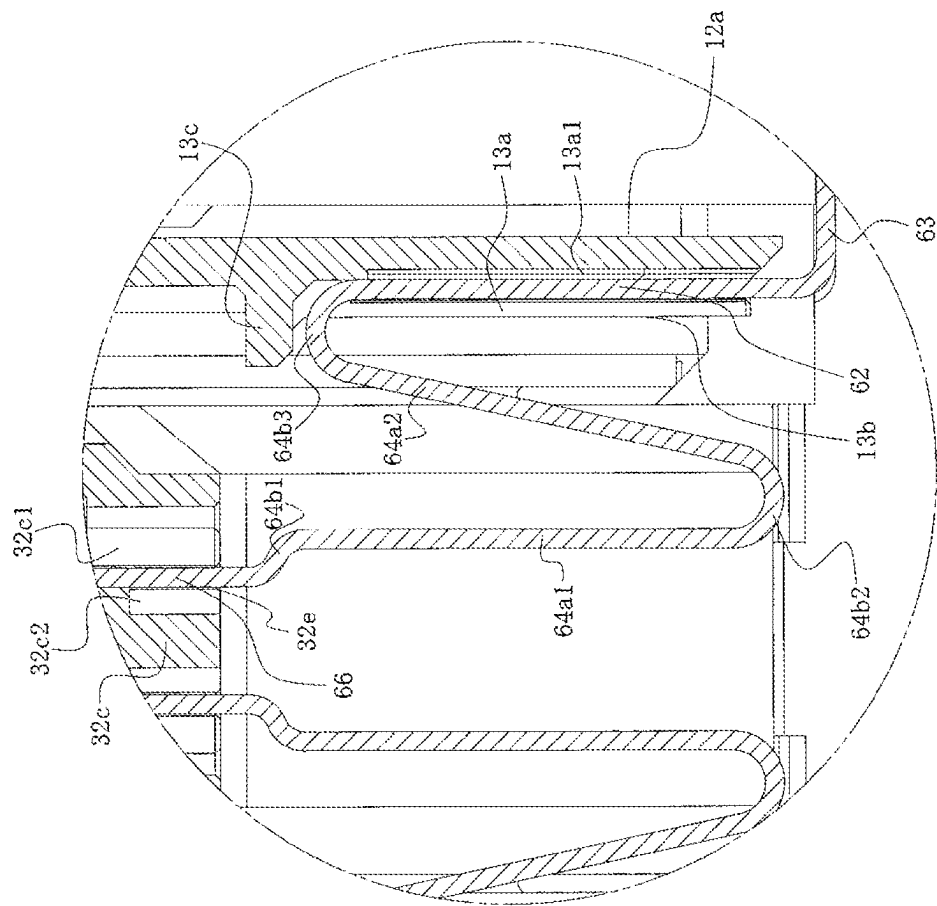


FIG. 9B

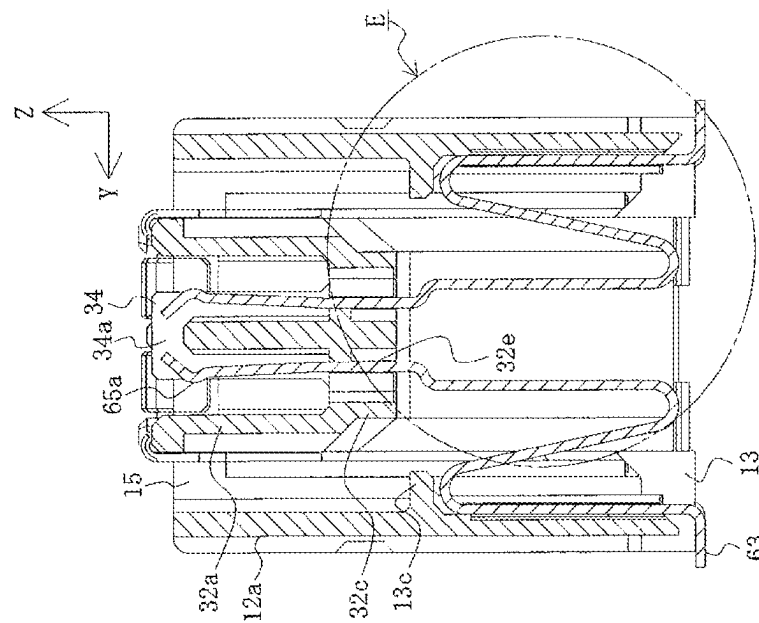


FIG. 9A

FIG. 10A

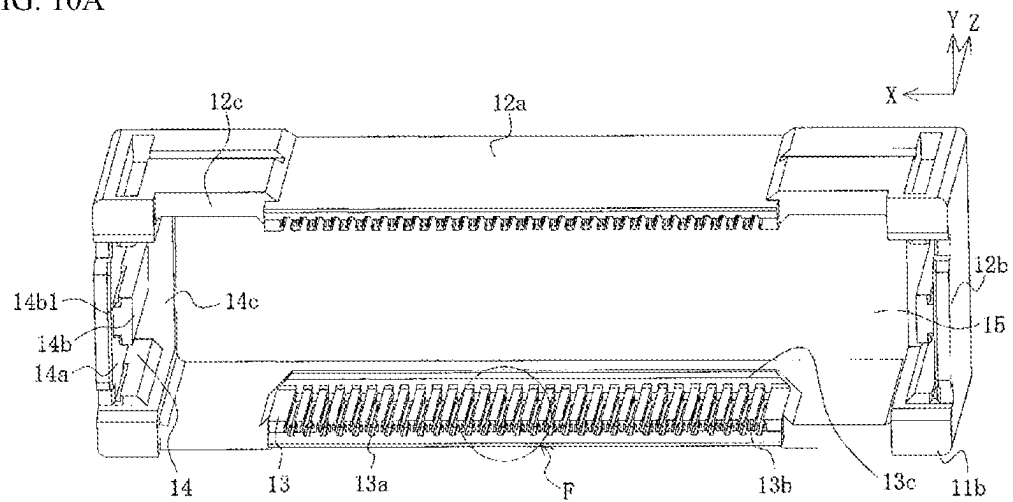


FIG. 10B

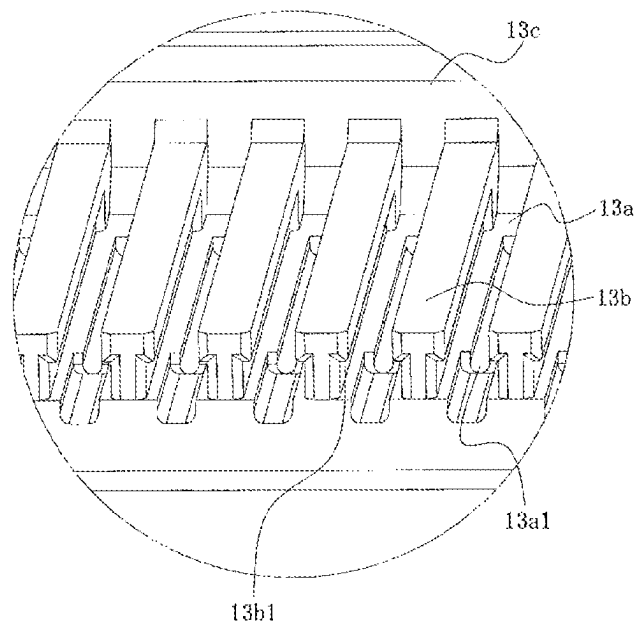
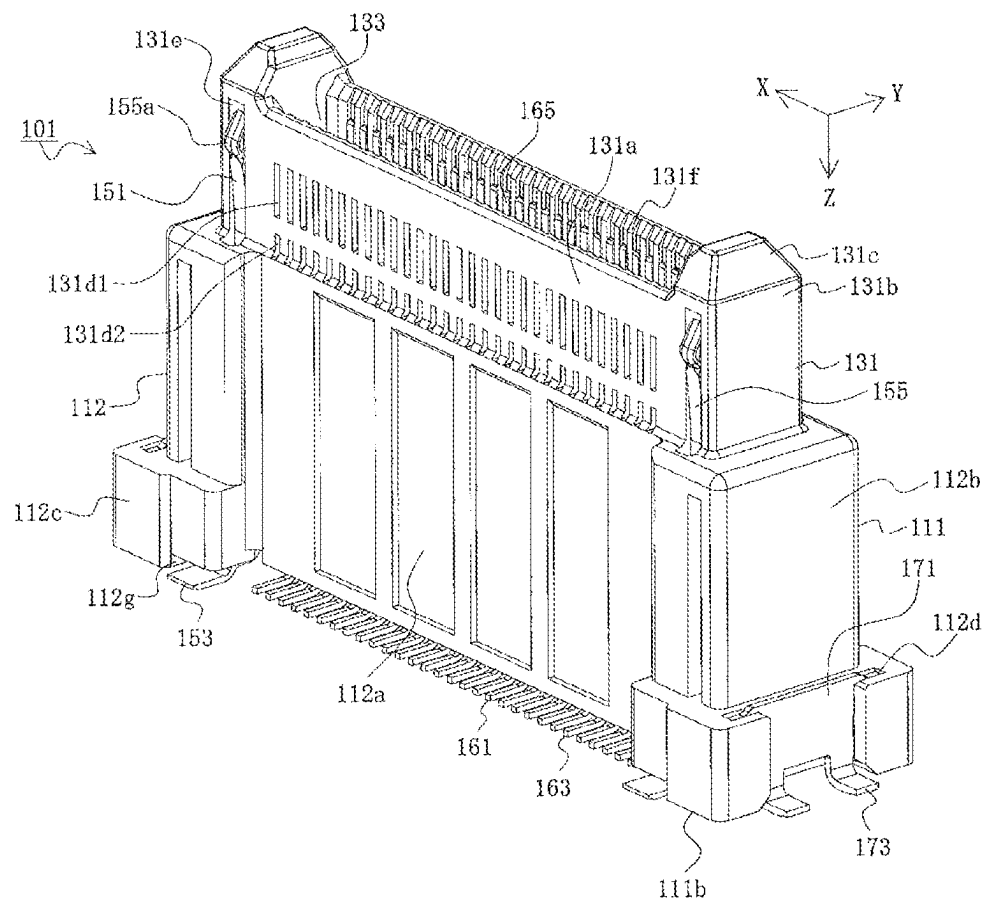


FIG. 11



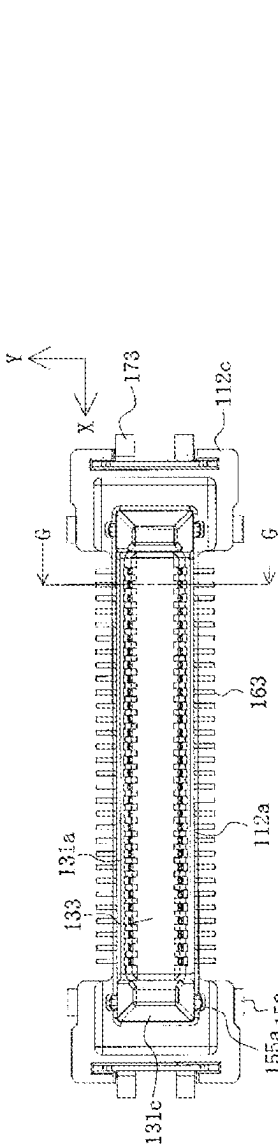


FIG. 12A

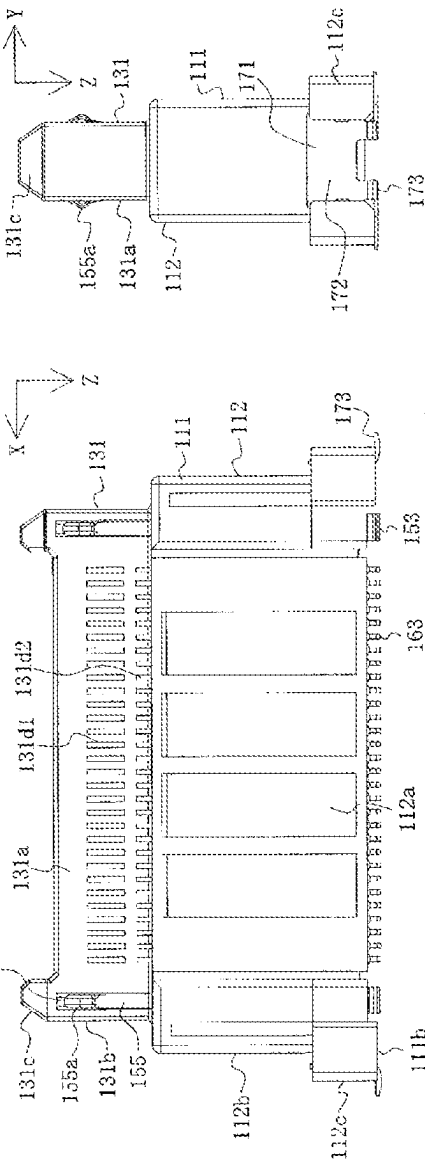


FIG. 12B

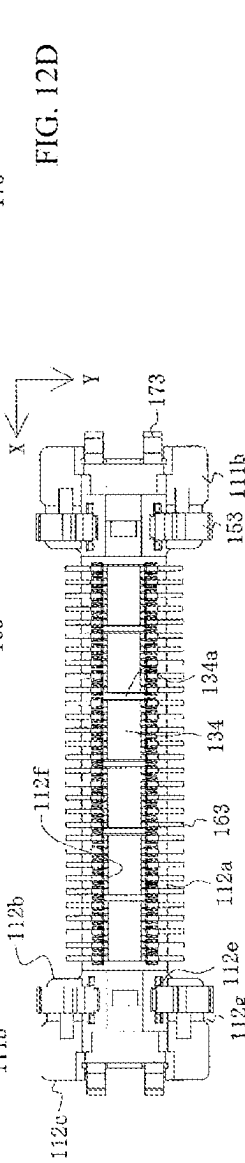


FIG. 12C

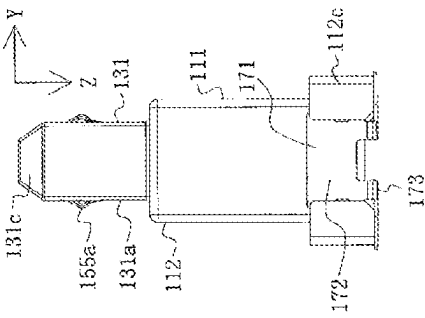


FIG. 12D

FIG. 13

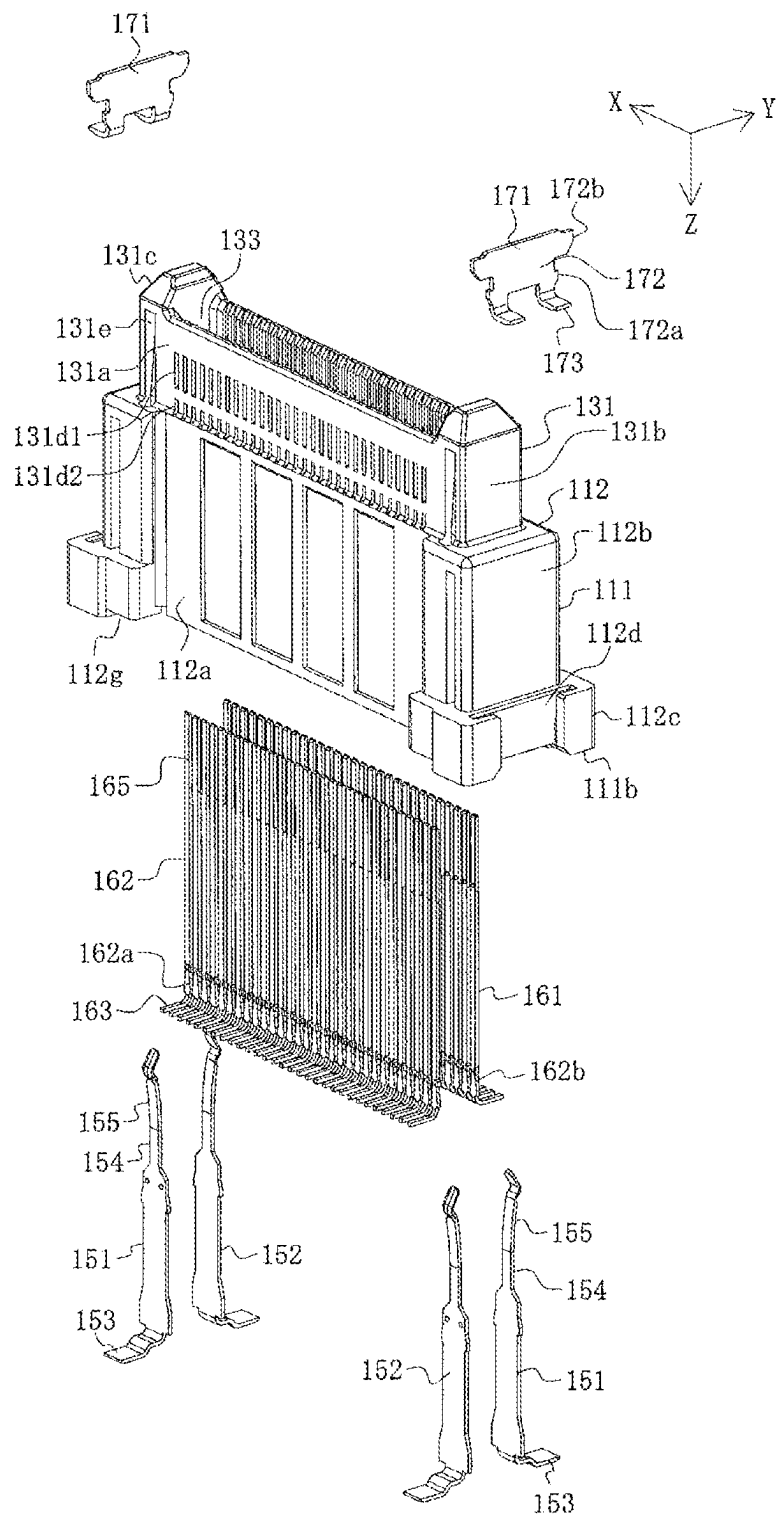


FIG. 14A

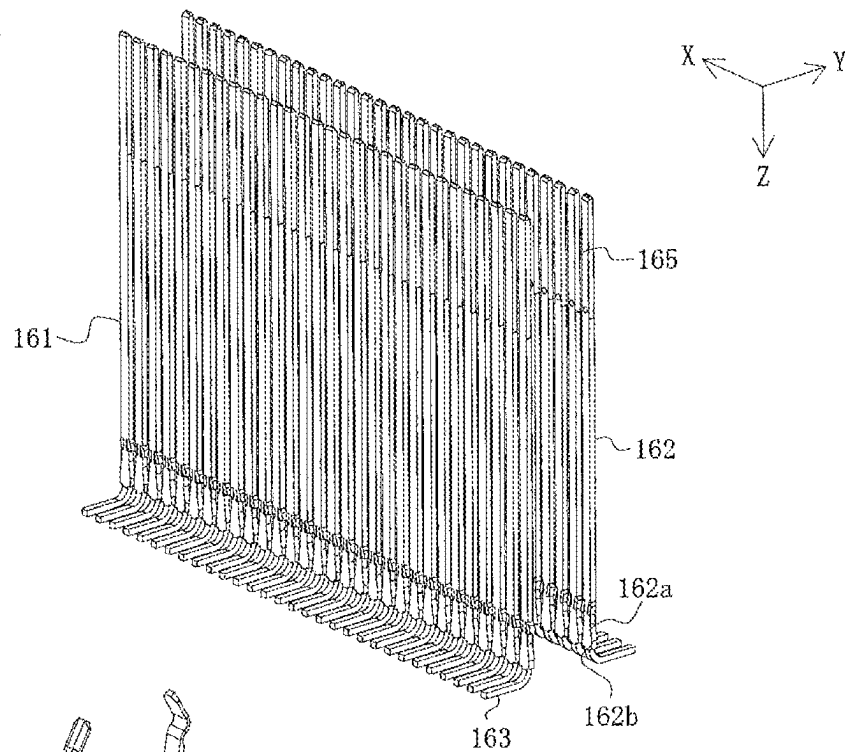


FIG. 14B

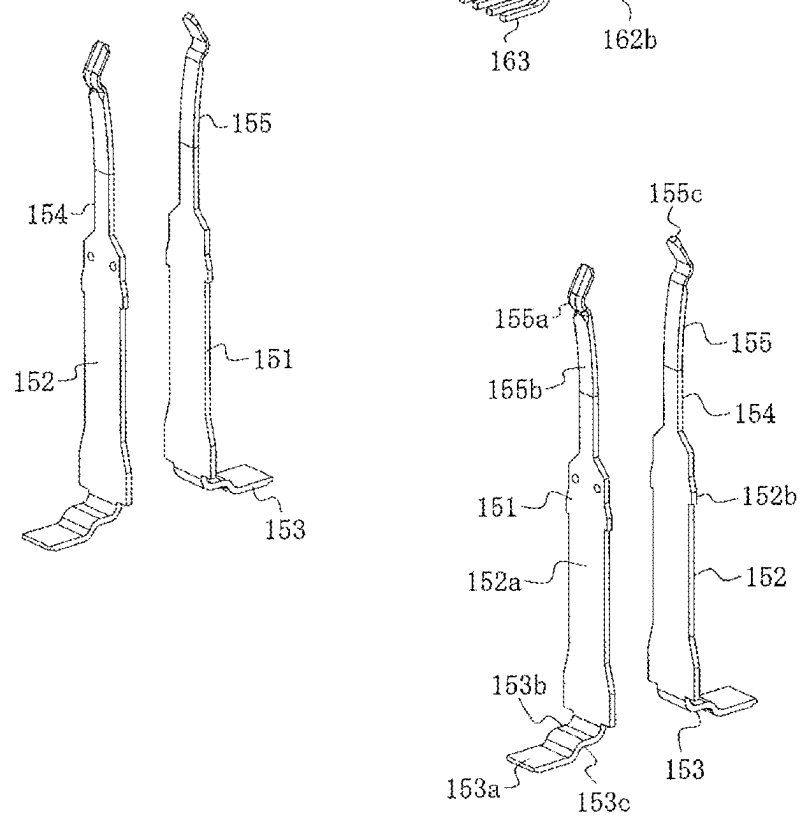


FIG. 15A

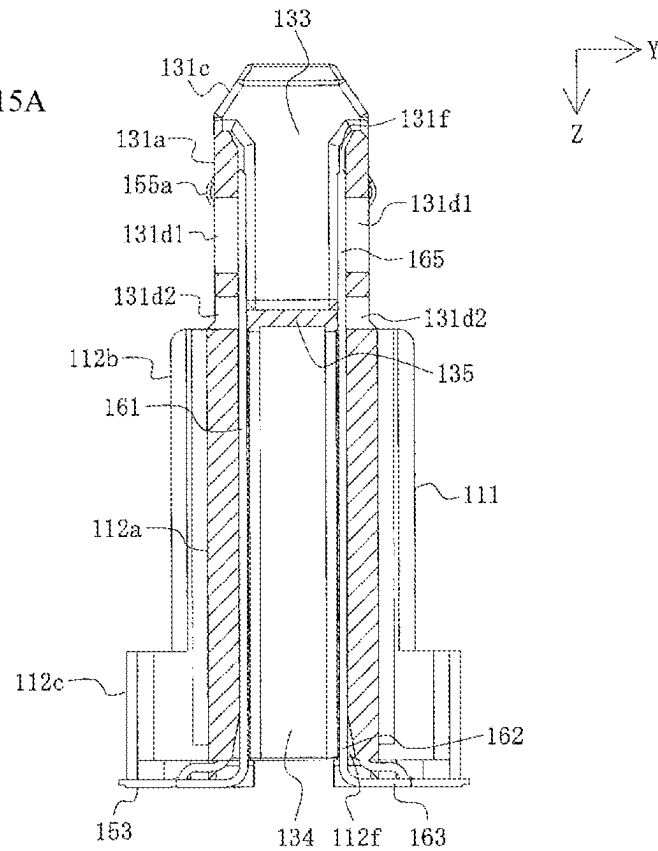


FIG. 15C

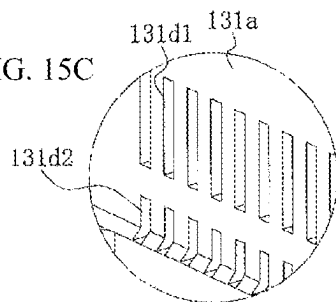
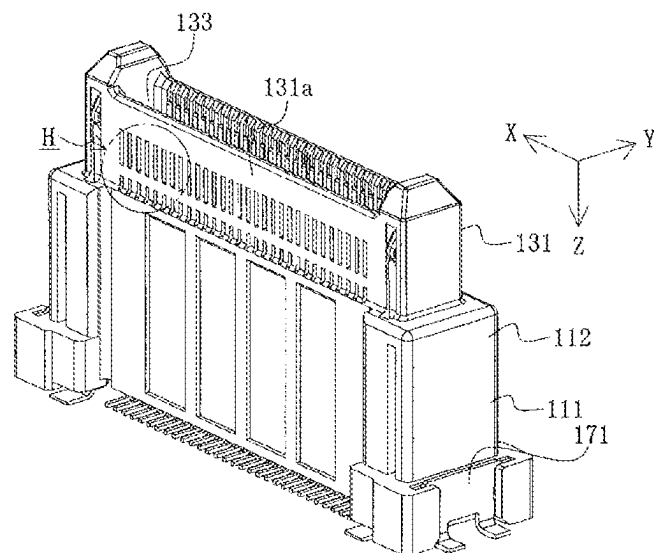


FIG. 15B



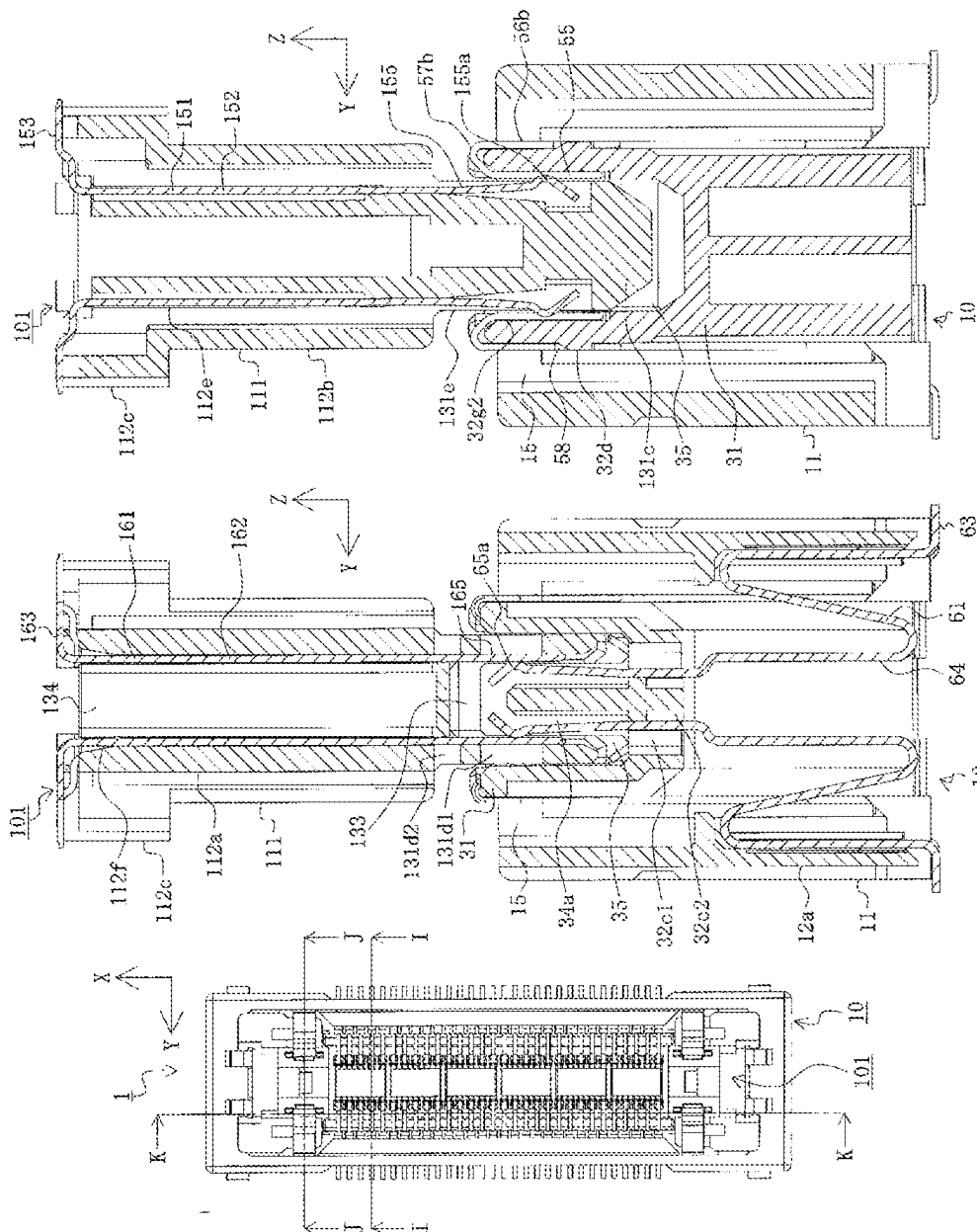


FIG. 16C

FIG. 16B

FIG. 16A

FIG. 17

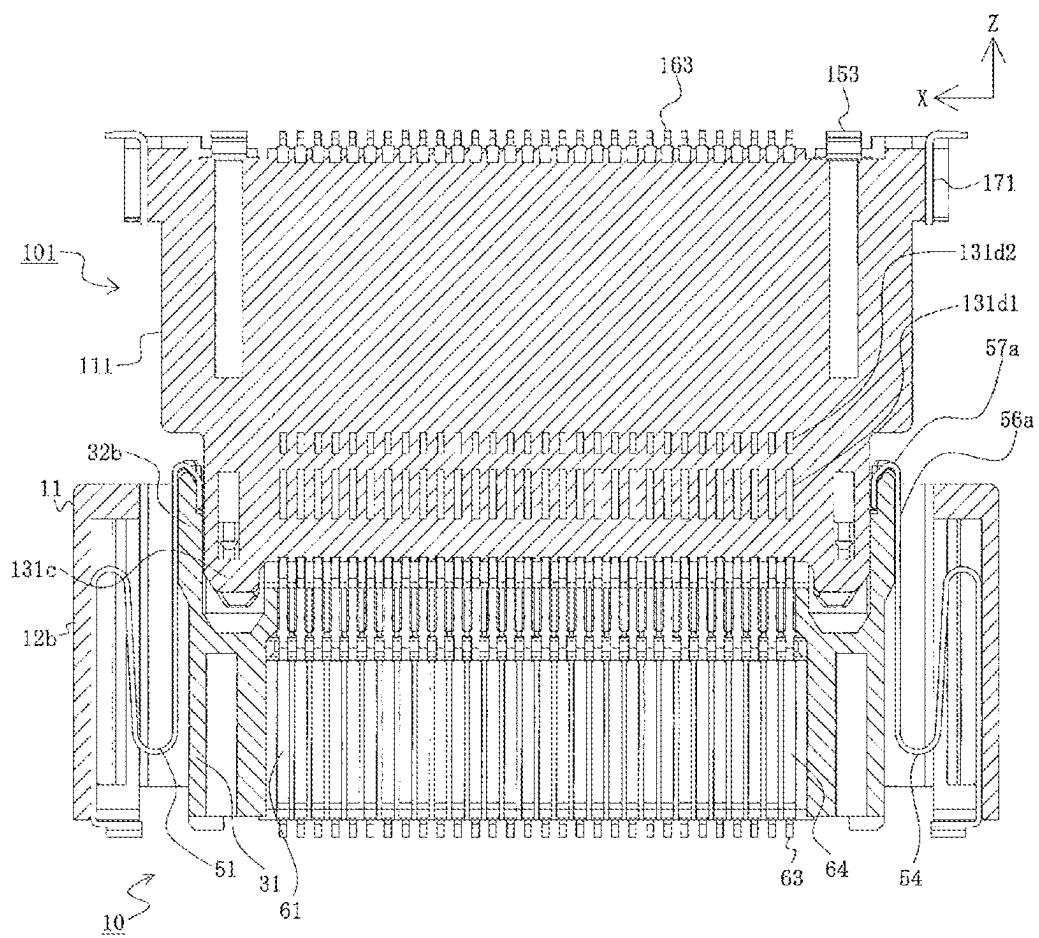


FIG. 18

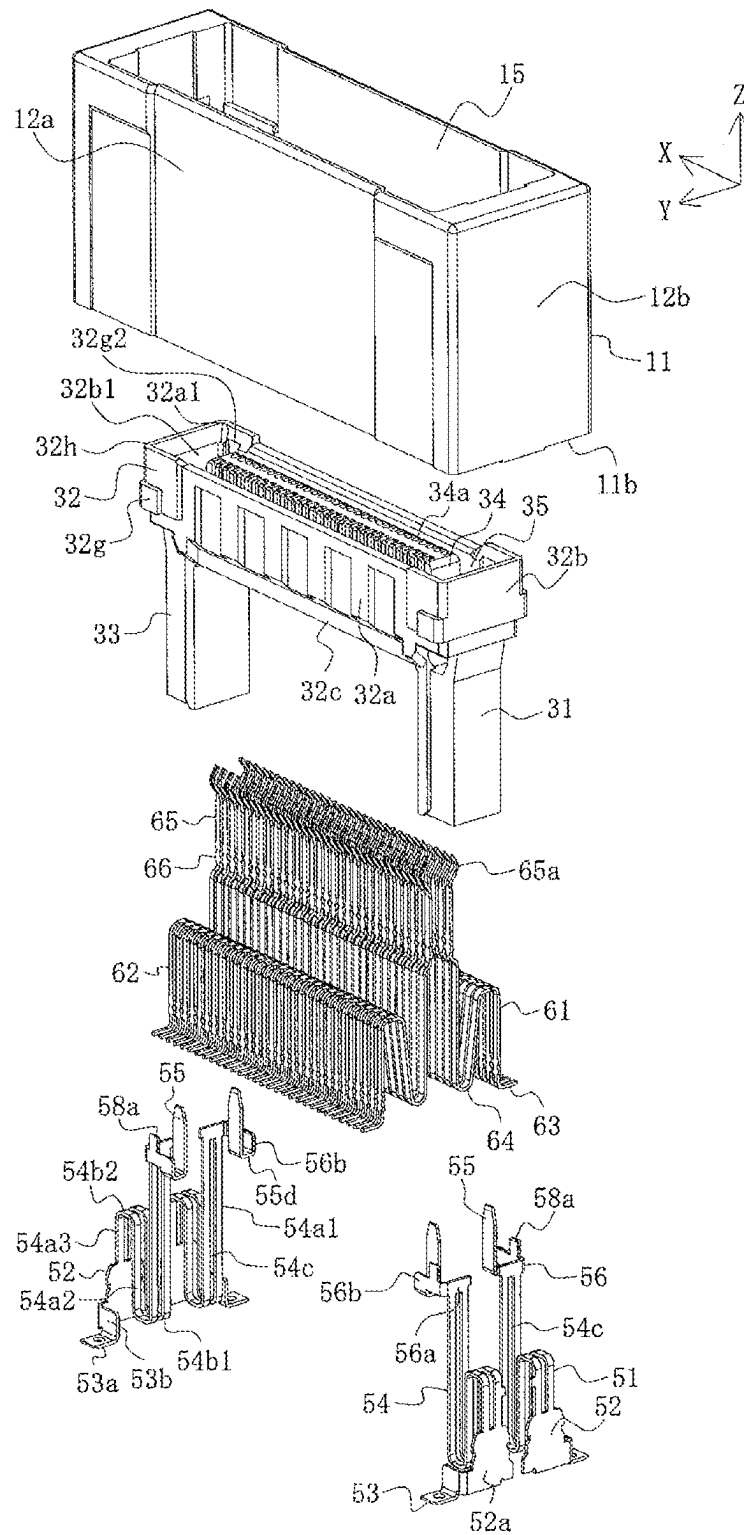


FIG. 19A

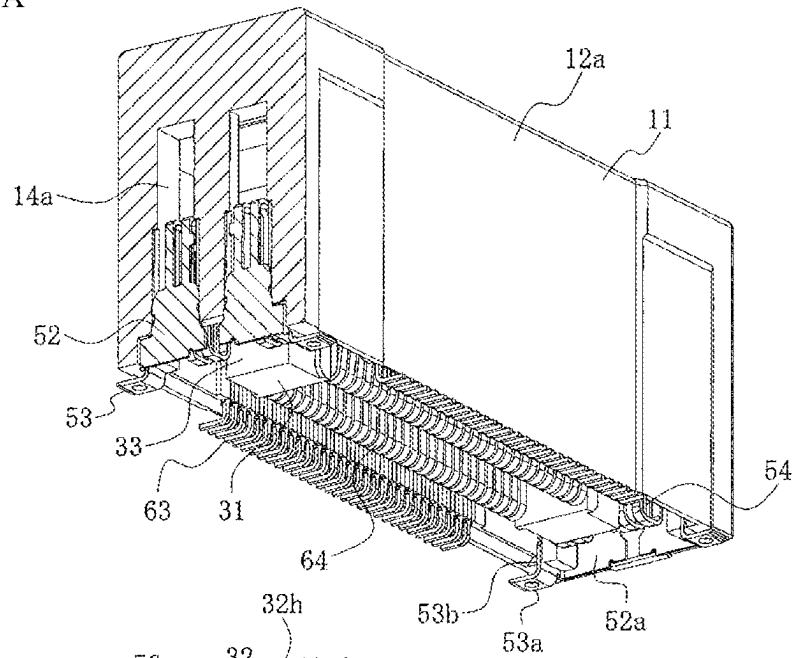
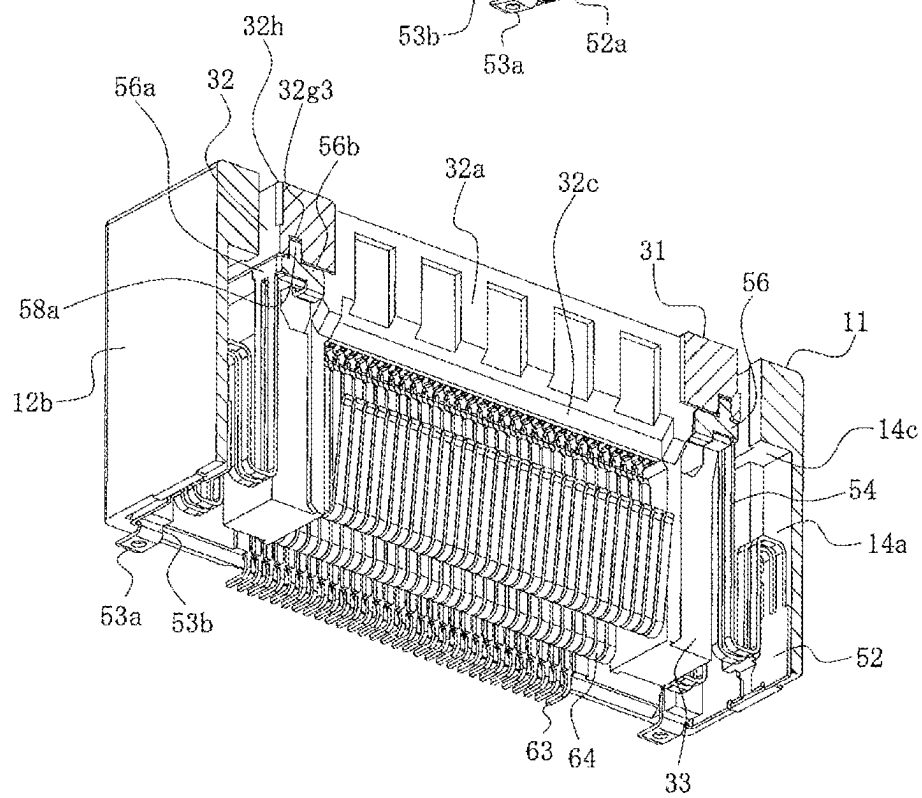


FIG. 19B



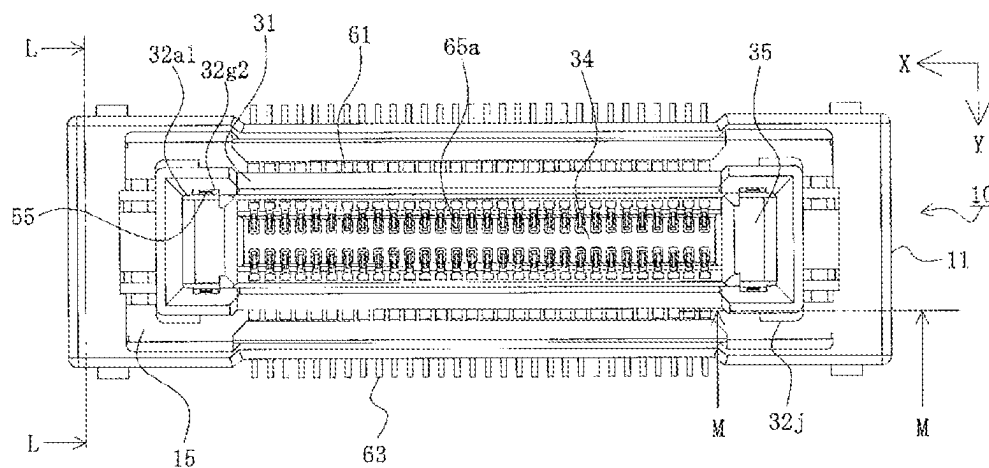


FIG. 20A

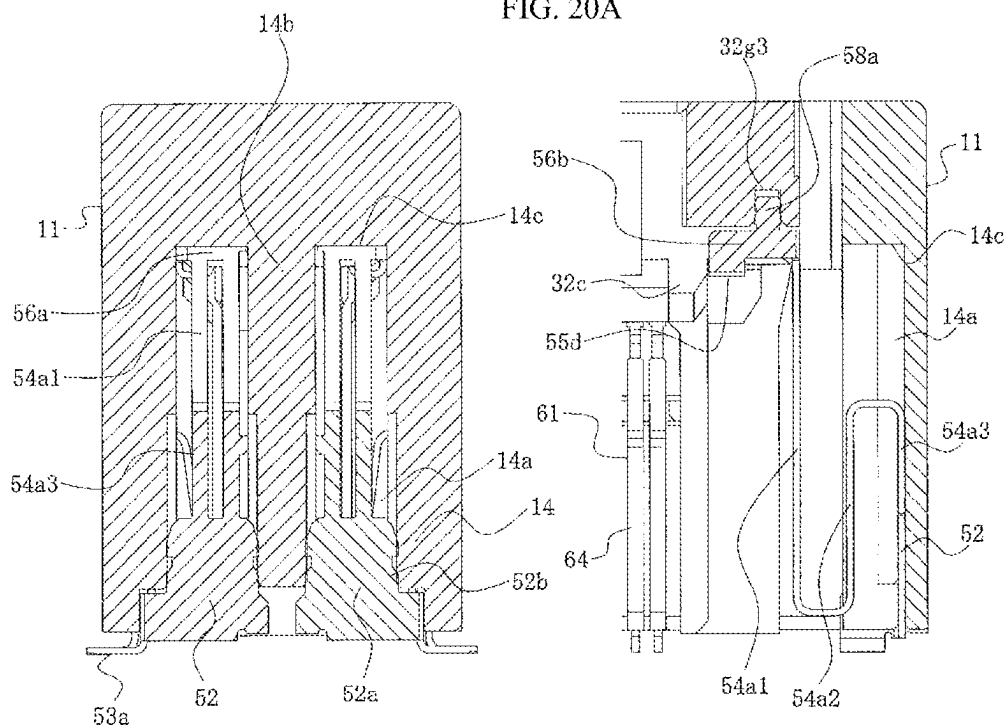
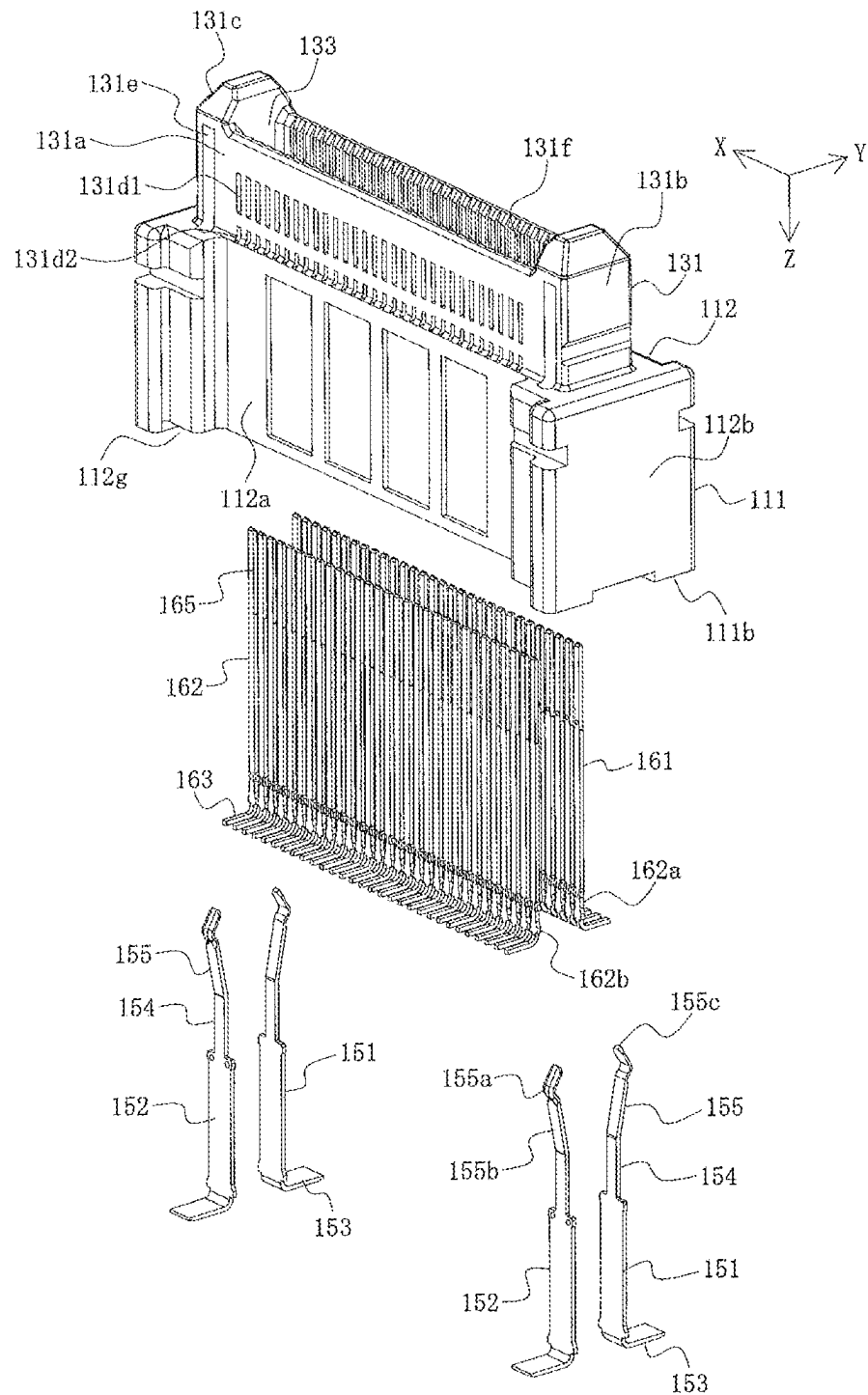


FIG. 20B

FIG. 20C

FIG. 21



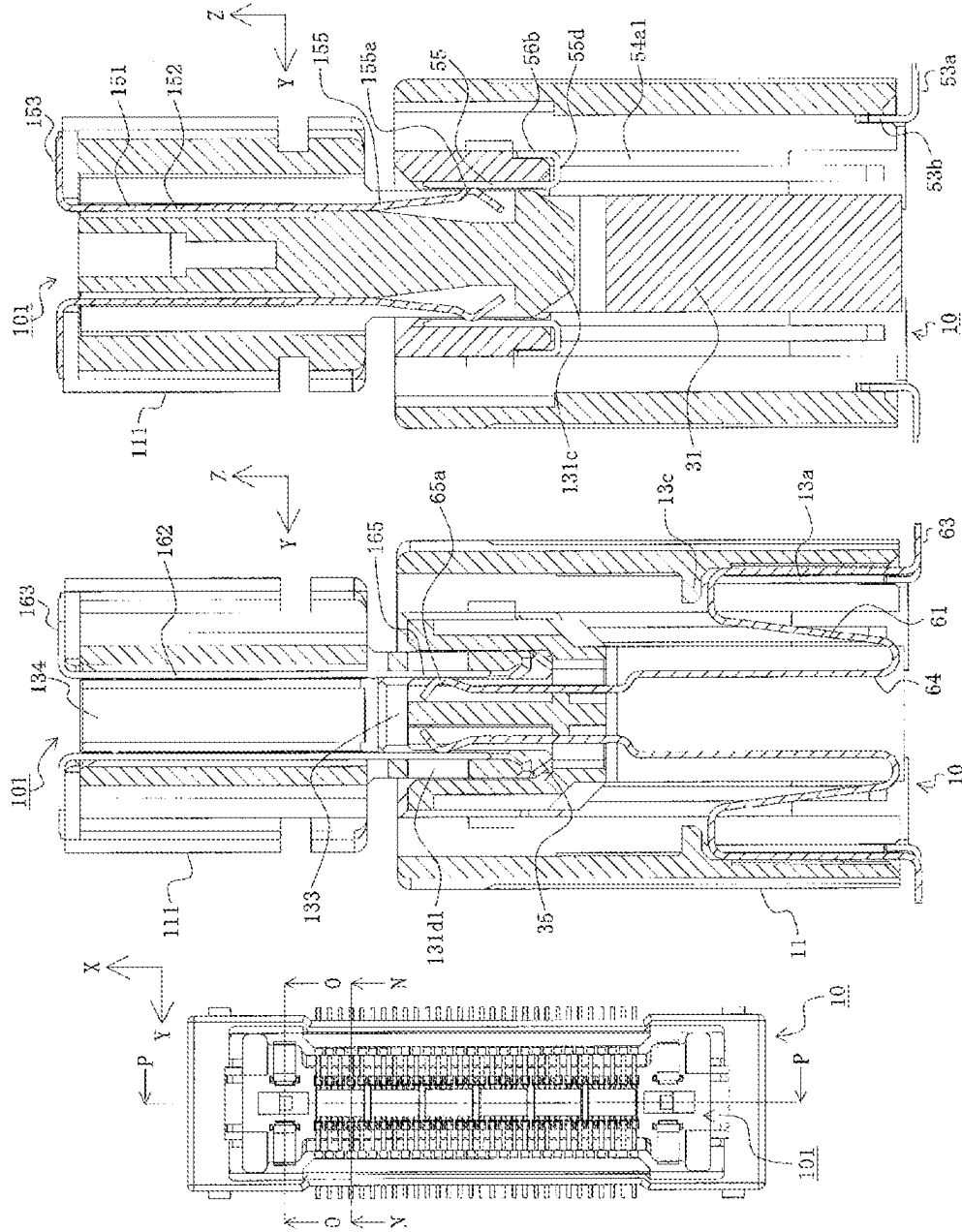


FIG. 22A

FIG. 22B

FIG. 22C

FIG. 23

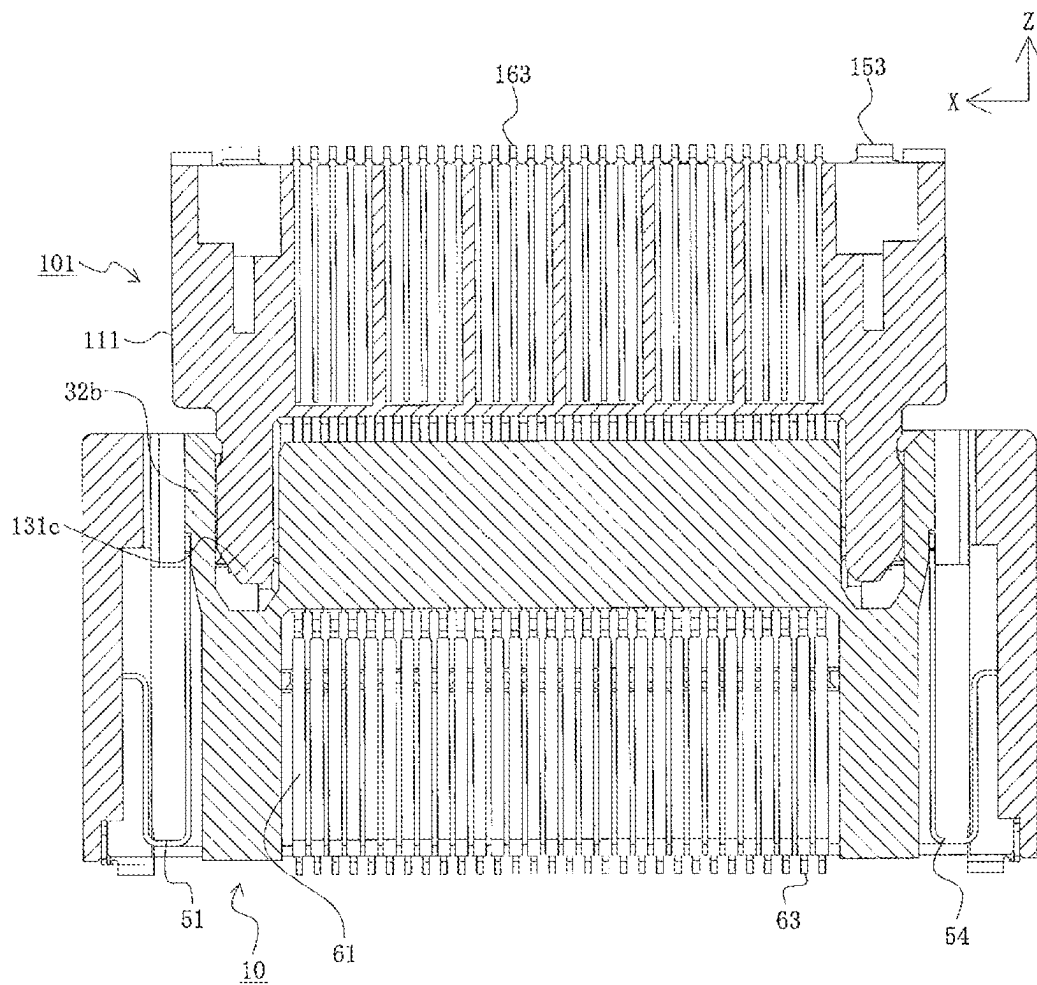


FIG. 24A

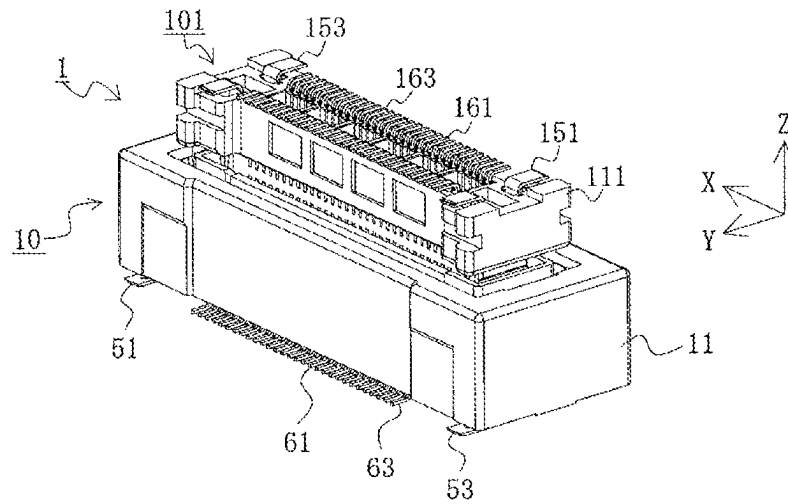


FIG. 24B

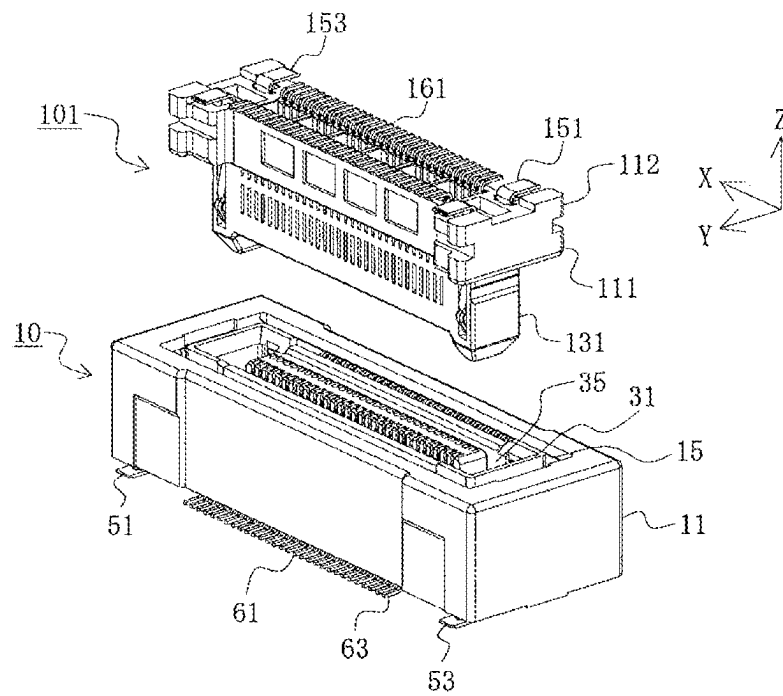
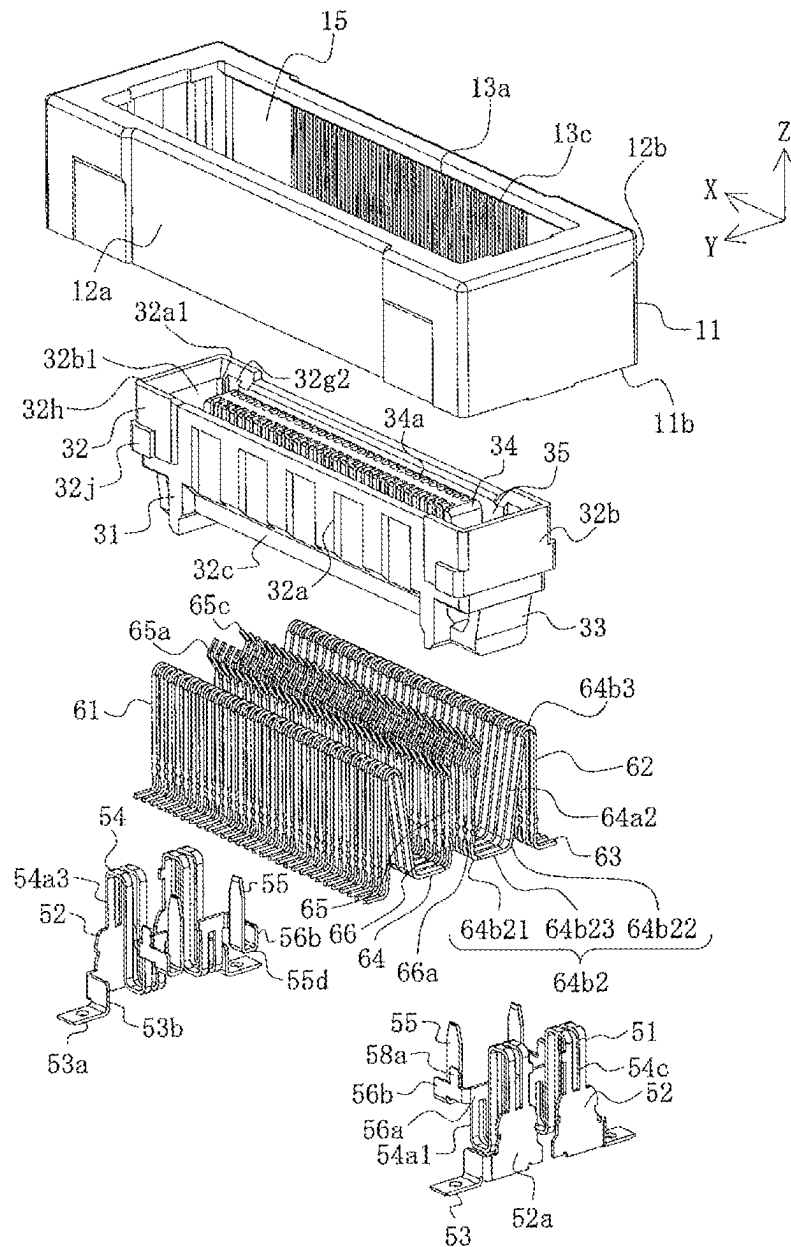


FIG. 25



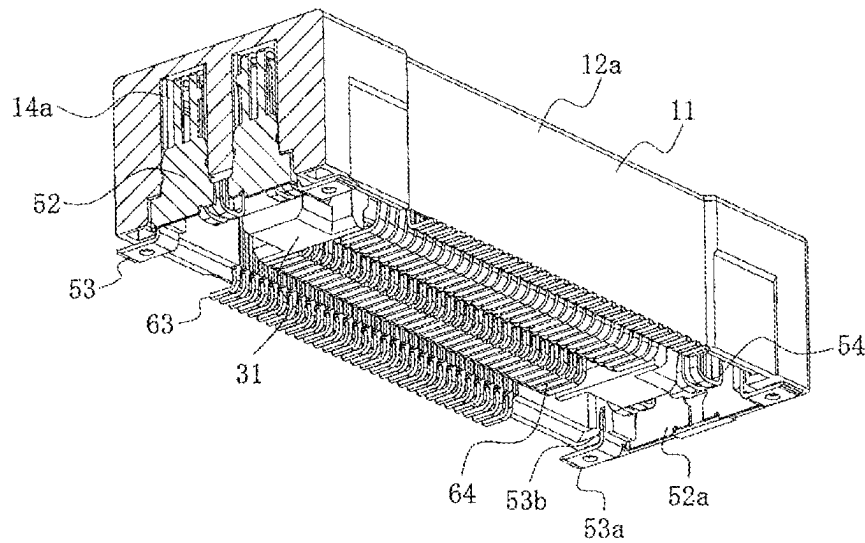


FIG. 26A

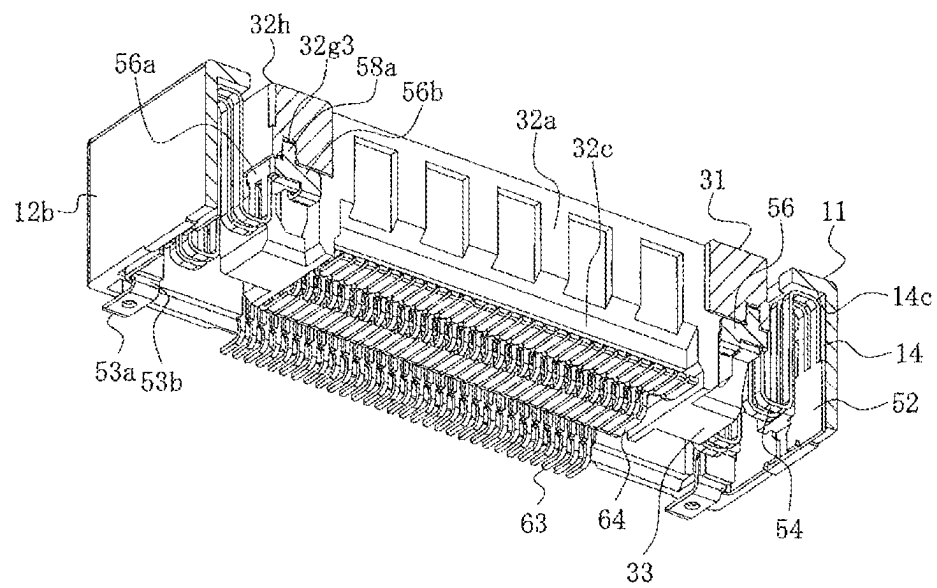


FIG. 26B

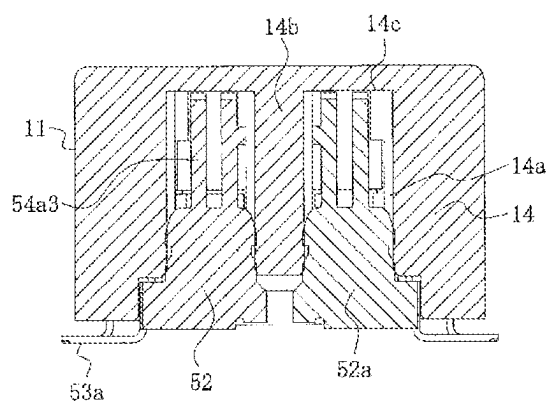
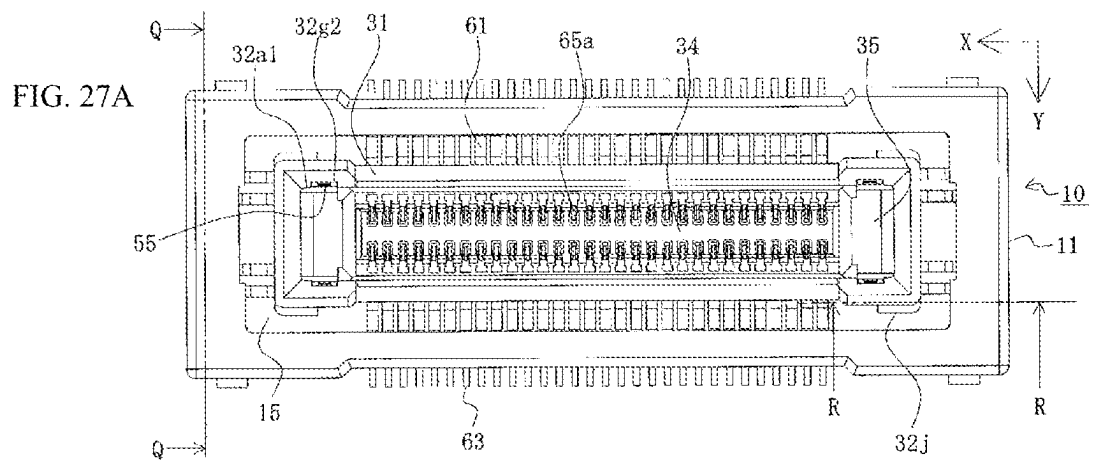


FIG. 27B

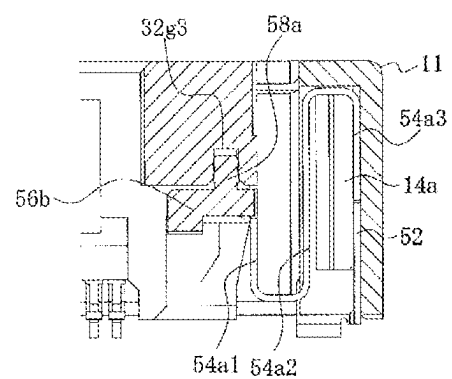
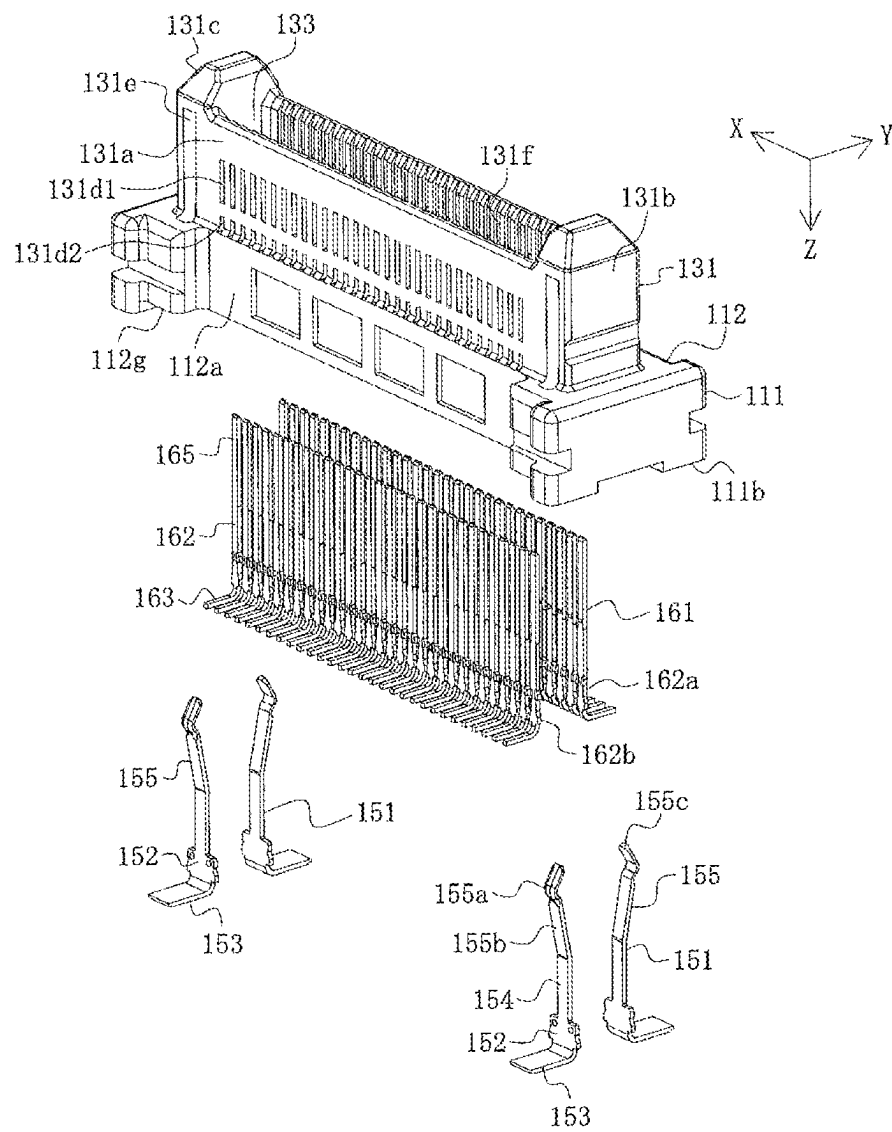


FIG. 27C

FIG. 28



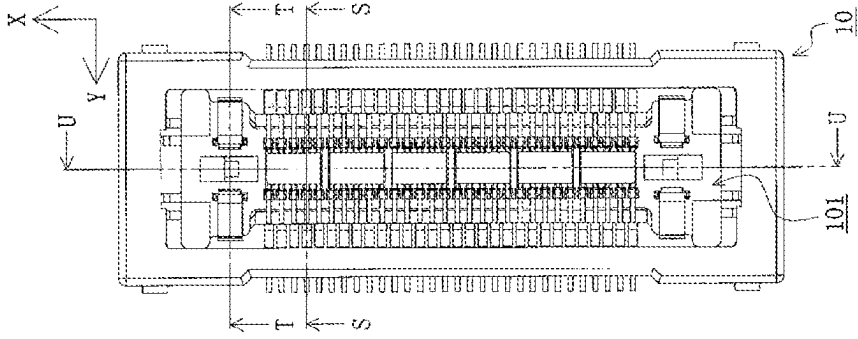


FIG. 29A

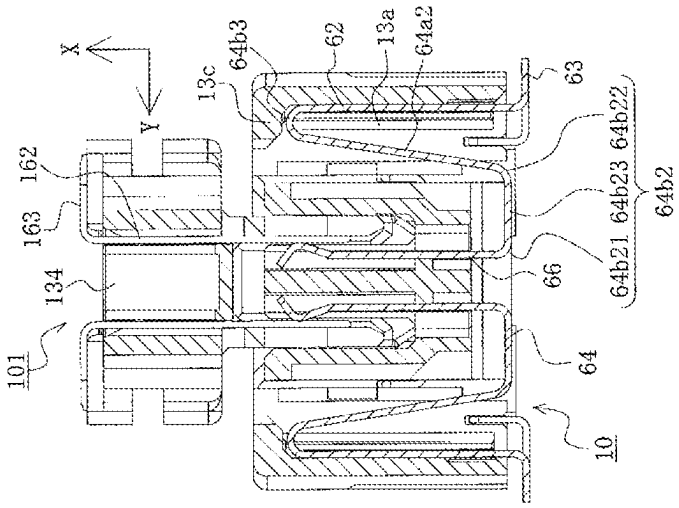


FIG. 29B

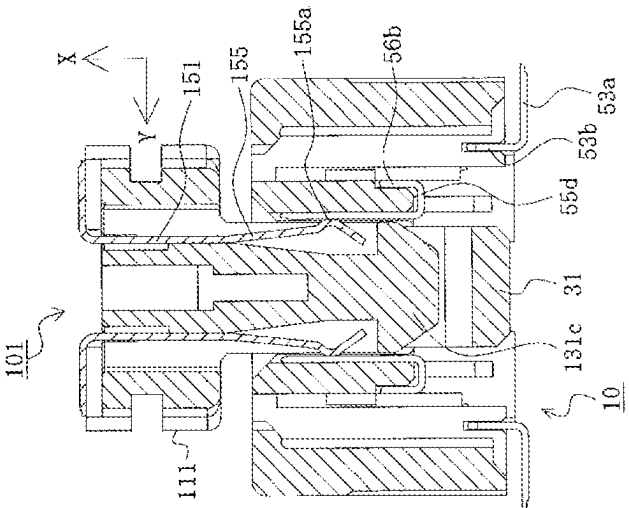


FIG. 29C

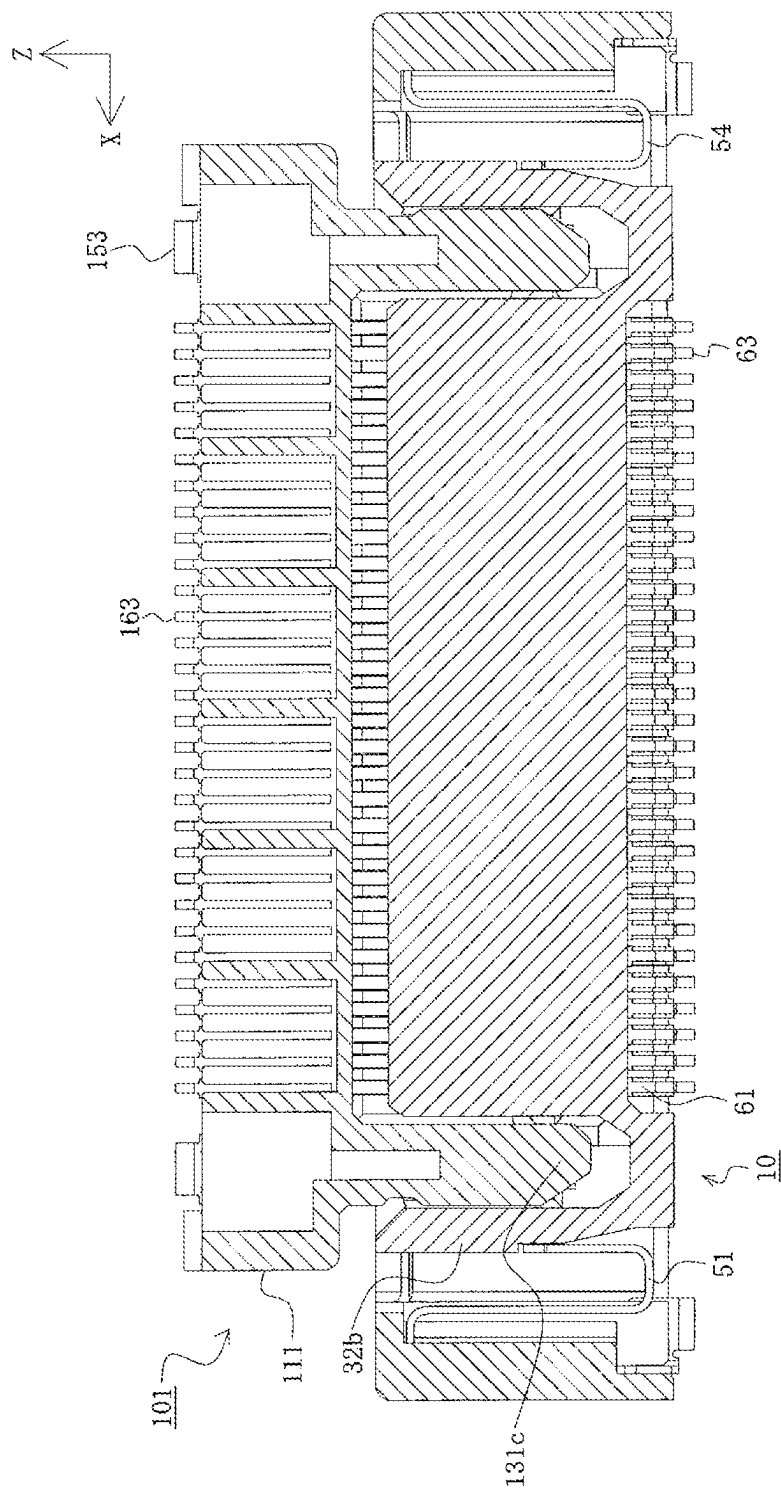


FIG. 30

FIG. 31

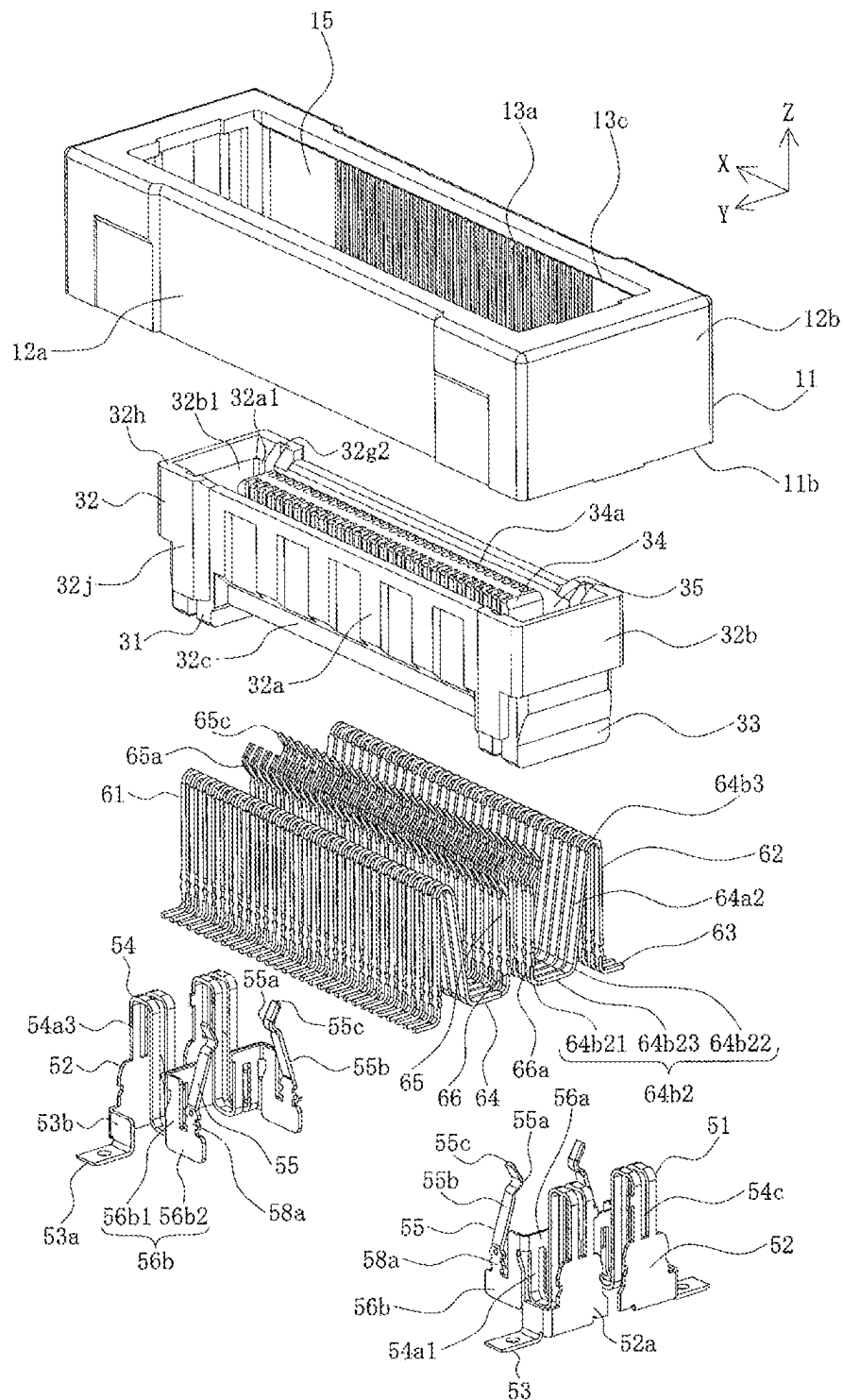


FIG. 32A

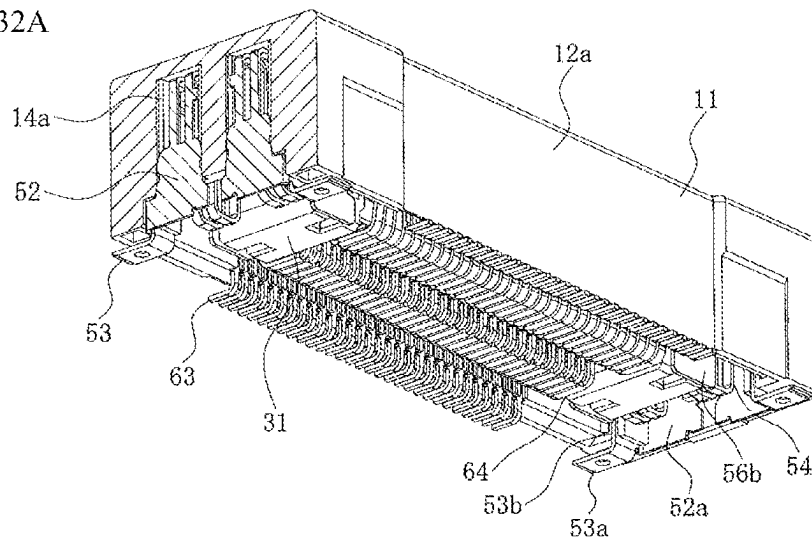


FIG. 32B

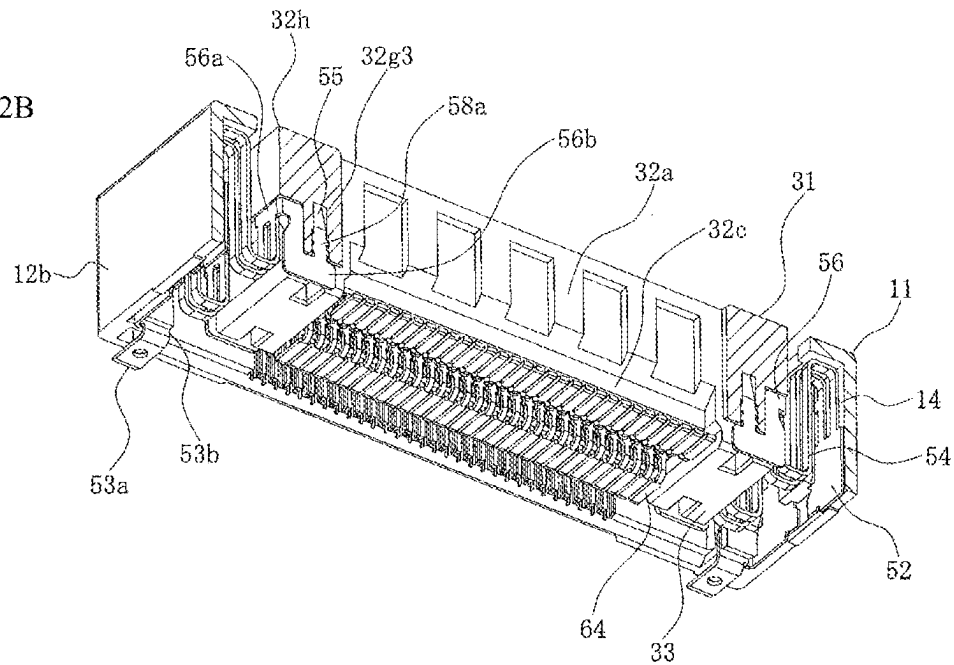


FIG. 33A

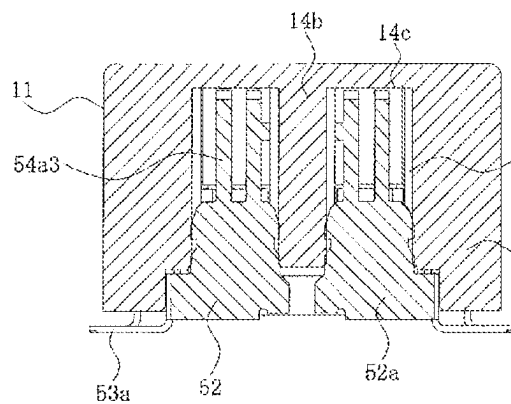
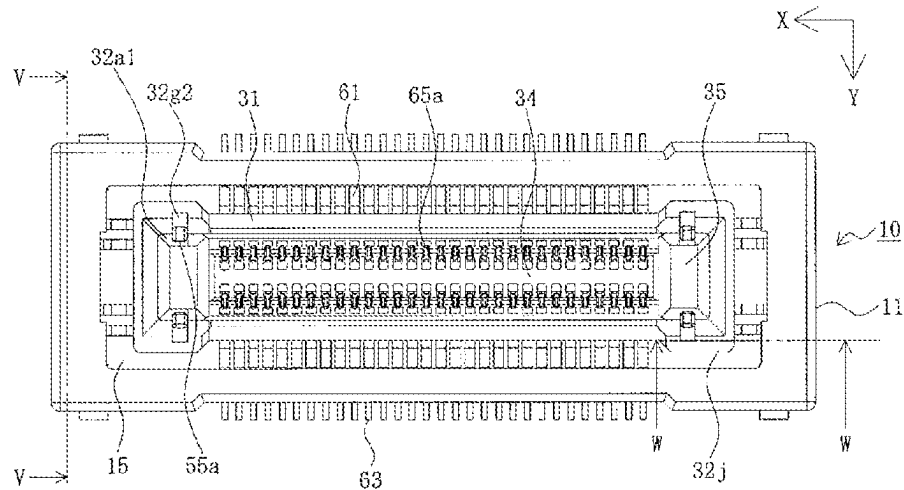


FIG. 33B

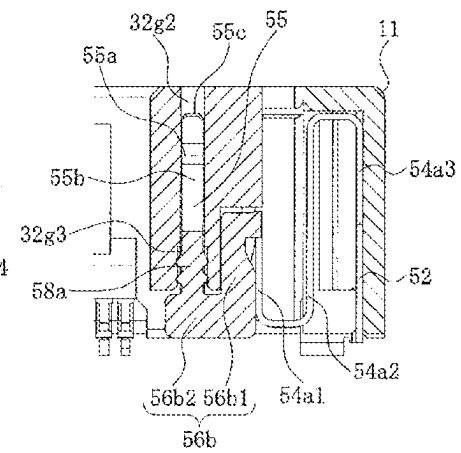
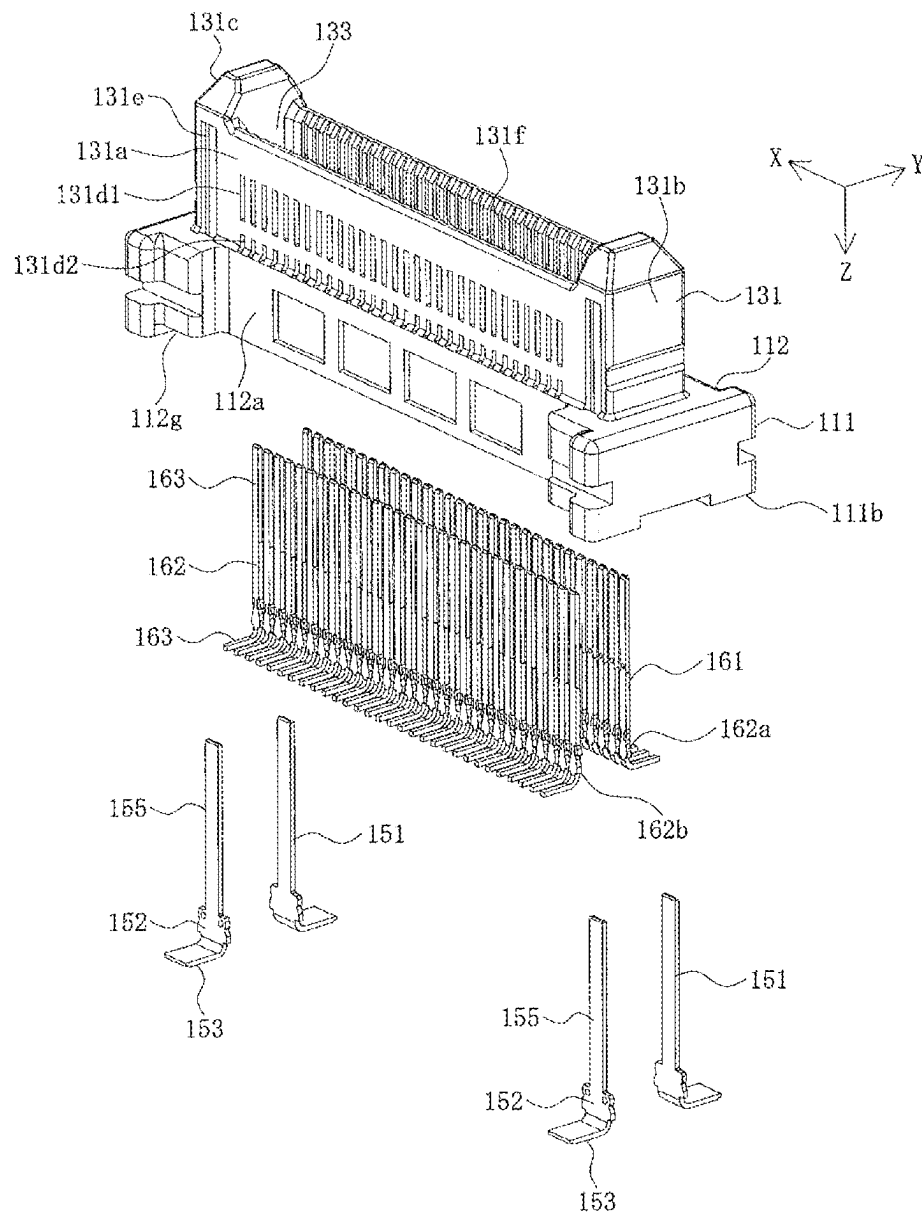


FIG. 33C

FIG. 34



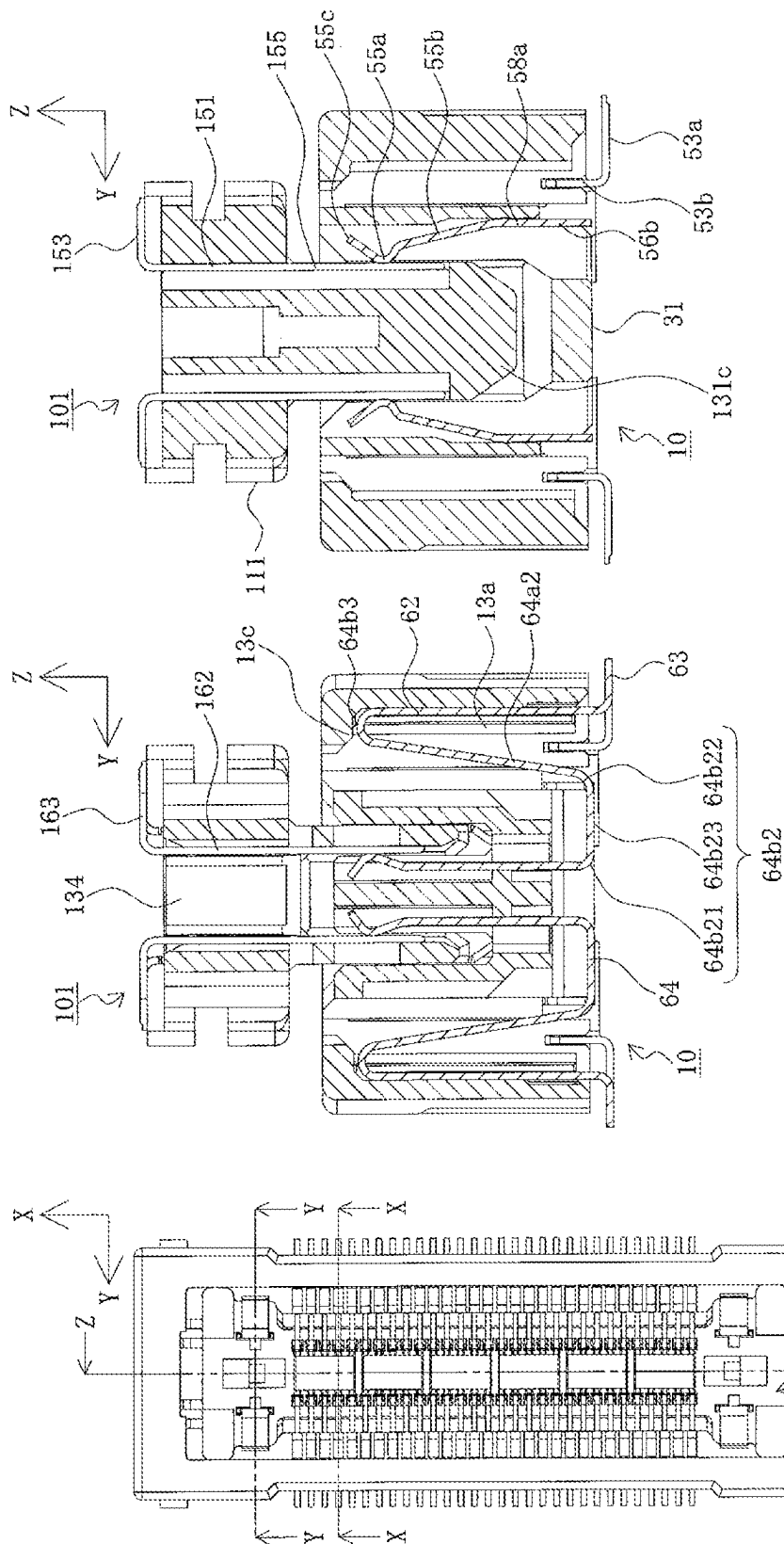


FIG. 35C

FIG. 35B

FIG. 35A

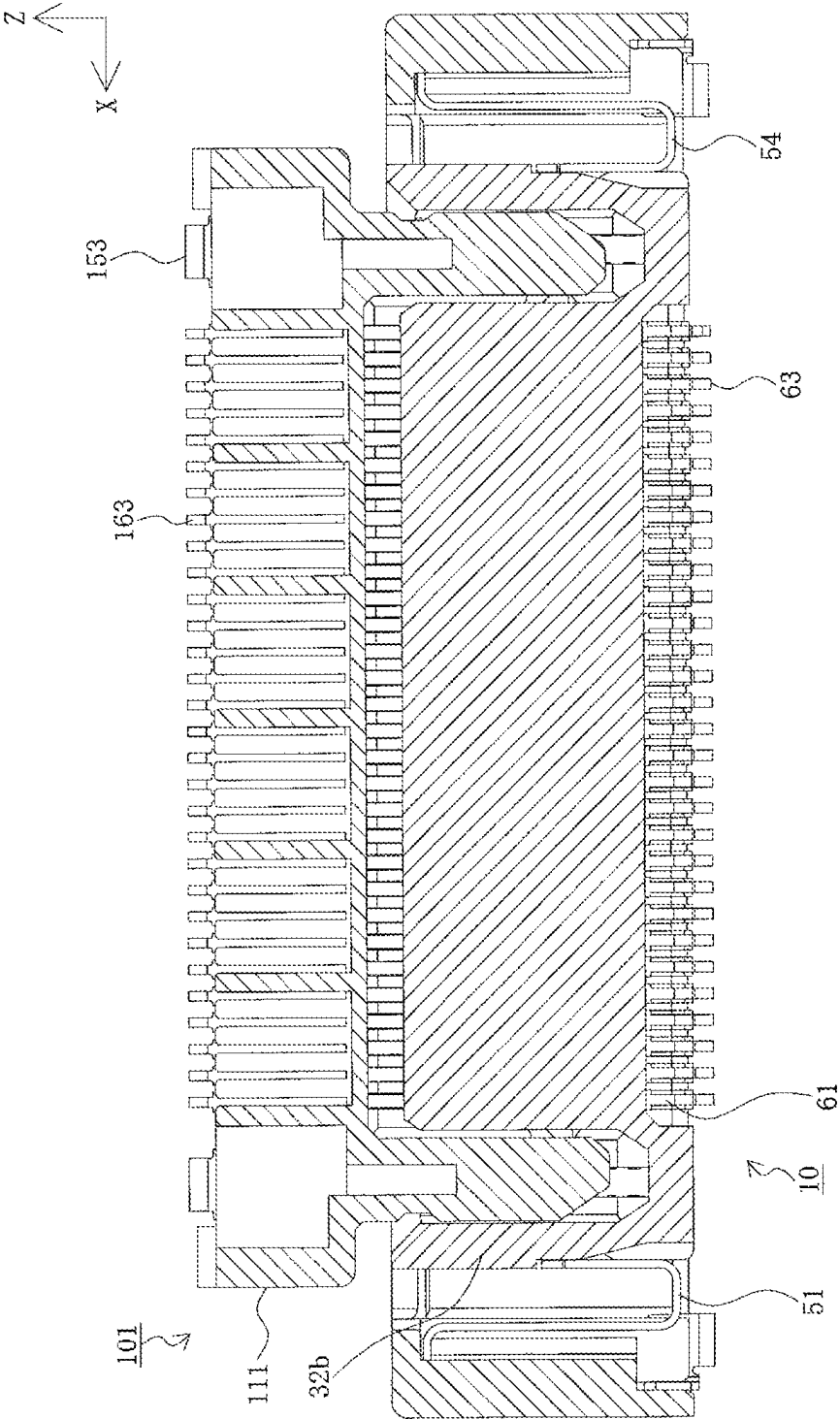
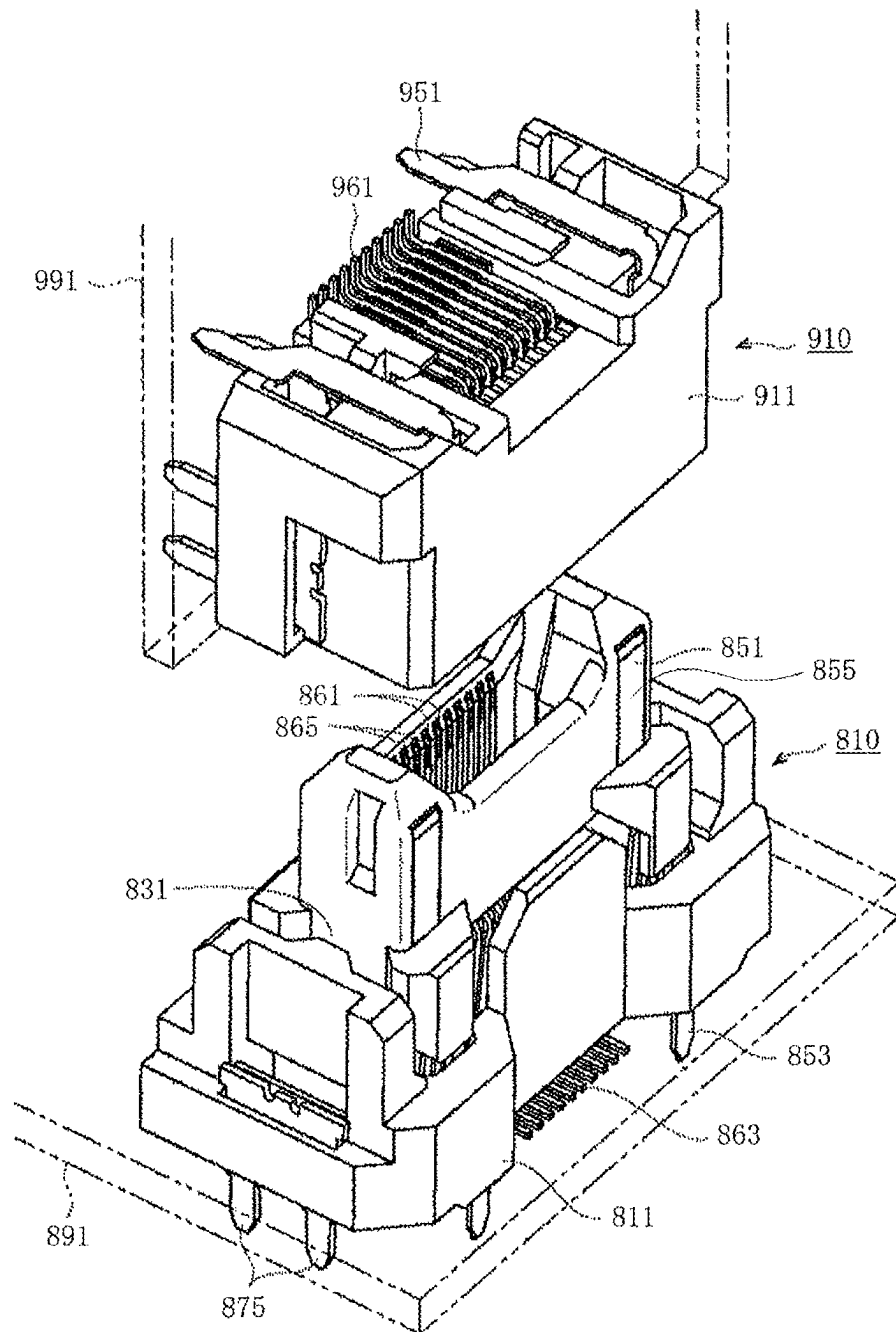


FIG. 36

FIG. 37



Prior art

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CONNECTOR

TECHNICAL FIELD

The present disclosure relates to a connector.

BACKGROUND ART

Conventionally, board-to-board connectors have been used to electrically connect pairs of circuit boards to each other. However, the relative positions of the circuit boards set in advance may shift. Therefore, in order to absorb misalignment of the circuit board, providing a floating structure for one or both connectors has been proposed (see, for example, Patent Document 1).

FIG. 37 is a perspective view illustrating a conventional board-to-board connector prior to mating.

In the diagram, **810** is a floating connector mounted on a surface of the first circuit board **891** and includes a fixed housing **811** fixed to the surface of the first circuit board **891** and a movable housing **831** connected in a movable manner to the fixed housing **811**. The floating connector **810** includes a plurality of signal terminals **861** and a power supply terminal **851**.

On the other hand, the counterpart connector **910** includes a counterpart housing **911** secured to a surface of a second circuit board **991** and a counterpart signal terminal **961** and a counterpart power supply terminal **951** attached to the counterpart housing **911**.

With the floating connector **810**, each signal terminal **861** has a contact portion **865** that contacts the counterpart signal terminal **961** of the counterpart connector **910**, a board connecting portion **863** that is connected to the surface of the first circuit board **891**, and an elastic deformation portion (not shown) formed between the contact portion **865** and the board connecting portion **863**. In addition, each power supply terminal **851** also has a contact portion **855** that contacts the counterpart power supply terminal **951** of the counterpart connector **910**, a board connecting portion **853** that is connected and fixed to the surface of the first circuit board **891**, and an elastic deformation portion (not shown) formed between the contact portion **855** and the board connecting portion **853**.

The contact portion **865** of the signal terminal **861** and the contact portion **855** of the power supply terminal **851** are attached to the movable housing **831**. The board connecting portion **863** of the signal terminal **861** and the board connecting portion **853** of the power supply terminal **851** are attached to the movable housing **831**, connected, and fixed to the surface of the first circuit board **891**. Therefore, the movable housing **831** can move relative to the fixed housing **811** fixed on the surface of the first circuit board **891** if the elastic deformation portion of the signal terminal **861** or the elastic deformation portion of the power supply terminal **851** are deformed. Note that the fixed housing **811** is fixed to the surface of the first circuit board **891** with a fixation fitting **875**.

Furthermore, in a state where the floating connector **810** and the counterpart connector **910** are mated, the movable housing **831** and the counterpart housing **911** mate together, the contact portion **865** of the signal terminal **861** is in contact with the counterpart signal terminal **961**, and the contact portion **855** of the power supply terminal **851** is in contact with the counterpart power supply terminal **951**. Accordingly, the first circuit board **891** and the second circuit board **991** are electrically connected. Also, even if

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alignment offset occurs between the floating connector **810** and the counterpart connector **910**, this can be reliably absorbed.

Patent Document 1: Japanese Unexamined Patent Application 2017-079214

SUMMARY

However, with the conventional connector, the elastic deformation portion of the signal terminal **861** and the elastic deformation portion of the power supply terminal **851** have the same form, and thus readily elastically deform in a similar direction so position offset between the floating connector **810** and the counterpart connector **910** may not be appropriately absorbed depending on the direction thereof. In addition, since the reinforcing member made of metal is not attached to the mating end of the movable housing **831** or the mating end of the counterpart housing **911**, the mating end of the movable housing **831** or the counterpart housing **911** may be injured or damaged during mating operation.

To prevent this manner of situation from happening, the elastic deformation portion of the signal terminal **861** or the power supply terminal **851** may be increased in size, the pitch thereof may be widened to increase the degree of freedom of deformation, the mating end of the movable housing **831** or the counterpart housing **911** may be enlarged, or a separate reinforcing fitting may be attached so that damage occurs less readily. However, in recent years, miniaturization of electronic components and electronic devices have led to size reduction of connectors as well. Therefore, dimensions and pitch of every component of terminals have become extremely fine and similarly, attaching a separate reinforcing member that would increase dimensions of every part of the housing, is difficult.

Here, to resolve the conventional problems described above, an object is to provide a connector with a high degree of freedom of displacement of the movable housing, that enables reliably maintaining a mated state, that enables size reduction, that has a simple construction, that has a small number of components, that is simple to manufacture, that is low cost, and that has high reliability.

To resolve the problems, a connector is provided with a fixed housing, a movable housing movable relative to the fixed housing, and power supply terminals retained in the fixed housing and the movable housing, wherein the fixed housing includes a long wall portion extending in the longitudinal direction of the connector and a short wall portion extending in the width direction of the connector and connected to the end portion of the long wall portion, the movable housing includes a long wall portion extending in the longitudinal direction of the connector and a short wall portion extending in the width direction of the connector and connected to the end portion of the long wall portion, the power supply terminals include a fixed-side retained portion retained in the short wall portion of the fixed housing, a movable-side retained portion retained in the movable housing, and a bent portion connected to the fixed-side retained portion and the movable-side retained portion, and the movable-side retained portion includes a first retained portion with at least a portion extending along the short wall portion and a second retained portion with at least a portion extending along the long wall portion.

With another connector, the first retained portion includes a first upper end cover portion covering an upper end portion of the short wall portion, and the second retained portion includes a second upper end cover portion covering an upper end portion of the long wall portion.

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With still another connector, the second retained portion is arranged inside the long wall portion and is connected to a contact portion that comes into contact with the counter-part power supply terminal.

With still another connector, the contact portion extends in the vertical direction along an inner surface of the long wall portion.

With still another connector, the contact portion includes a contact arm portion that extends in the vertical direction at an incline with respect to an inner surface of the long wall portion, and a contact protruding portion that is connected to an upper end of the contact arm portion and bulges inward in the width direction of the connector.

With still another connector, a second retained portion of the movable-side retained portion includes an engaged portion that is engaged to the movable housing.

With still another connector, the engaged portion is inserted and engaged from below in an engagement recess portion in which the lower end is open to the downward facing surface of the movable housing.

With still another connector, the bent portion is bent in a substantially S-shape when viewed from the width direction of the connector, having a first end connected to an upper end of the fixed-side retained portion and a second end connected to a lower end of a first retained portion of the movable-side retained portion.

With still another connector, the power supply terminal includes a board connecting portion connected to the fixed-side retained portion, and the board connecting portion includes a relay portion extending from the fixed-side retained portion toward the center of the connector in a longitudinal direction and a main body portion connected to the relay portion that extends towards the outside of the connector in the width direction.

With still another connector, the power supply terminals are arranged in pairs at both ends of the connector in the longitudinal direction lined up in the width direction of the connector respectively at both ends in the longitudinal direction.

According to the present disclosure, the connector is capable of ensuring a high degree of freedom of displacement of the movable housing and maintaining a mated state. In addition, the size can be reduced, the structure is simple, the number of components is low, manufacturing is simple, cost can be reduced, and reliability is high.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are perspective views illustrating the first connector and the second connector of Embodiment 1. FIG. 1A illustrates a state of being mated. FIG. 1B illustrates a state prior to being mated.

FIG. 2 is a perspective view illustrating the first connector according to Embodiment 1.

FIGS. 3A-3D are a four-plane diagram of the first connector according to Embodiment 1, where FIG. 3A is a plan view, FIG. 3B is a side view, FIG. 3C is a bottom view, and FIG. 3D is a front view.

FIG. 4 is an exploded view of the first connector according to Embodiment 1.

FIGS. 5A and 5B are perspective views illustrating a terminal of the first connector according to Embodiment 1. FIG. 5A illustrates a signal terminal. FIG. 5B illustrates a power supply terminal.

FIGS. 6A and 6B are perspective views of the relationship between the movable housing and the signal terminal of the first connector according to Embodiment 1 as viewed diagonally

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from below. FIG. 6A is an overall diagram. FIG. 6B is a sectional view taken along the line A-A in FIG. 6A.

FIG. 7 is a perspective view of one signal terminal assembly of the first connector according to Embodiment 1.

FIGS. 8A-8D are diagrams illustrating an attaching portion of the power supply terminal of the first connector onto the movable housing according to Embodiment 1. FIG. 8A is a perspective view of the first connector as viewed diagonally from above. FIG. 8B is an enlarged view of part B of FIG. 8A. FIG. 8C is a plan view of the first connector. FIG. 8D is an enlarged view of part C of FIG. 8C.

FIGS. 9A and 9B are diagrams illustrating an attaching portion of the fixed housing of the first connector signal terminal to the movable housing according to Embodiment 1. FIG. 9A is a sectional view taken along the line D-D of FIG. 3A. FIG. 9B is an enlarged view of part E of FIG. 9A.

FIGS. 10A and 10B are diagrams illustrating a terminal mounting portion of a fixed housing of the first connector according to Embodiment 1. FIG. 10A is a perspective view of the fixed terminal of the first connector as viewed diagonally from below. FIG. 10B is an enlarged view of part F of FIG. 10A.

FIG. 11 is a perspective view of the second connector according to Embodiment 1.

FIGS. 12A-12D are four-plane diagrams of the second connector according to Embodiment 1, where FIG. 12A is a plan view, FIG. 12B is a side view, FIG. 12C is a bottom view, and FIG. 12D is a front view.

FIG. 13 is an exploded view illustrating the second connector according to Embodiment 1.

FIGS. 14A and 14B are perspective views illustrating a terminal of the second connector according to Embodiment 1. FIG. 14A illustrates a signal terminal. FIG. 14B illustrates a power supply terminal.

FIGS. 15A-15C are diagrams illustrating an attaching portion of the second connector signal terminal to the fixed housing according to Embodiment 1. FIG. 15A is a sectional view taken along the line G-G of FIG. 12A. FIG. 15B is a perspective view. FIG. 15C is an enlarged view of part H of FIG. 15B.

FIGS. 16A-16C are three views that illustrate the first connector and the second connector according to Embodiment 1 in a mated state. FIG. 16A is a plan view from the second connector side. FIG. 16B is a sectional view taken along the line I-I of FIG. 16A. FIG. 16C is a sectional view taken along the line J-J of FIG. 16A.

FIG. 17 is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 1 in a mated state and is a sectional view taken along the line K-K of FIG. 16A.

FIG. 18 is an exploded view illustrating the first connector according to embodiment 2.

FIGS. 19A and 19B are perspective views illustrating the relationship of the movable housing and the signal terminal of the first connector according to Embodiment 2 as viewed diagonally from below. FIG. 19A is a perspective view including a cross section of the first connector as viewed diagonally from below. FIG. 19B is a perspective view including a side cross section of the first connector as viewed diagonally from below.

FIGS. 20A-20C illustrate a mounting portion of the first connector power supply terminal onto the movable housing according to Embodiment 2. FIG. 20A is a plan view of the first connector. FIG. 20B is a sectional view taken along the line L-L of FIG. 20A. FIG. 20C is a sectional view taken along the line M-M of FIG. 20A.

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FIG. 21 is an exploded view of the second connector according to Embodiment 2.

FIGS. 22A-22C are three views illustrating a state of the first connector and the second connector according to Embodiment 2 in a mated state. FIG. 22A is a plan view as viewed from the second connector side. FIG. 22B is a sectional view taken along the line N-N of FIG. 22A. FIG. 22C is a sectional view taken along the line O-O of FIG. 22A.

FIG. 23 is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 2 in a mated state and is a sectional view taken along the line P-P in FIG. 22A.

FIGS. 24A and 24B are perspective views illustrating the first connector and the second connector of Embodiment 3. FIG. 24A illustrates a state of being mated. FIG. 24B illustrates a state prior to being mated.

FIG. 25 is an exploded view of the first connector according to Embodiment 3.

FIGS. 26A and 26B are perspective views illustrating the relationship of the movable housing and the signal terminal of the first connector according to Embodiment 3 as viewed diagonally from below. FIG. 26A is a perspective view including a cross section of the first connector as viewed diagonally from below. FIG. 26B is a perspective view including a side cross section of the first connector as viewed diagonally from below.

FIGS. 27A-27C illustrate a mounting portion of the first connector power supply terminal onto the movable housing according to Embodiment 3. FIG. 27A is a plan view of the first connector. FIG. 27B is a sectional view taken along the line Q-Q of FIG. 27A. FIG. 27C is a sectional view taken along the line R-R of FIG. 27A.

FIG. 28 is an exploded view illustrating the second connector according to Embodiment 3.

FIGS. 29A-29C are three views that illustrate the first connector and the second connector according to Embodiment 3 in a mated state. FIG. 29A is a plan view from the second connector side. FIG. 29B is a sectional view taken along the line I-I of FIG. 29A. FIG. 29C is a sectional view taken along the line J-J of FIG. 29A.

FIG. 30 is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 3 in a mated state and is a sectional view taken along the line U-U of FIG. 29A.

FIG. 31 is an exploded view illustrating the first connector according to Embodiment 4.

FIGS. 32A and 32B are perspective views illustrating the relationship of the movable housing and the signal terminal of the first connector according to Embodiment 4 as viewed diagonally from below. FIG. 32A is a perspective view including a cross section of the first connector as viewed diagonally from below. FIG. 32B is a perspective view including a side cross section of the first connector as viewed diagonally from below.

FIGS. 33A-33C illustrate a mounting portion of the first connector power supply terminal onto the movable housing according to Embodiment 4. FIG. 33A is a plan view of the first connector. FIG. 33B is a sectional view taken along the line Q-Q of FIG. 33A. FIG. 33C is a sectional view taken along the line R-R of FIG. 33A.

FIG. 34 is an exploded view illustrating the second connector according to embodiment 4.

FIGS. 35A-35C are three views that illustrate the first connector and the second connector according to Embodiment 4 in a mated state. FIG. 35A is a plan view from the second connector side. FIG. 35B is a sectional view taken

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along the line I-I of FIG. 35A. FIG. 35C is a sectional view taken along the line J-J of FIG. 35A.

FIG. 36 is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 4 in a mated state and is a sectional view taken along the line Z-Z of FIG. 35A.

FIG. 37 is a perspective view illustrating a conventional board to board connector prior to mating.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Embodiments will hereinafter be described in detail with reference to the drawings.

FIG. 1 is a perspective view illustrating the first connector and the second connector according to Embodiment 1. FIG. 2 is a perspective view of the first connector according to Embodiment 1. FIGS. 3A-3D are four views of the first connector according to Embodiment 1. FIG. 4 is an exploded view of the first connector according to Embodiment 1. FIGS. 5A and 5B are perspective views of the first connector terminals according to Embodiment 1. FIGS. 6A and 6B are perspective views diagonally from below of the relationship between the movable housing and the signal terminal of the first connector according to Embodiment 1. FIG. 7 is a perspective view of one signal terminal assembly of the first connector according to Embodiment 1. FIGS. 8A-8D illustrate a mounting portion of the power supply terminal of the first connector to the movable housing according to Embodiment 1. FIGS. 9A and 9B illustrate a mounting portion of the signal terminal of the first connector to the fixed housing and the movable housing in Embodiment 1. FIGS. 10A and 10B illustrate a terminal mounting portion of the fixed housing of the first connector according to Embodiment 1.

Note, FIG. 1A is a diagram illustrating a mated state and FIG. 1B is a diagram illustrating a state prior to mating. FIG. 3A is a top view, FIG. 3B is a side view, FIG. 3C is a bottom view, and FIG. 3D is a front view. FIG. 5A is a diagram illustrating a signal terminal, and FIG. 5B is a diagram illustrating a power supply terminal. FIG. 6A is a view illustrating the whole, and FIG. 6B is a sectional view taken along the line A-A in FIG. 6A. FIG. 8A is a perspective view of the first connector as seen diagonally from above, FIG. 8B is an enlarged view of part B in FIG. 8A, FIG. 8C is a top view of the first connector, and FIG. 8D is an enlarged view of part C in FIG. 8C. In FIG. 9A is a sectional view taken along the line D-D in FIG. 3A and FIG. 9B is an enlarged view of part E in FIG. 9A. FIG. 10A is a perspective view of the fixed terminal of the first connector as seen diagonally from below, and FIG. 10B is an enlarged view of part F in FIG. 10A.

In the diagrams, 10 represents a first connector as one of a board to board connector 1, which are connectors of Embodiment 1, in other words, the present Embodiment. The first connector 10 is a surface mounting type receptacle connector mounted on a surface of a first board that is a board (not shown) serving as a mounting member and is mated together with a second connector 101 serving as a counterpart connector. The first connector 10 is a so-called floating type connector, and includes a first fixed housing 11 as a fixed housing fixedly attached to a surface of the first board, and a first movable housing 31 as a movable housing movable with respect to the first fixed housing 11.

Furthermore, the second connector 101 is the other of the pair of board to board connectors 1 and is a surface mounting type plug connector mounted on a surface of a second

board (not shown) serving as a mounting member. The second connector **101** is not a so-called floating type connector, and includes only a second housing **111** as a counterpart housing that is a fixed housing fixedly attached to a surface of the second board, and does not include a movable housing that is able to move.

Note that the first connector **10** and the second connector **101** of the connector pair according to the present embodiment are preferably used to electrically connect the first board to the second board but can also be used to electrically connect other members. For example, the first substrate and the second substrate are each a printed circuit board, a flexible flat cable (FFC), a flexible circuit board (FPC), or the like as used in electronic devices or the like, but may be any type of substrate.

Note, in the present embodiment, expressions indicating direction such as up, down, left, right, front, rear, and the like used to describe a configuration and operation of each part of the connector pair first connector **10** and the second connector **101** are relative rather than absolute and are appropriate when each part of the first connector **10** and the second connector **101** are in positions illustrated in the drawings. However, these directions should be interpreted as changing in accordance with a change in position when the position thereof is changed.

As illustrated in FIGS. 3A-3D, the first connector **10** is configured so as to be left-right symmetrical in the width direction (Y-axis direction), and is also configured to be left-right symmetrical when viewed in the longitudinal direction (X-axis direction). The first fixed housing **11** and the first movable housing **31** of the first connector **10** are each members integrally formed by an insulator (dielectric) such as a synthetic resin.

The first fixed housing **11** is a substantially square columnar member, and includes a pair of rectangular flat plate-shaped long wall portions **12a** that are mutually parallel extending in the longitudinal direction of the first connector **10**, short wall portions **12b** that are a mutually parallel pair of rectangular thick plate-like members extending in the width direction of the first connector **10**, are shorter than the long wall portions **12a**, and connect the end portions of the long wall portions **12a** in the longitudinal direction, and a movable housing cavity for stowing **15** demarcated by the four surfaces of the long wall portions **12a** and the short wall portions **12b** and is a hollow square columnar shape that is open at the top and bottom (both ends in the Z-axis direction). The lower surface of the first fixed housing **11** is a mounting surface **11b** facing the surface of the first board. Furthermore, as illustrated in FIGS. 10A and 10B, a signal terminal mounting portion **13** is formed on the surface of each long wall portion **12a** on the movable housing cavity for stowing **15** side (negative direction end of the Z-axis), that is, on the inner surface, and a power supply terminal mounting portion **14** is formed on the inner surface at the lower end of each short wall portion **12b**.

Note that at the lower end of each long wall portion **12a**, a lower end notch **12c** is formed at locations adjacent on both sides in the longitudinal direction of the signal terminal mounting portion **13**. When the first movable housing **31** is stowed in the movable housing cavity for stowing **15**, foot portions **33a** extending outward in the width direction (Y-axis direction) from the lower end of a pair of leg portions **33** positioned at both ends of the first movable housing **31** in the longitudinal direction (X-axis direction) are stowed in the lower end notch **12c**. Accordingly, the movable range of the first movable housing **31** with respect

to the first fixed housing **11** is appropriately limited in the longitudinal direction and the vertical direction.

A signal terminal fixed-side retention recess **13a** is formed in the signal terminal mounting portion **13** that retains the fixed-side retained portion **62** of a first signal terminal **61** that is a signal terminal out of the first terminals the first connector **10** is provided with. Each signal terminal fixed-side retention recess **13a** is an elongated groove shaped recessed portion extending in the vertical direction (Z-axis direction), and the fixed-side retained portion **62** of each of the first signal terminals **61** is stowed therein. In addition, there is a signal terminal fixed-side retention wall **13b** formed so as to protrude relatively toward the center of the first fixed housing **11** in the width direction on both sides of each signal terminal fixed-side retention recess **13a**. Furthermore, a signal terminal fixed-side retention groove **13b1** extending upward from the lower end is formed on both side surfaces of each signal terminal fixed-side retention recess **13a**, that is, the side wall surface of the signal terminal fixed-side retention wall **13b**. In addition, the bottom surface of each signal terminal fixed-side retention recess **13a** has a signal terminal fixed-side retention portion thickness reduction **13a1** formed as a long groove recessed portion thickness reduction extending in the vertical direction. The signal terminal fixed-side retention portion thickness reduction **13a1** is formed to match the impedance of the first signal terminal **61**. Furthermore, the upper end of the signal terminal mounting portion **13** protrudes toward the center in the width direction of the first fixed housing **11** where a brim portion **13c** is formed, demarcating the upper end of the signal terminal fixed-side retention recess **13a**. Note that in the example illustrated in the drawings, there are thirty signal terminal fixed-side retention recesses **13a** formed respectively for the signal terminal mounting portions **13** on the left and the right of the first fixed housing **11**. This number is changed according to the number of first signal terminals **61**.

Additionally, power supply terminal fixed-side retention recesses **14a** are formed in the power supply terminal mounting portion **14** for retaining power supply terminals of the first terminals provided in the first connector **10**, in other words, for retaining a fixed-side retained portion **52** of first power supply terminals **51** that are power supply terminals. Each power supply terminal fixed-side retention recess **14a** is a narrow long groove-shaped recess extending in the vertical direction, in which the fixed-side retained portion **52** of each first power supply terminal **51** is stowed and retained. In addition, power supply terminal fixed-side retention walls **14b** are formed on both sides of each power supply terminal fixed-side retention recess **14a** protruding relatively toward the center of the first fixed housing **11** in the longitudinal direction. A power supply terminal fixed-side retention groove **14b1** extending upward from the lower end is formed on both side surfaces of each power supply terminal fixed-side retention recess **14a**, that is, the side wall surfaces of the power supply terminal fixed-side retention walls **14b**. Furthermore, the upper end of the power supply terminal mounting portion **14** protrudes toward the center of the first fixed housing **11** in the longitudinal direction forming a brim portion **14c** that demarcates the upper end of the power supply terminal fixed-side retention recess **14a**. Note that in the example illustrated in the diagrams, two power supply terminal fixed-side retention recesses **14a** are formed in the power supply terminal mounting portion **14** at both ends in the longitudinal direction of the first fixed housing **11** and the number thereof is changed according to the number of first power supply terminals **51**.

The first movable housing 31 includes a substantially trough shaped main body portion 32 extending in the longitudinal direction of the first connector 10 and a pair of leg portions 33 extending downward from the lower end at both ends in the longitudinal direction of the main body portion 32. Note, the foot portion 33a extending outward in the width direction is connected to the lower end of each leg portion 33. The lower surface of the foot portion 33a is a mounting surface 11b facing the surface of the first board. The main body portion 32 includes a pair of rectangular thick plate-shaped long wall portions 32a parallel to each other and extending in the longitudinal direction of the first connector 10, short wall portions 32b that are a mutually parallel pair of rectangular shaped thick plate-shaped members extending in the width direction of the first connector 10, are shorter than the long wall portions 32a, and connect the end portions of the long wall portions 32a in the longitudinal direction, a counterpart housing cavity for stowing 35 that is demarcated by the four surfaces of the long wall portions 32a and the short wall portions 32b and has a hollow square columnar shape with an upper end (end in the positive direction of the Z-axis) that is open, and a bottom portion 32c that is connected to the lower end of the long wall portions 32a and the lower end of the short wall portions 32b and that blocks the lower end of the counterpart housing cavity for stowing 35 (end in the negative direction of the Z-axis). Note, engaging protrusions 32d that protrude outward are formed on the outer surface near both ends in the longitudinal direction of the long wall portions 32a to engage with movable-side retained portions 56 of the first power supply terminals 51.

As illustrated in FIGS. 1A and 1B, in a state where the first connector 10 and the second connector 101 are mated, at least a portion of an upper portion 131 of the second housing 111 of the second connector 101 is stowed in the counterpart housing cavity for stowing 35. A thick plate-shaped central wall portion 34 extending in the longitudinal direction of the first connector 10 is disposed within the counterpart housing cavity for stowing 35. The central wall portion 34 is a member that extends upward from the upper surface of the bottom portion 32c, as illustrated in FIG. 6B, and the outer peripheral surface thereof is formed so as to be spaced apart from the inner surface of the long wall portions 32a and the inner surface of the short wall portions 32b. Furthermore, as illustrated in FIG. 9A, a plurality of inverted U-shaped signal terminal contact portion grooves for stowing 34a, continuous from the upper end and across both sides, are formed in the central wall portion 34. In addition, a signal terminal movable-side retention recess 32e that extends in the vertical direction and penetrates from the lower surface to an upper surface of the bottom portion 32c is formed in a position corresponding to each signal terminal contact portion groove for stowing 34a of the bottom portion 32c. Contact portions 65 and movable-side retained portions 66 of the first signal terminals 61 are stowed respectively in the left and right portions extending in the vertical direction of each signal terminal contact portion groove for stowing 34a and the signal terminal movable-side retention recess 32e connected thereto.

In addition, as illustrated in FIG. 9B, a signal terminal movable-side retention portion first thickness reduction 32c1 and a signal terminal movable-side retention portion second thickness reduction 32c2 are formed in each signal terminal movable-side retention recess 32e as thickness reductions that are long narrow groove shaped recesses extending in the vertical direction more to the outside in the width direction and to the inside in the width direction of the

first connector 10 than the movable-side retained portion 66. The signal terminal movable-side retention portion first thickness reduction 32c1 and the signal terminal movable-side retention portion second thickness reduction 32c2 are formed to match the impedance of the first signal terminal 61. In the example illustrated in the drawing, 30 signal terminal contact portion grooves for stowing 34a are formed, and 30 signal terminal movable-side retention recesses 32e are formed on the left and the right, and this number is changed according to the number of first signal terminals 61.

Furthermore, as illustrated in FIG. 4, power supply terminal movable-side recesses for stowing 32g are formed in the four corner portions 32h of the main body portion 32 for stowing a part of the movable-side retained portion 56 of the first power supply terminal 51. The power supply terminal movable-side recesses for stowing 32g include first recesses for stowing 32g1 formed near the upper end of an inner surface 32b1 of the short wall portions 32b and second recesses for stowing 32g2 formed near the upper end of an inner surface 32a1 of the long wall portions 32a. As illustrated in FIGS. 8A-8D, a first upper end cover portion 57a and first upper end cover extension portion 57a1 as well as a second upper end cover portion 57b and contact portion 55 of the movable-side retained portion 56 of the first power supply terminal 51 are stowed in the first recess for stowing 32g1 and second recess for stowing 32g2. Note that the surfaces of the first upper end cover extension portion 57a1 and contact portion 55 stowed in the first recess for stowing 32g1 and the second recess for stowing 32g2 are preferably more indented into the short wall portions 32b and the long wall portions 32a than the inner surface 32b1 of the short wall portions 32b and the inner surface 32a1 of the long wall portions 32a. In addition, the top surface of the first upper end cover portion 57a and the second upper end cover portion 57b (upper end surface) is preferably positioned above an upper end surface 32b2 of the short wall portions 32b and an upper end surface 32a2 of the long wall portions 32a.

Each of the first signal terminals 61 are long narrow rod shaped members integrally formed and provided elasticity by performing processing such as punching and bending of a conductive metal plate and as illustrated in FIG. 7, include a fixed-side retained portion 62, a tail portion 63 connected to the lower end of the fixed-side retained portion 62 as a board connecting portion, a bent portion 64 connected to the upper end of the fixed-side retained portion 62, a movable-side retained portion 66 connected to the upper end of the bent portion 64, and a contact portion 65 connected to the upper end of the movable-side retained portion 66. In the example illustrated in the drawings, there are thirty first signal terminals 61 lined up extending in the longitudinal direction of the first connector 10 configured on each of a pair of rows on the left and right having a prescribed pitch (for example, roughly 0.3 to 0.5 mm). The pitch and the number of the first signal terminals 61 can be changed as appropriate. In addition, the first signal terminals 61 in each of the left and right rows are arranged so as to be symmetrical as viewed in the longitudinal direction of the first connector 10. Furthermore, each first signal terminal 61 is formed of a plate material extending in the longitudinal direction of the first connector 10 in a parallel plane and is bent within a cross-sectional surface of the first connector 10 as illustrated in FIGS. 9A and 9B.

A tail portion 63 is bent at approximately 90 degrees relative to the fixed-side retained portion 62 that extends in the vertical direction and is connected thereto and extends

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outward in the width direction of the first fixed housing 11. Furthermore, the lower surface thereof is substantially flush with the mounting surface 11b, which is the lower surface of the first fixed housing 11, and is connected and fixed to a connection pad formed on the surface of the first board (not shown) by means of soldering or the like. This connection pad is typically connected to a conductive trace for transmitting signals on the first board. In addition, as illustrated in FIG. 8C, the tail portions 63 are disposed so that at least the tips thereof are positioned to the outside of the long wall portions 12a of the first fixed housing 11 in the width direction so as to form a row extending along the long wall portions 12a extending in the longitudinal direction of the first connector 10. Accordingly, in the work of mounting the first connector 10 to the surface of the first board, the operator can readily visually confirm the connection state of the tail portions 63 to connection pads formed on the surface of the first board. Furthermore, the length of tips of the tail portions 63 that protrude outside the long wall portions 12a of the first fixed housing 11 is set small to enable shortening the length thereof. This enables preventing an increase in the impedance of the first signal terminal 61.

As illustrated in FIGS. 6A and 6B, the bent portion 64 is a portion that is bent so as to meander when viewed from the longitudinal direction (X-axis direction) of the first connector 10, and functions as an elastic deformation portion. The bent portion 64 includes a first straight portion 64a1 and a second straight portion 64a2 as straight portions, and includes a first bent portion 64b1 and a second bent portion 64b2 as a bent portion that is bent nearly 180 degrees as well as a third bent portion 64b3. The first bent portion 64b1 has an upper end thereof connected to the lower end of the movable-side retained portion 66 and a lower end thereof connected to the first straight portion 64a1 and is bent into a substantially S shape as viewed from the side. The first straight portion 64a1 is a portion that extends vertically and the two ends are connected to the first bent portion 64b1 and the second bent portion 64b2. Furthermore, the second straight portion 64a2 extends diagonally with respect to the width direction of the first fixed housing 11 and the two ends thereof are connected to the second bent portion 64b2 and the third bent portion 64b3. Note that a first end of the third bent portion 64b3 is connected to the second straight portion 64a2, and a second end is connected to the upper end of the fixed-side retained portion 62.

In this manner, the bent portion 64 has a first end connected to the fixed-side retained portion 62 retained in the signal terminal fixed-side retention recess 13a of the first fixed housing 11 and a second end connected to the movable-side retained portion 66 retained in the signal terminal movable-side retention recess 32e of the first movable housing 31, has a bent shape as viewed from the side, includes a first straight portion 64a1 and a second straight portion 64a2, and a first bent portion 64b1, a second bent portion 64b2, and a third bent portion 64b3 that connect the straight shapes; therefore, the portion not restrained by the first fixed housing 11 or first movable housing 31, in other words, the length of the elastically deformable portion is long. Therefore, the bent portion 64 that is present between the signal terminal fixed-side retention recess 13a and the signal terminal movable-side retention recess 32e can flexibly deform enabling the first movable housing 31 to move flexibly with respect to the first fixed housing 11. Also, as illustrated in FIGS. 9A and 9B, the bent portion 64 is positioned on the lower side of the brim portion 13c that demarcates the upper end of the first fixed housing 11 and the signal terminal fixed-side retention recess 13a and the

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lower side of the bottom portion 32c of the first movable housing 31 main body portion 32 stowed in the movable housing cavity for stowing 15 of the first fixed housing 11 and so can deform without interference with the first fixed housing 11 or first movable housing 31. Accordingly, relative to the first fixed housing 11, the first movable housing 31 can move in the span direction of the first signal terminal 61, or in other words, the width direction of the first connector 10, and the pitch direction of the first signal terminal 61, or in other words, the longitudinal direction of the first connector 10. Furthermore, the first movable housing 31 can be inclined in the span direction of the first signal terminal 61 with respect to the first fixed housing 11.

The contact portion 65 of each of the first signal terminals 61 is a portion that comes into contact with the second signal terminal 161 of the second connector 101, which is the counterpart signal terminal, is a straight rod or strip having a rectangular cross-sectional shape with substantially uniform dimensions in the width direction (X-axis direction) and thickness direction (Y-axis direction), and functions as a cantilever with a lower end connected to the movable-side retained portion 66 as a base end (fixed end) and the upper end as a free end. Specifically, the contact portion 65 includes a contact arm portion 65b, the lower end of which is connected to the movable-side retained portion 66, that initially has no external force applied, that is overall slightly inclined, and which has an upper end positioned more to the outside of the first movable housing 31 than the movable-side retained portion 66, a contact protrusion portion 65a that is connected to the upper end of the contact arm portion 65b and bulges outwards in the width direction towards the first movable housing 31, and a contact tip portion 65c that is the upper end of the contact protrusion portion 65a and faces inward in the width direction of the first movable housing 31. As illustrated in FIGS. 9A and 9B, the contact portion 65 stowed in the signal terminal contact portion groove for stowing 34a of the central wall portion 34 of the first movable housing 31 is set so that at least a portion of the contact protrusion portion 65a protrudes outside of the side surface of the central wall portion 34.

In addition, the movable-side retained portion 66 connected to the lower end of the contact portion 65 includes a straight rod-shaped or strip-shaped main body portion 66a having a rectangular cross-sectional shape with uniform dimensions in the width direction (X-axis direction) and thickness direction (Y-axis direction) thereof and a plurality of protrusions 66b protruding outward in the width direction from both surfaces in the width direction (X-axis direction) of the main body portion 66a. The main body portion 66a is configured such that the width dimension thereof is smaller than the width dimension of the signal terminal movable-side retention recess 32e, and the distance between the tips of the left and right protrusions 66b is set to be greater than the width dimension of the signal terminal movable-side retention recess 32e. As a result, the movable-side retained portion 66 of the first signal terminal 61 inserted from below into the signal terminal movable-side retention recess 32e is stably retained by the left and right protrusions 66b penetrating into the left and right wall surfaces of the signal terminal movable-side retention recess 32e. In addition, since the first bent portion 64b1 of the bent portion 64 is connected to the lower end of the movable-side retained portion 66, the first straight portion 64a1 of the bent portion 64 is in a state of being offset with respect to the movable-side retained portion 66. Therefore, while holding the first straight portion 64a1, the operator can easily insert the

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movable-side retained portion **66** into the signal terminal movable-side retention recess **32e** from below.

Furthermore, the fixed-side retained portion **62** includes a straight rod-shaped or strip-shaped main body portion **62a** having a rectangular cross-sectional shape and uniform dimensions in the width direction (X-axis direction) and the thickness direction (Y-axis direction), and a plurality of protrusions **62b** protruding outward in the width direction from both side surfaces in the width direction (X-axis direction) of the main body portion **62a**. Also, the width dimension of the main body portion **62a** is smaller than the width dimension of the portion of the signal terminal fixed-side retention recess **13a** corresponding to the signal terminal fixed-side retention groove **13b1**. The distance between the tips of the left and right protrusions **62b** is set larger than the width dimension of the portion of the signal terminal fixed-side retention recess **13a** corresponding to the signal terminal fixed-side retention groove **13b1**. As a result, the fixed-side retained portion **62** of the first signal terminal **61** inserted from below the signal terminal fixed-side retention recess **13a** is stably retained by the left and right protrusions **62b** penetrating into the left and right wall surfaces of the signal terminal fixed-side retention groove **13b1**.

Although the first signal terminal **61** is used to transmit the signal, in recent years, a high-frequency signal can be transmitted at, for example, 8 (GHz) (16 (Gbps)) as the speed of various types of signals has increased. If this manner of radio-frequency signals are transmitted, the impedance needs to be matched. Therefore, with the present embodiment, the signal terminal fixed-side retention portion thickness reduction **13a1** that is a space containing air having a lower dielectric constant than the first fixed housing **11** is established between the fixed-side retained portion **62** and the bottom surface of the signal terminal fixed-side retention recess **13a** of the first fixed housing **11** that is a dielectric. The dielectric constant of the dielectric surrounding the fixed-side retained portion **62** can be adjusted, and the signal terminal movable-side retention portion first thickness reduction **32c1** and the signal terminal movable-side retention portion second thickness reduction **32c2** that are spaces containing air having a lower dielectric constant than the first movable housing **31** are established between the movable-side retained portion **66** and the bottom portion **32c** of the first movable housing **31** main body portion **32** that is a dielectric. The dielectric constant of the dielectric surrounding the movable-side retained portion **66** was made adjustable, enabling the impedance of the first signal terminal **61** to be adjusted.

Furthermore, the pair of first signal terminals **61** adjacent to each other in the width direction (X-axis direction) in each column on the left and right can be used as a differential pair, enabling transmitting a high frequency signal. In such a case, the impedance of the first signal terminals **61** adjacent to each other in the width direction needs to be matched. Here, with the present embodiment, the dimension of the bent portion **64** in the width direction is set larger than that of the fixed-side retained portion **62**, the movable-side retained portion **66**, or the contact portion **65**, into which a portion of the first fixed housing **11** or the first movable housing **31** is interposed between first signal terminals **61** adjacent in the width direction. Shortening of the distance between adjacent bent portions **64** with no first fixed housing **11** or first movable housing **31** interposed therebetween enables adjusting the dielectric constant of the space surrounding the periphery of the bent portion **64**. This enables adjusting the impedance between first signal terminals **61** adjacent in the width direction.

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Each of the first power supply terminals **51** are members integrally formed and provided elasticity by performing processing such as punching and bending of a conductive metal plate and as illustrated in FIG. 5B, include a fixed-side retained portion **52**, a tail portion **53** connected to the lower end of the fixed-side retained portion **52** as a board connecting portion, a bent portion **54** connected to the upper end of the fixed-side retained portion **52**, a movable-side retained portion **56** connected to the upper end of the bent portion **54**, and a contact portion **55** connected to the movable-side retained portion **56**. In the example illustrated in the drawings, the first power supply terminals **51** are arranged in pairs at both ends in the longitudinal direction of the first connector **10**, and arranged such that two are lined up in the width direction of the first connector **10** respectively at both ends in the longitudinal direction of the first connector **10**. Thus, the first connector **10** is provided with power supply lines respectively at both ends in the longitudinal direction in which positive and negative power supply current flows. Also, the left and right first power supply terminals **51** at each of both ends in the longitudinal direction of the first connector **10** are configured so as to be symmetrical in the longitudinal direction of the first connector **10**. The first power supply terminal **51** is formed, in general, with plate material extending in the width direction and the vertical direction of the first connector **10** bent within a plane parallel to the first connector **10** in the longitudinal direction.

The fixed-side retained portion **52** is a portion retained in the short wall portion **12b** of the first fixed housing **11**, and includes a flat plate-shaped main body portion **52a** extending in the width direction and the vertical direction of the first connector **10**, and a plurality of protrusions **52b** protruding outward in the width direction from both side surfaces in the width direction (Y-axis direction) of the main body portion **52a**. Also, the width dimension of the main body portion **52a** is smaller than the width dimension of the portion of the power supply terminal fixed-side retention recess **14a** corresponding to the power supply terminal fixed-side retention groove **14b1**. The distance between the tips of the left and right protrusions **52b** is set larger than the width dimension of the portion of the power supply terminal fixed-side retention recess **14a** corresponding to the power supply terminal fixed-side retention groove **14b1**. Thus, the fixed-side retained portion **52** of the first power supply terminal **51** inserted from below into the power supply terminal fixed-side retention recess **14a** is stably retained based on the protrusions **52b** on the left and right penetrating the left and right wall surfaces of the power supply terminal fixed-side retention groove **14b1**.

The movable-side retained portion **56** is a portion retained by the short wall portion **32b** of the first movable housing **31**, and includes a first main body portion **56a** as a flat plate-shaped first retained portion extending in the width direction and the vertical direction of the first connector **10**, and a second main body portion **56b** as a flat plate-shaped second retained portion that is bent by approximately 90 degrees on an outer end edge in the width direction of the first connector **10** and connected to the first body portion **56a**, and extends toward the center of the first connector **10** in the longitudinal direction. At least a portion of the first main body portion **56a** extends along the short wall portion **32b**, and at least a portion of the second main body portion **56b** extends along the long wall portion **32a**. Note that an engagement opening **58** is formed in the second main body portion **56b** as an engaged portion to be engaged to the first movable housing **31**. In particular, the engagement opening **58** engages with

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the engaging protrusion **32d**, which is an engaging part formed on the outer surface of the long wall portion **32a** of the first movable housing **31**.

The first upper end cover portion **57a** is connected to the upper end of the first main body portion **56a**, and the second upper end cover portion **57b** is connected to the upper end of the second main body portion **56b**. The first upper end cover portion **57a** and the second upper end cover portion **57b** are members covering the upper end portion of the short wall portion **32b** and the upper end portion of the long wall portion **32a** near the corner portion **32h** of the first movable housing **31**, and are formed with an approximately 180 degree bend so that the tip (lower end) faces downward. Furthermore, a flat plate-shaped first upper end cover extension portion **57a1** is connected to the lower end of the first upper end cover portion **57a**, and a flat plate-shaped contact portion **55** is connected to the lower end of the second upper end cover portion **57b**. The contact portion **55** is a rectangular plate material that is long in the vertical direction, and functions as a contact surface in contact with a second power supply terminal **151** of the second connector **101**, which is a counterpart power supply terminal. This contact surface is parallel with the inner surface **32a1** of the long wall portion **32a** but is preferably more indented into the long wall portion **32a** than the inner surface **32a1**. In addition, the top surface of the first upper end cover portion **57a** and the second upper end cover portion **57b** (upper end surface) is preferably positioned above an upper end surface **32b2** of the short wall portions **32b** and an upper end surface **32a2** of the long wall portions **32a**.

In this manner, the movable-side retained portion **56** covers the long wall portions **32a** close to both ends of the first connector **10** in the longitudinal direction, the inner and outer side surfaces of the short wall portions **32b**, the upper end surface **32a2**, and the upper end surface **32b2** of the first movable housing **31** and therefore effectively protects the first movable housing **31**, improving robustness. In other words, in addition to the function of contacting the second power supply terminal **151**, the movable-side retained portion **56** effectively exhibits a function as a reinforcing metal fitting that reinforces near the corner portions **32h** of the first movable housing **31**.

Furthermore, the bent portion **54** is a portion bent in a meander like substantially S shape as viewed in the width direction (Y-axis direction) of the first connector **10** and functions as an elastic deformation portion. Furthermore, the bent portion **54** includes a first straight portion **54a1**, a second straight portion **54a2**, and a third straight portion **54a3** as straight portions as well as a first bent portion **54b1** and a second bent portion **54b2** as bent portions bent close to 180 degrees. The first straight portion **54a1** has an upper end connected to the lower end of the first main body portion **56a** of the movable-side retained portion **56** and a lower end that is a portion extending in the vertical direction and is connected to the first bent portion **54b1**. In addition, the second straight portion **54a2** is a portion that extends vertically and the two ends are connected to the first bent portion **54b1** and the second bent portion **54b2**. Furthermore, the third straight portion **54a3** has an upper end connected to the second bent portion **54b2** and a lower end that is a portion extending in the vertical direction and is connected to the upper end of the fixed-side retained portion **52** main body portion **52a**. In addition, the bent portion **54** is split into two in the width direction (Y-axis direction) by a slit **54c** that extends the full length thereof.

In this manner, the bent portion **54** has a first end connected to the fixed-side retained portion **52** retained in

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the power supply terminal fixed-side retention recess **14a** of the first fixed housing **11** and a second end connected to the movable-side retained portion **56** retained on both ends of the first movable housing **31** main body portion **32** in the longitudinal direction, has a bent shape as viewed from the side, and includes a first straight portion **54a1**, a second straight portion **54a2**, and a third straight portion **54a3**, as well as a first bent portion **54b1** and second bent portion **54b2** that connect the straight shapes; therefore, the portion not restrained by the first fixed housing **11** or the first movable housing **31**, in other words, the length of the elastically deformable portion, is long. Therefore, the bent portion **54** present between the power supply terminal fixed-side retention recess **14a** and both ends of the main body portion **32** in the longitudinal direction can flexibly deform enabling the first movable housing **31** to flexibly move relative to the first fixed housing **11**. Also, the bent portion **54** is positioned on the lower side of the brim portion **14c** that demarcates the upper end of the power supply terminal fixed-side retention recess **14a** of the first fixed housing **11** and the lower side of the first movable housing **31** main body portion **32** stowed in the movable housing cavity for stowing **15** of the first fixed housing **11**; therefore, deformation is feasible between the first fixed housing **11** and first movable housing **31**. In addition, the bent portion **54** is split into two in the width direction thereof by the slit **54c**. Therefore, the first movable housing **31** can move not only in the span direction of the first power supply terminal **51**, in other words, the longitudinal direction of the first connector **10**, relative to the first fixed housing **11**, but also in the pitch direction of the first power supply terminal **51**, in other words, the width direction of the first connector **10**. Furthermore, the first movable housing **31** can be inclined in the pitch direction of the first power supply terminal **51** with respect to the first fixed housing **11**.

In addition, with the first signal terminal **61**, the bent portion **64** is bent as viewed from the X-axis direction and primarily is flexible in the Y-axis direction, in other words, the width direction of the first connector **10**. On the other hand, with the first power supply terminal **51**, the bent portion **54** is bent as viewed from the Y-axis direction and primarily is flexible in the X-axis direction, in other words, the longitudinal direction of the first connector **10**. Therefore, through cooperative elastic deformation of the first signal terminal **61** and the first power supply terminal **51**, the first movable housing **31** can be stably displaced in any direction relative to the first fixed housing **11**.

Furthermore, with the first power supply terminal **51**, while the bent portion **54** is primarily flexible in the longitudinal direction of the first connector **10**, the contact surface of the contact portion **55** faces in the width direction of the first connector **10**, differing from the deformation direction of the bent portion **54** and is thus able to stably maintain a contact state with the second power supply terminal **151** of the second connector **101**.

Furthermore, the tail portion **53** is bent at approximately 90 degrees with respect to the main body portion **52a** of the fixed-side retained portion **52** that extends in the vertical direction and extends towards the center of the first fixed housing **11** in the longitudinal direction. Note, the tail portion **53** includes a relay portion **53b** connected to the main body portion **52a** of the fixed-side retained portion **52** and a main body portion **53a** that is connected to the relay portion **53b** via a stepped portion **53c** and extends outwardly in the width direction of the first fixed housing **11**. The lower surface of the main body portion **53a** is substantially flush with the mounting surface **11b**, which is the lower surface of

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the first fixed housing **11**, and is connected and fixed to a connection pad formed on the surface of the first board (not shown) by means of soldering or the like. This connection pad is connected to a conductive trace for transmitting power supply current on the first board. The relay portion **53b** connected to the main body portion **53a** via the stepped portion **53c** is positioned higher than the main body portion **53a** and the lower surface thereof is separated from the surface of the first board. Also, the tail portions **53** extend from the lower end of the fixed-side retained portion **52** main body portions **52a** toward the center of the first fixed housing **11** in the longitudinal direction and are therefore positioned to the inside at both ends of the first fixed housing **11** in the longitudinal direction and are not positioned to the outside. Therefore, the mounting surface of the first connector **10** does not increase in size in the longitudinal direction on the surface of the first board.

In addition, as illustrated in FIG. **8C** and as viewed from above, with the tail portion **53**, the tip of the main body portion **53a** is positioned at least more to the outside than the long wall portion **12a** of the first fixed housing **11** and is positioned along the long wall portion **12a** in the same row as the tail portions **63** of the first signal terminals **61** that extend in the longitudinal direction of the first connector **10**. Therefore, in the work of mounting the first connector **10** on the surface of the first board, the tail portions **53** of the first power supply terminals **51** can be connected to connection pads formed on the surface of the first board by means of work similar to that of the tail portions **63** of the first signal terminals **61**, improving work efficiency. In addition, similar to the tail portions **63** of the first signal terminals **61**, the operator can readily visually confirm the connection state of the tail portions **53** of the first power supply terminals **51** on the connection pads formed on the surface of the first board.

Next, the configuration of the second connector **101** will be described.

FIG. **11** is a perspective view of the second connector according to Embodiment 1. FIGS. **12A-12D** are four views of the second connector according to Embodiment 1. FIG. **13** is an exploded view of the second connector according to Embodiment 1. FIGS. **14A** and **14B** are a perspective view of the second connector terminals according to Embodiment 1. FIGS. **15A-15C** illustrate a mounting portion of the signal terminal of the second connector to the fixed housing in Embodiment 1. FIG. **12A** is a top view, FIG. **12B** is a side view, FIG. **12C** is a bottom view, and FIG. **12D** is a front view. FIG. **14A** is a diagram illustrating a signal terminal, and FIG. **14B** is a diagram illustrating a power supply terminal. FIG. **15A** is a sectional view taken along the line G-G of FIG. **12A**, FIG. **15B** is a perspective view, and FIG. **15C** is an enlarged view of part H in FIG. **15B**.

In the present embodiment, as illustrated in FIGS. **12A-12D**, the second connector **101** is configured so as to be left-right symmetrical when viewed in the width direction (Y-axis direction), and is also configured to be left-right symmetrical when viewed in the longitudinal direction (X-axis direction). Furthermore, the second housing **111** of the second connector **101** is a member that is integrally formed using an insulating material (dielectric body) such as synthetic resin.

The second housing **111** has a lower portion **112** and an upper portion **131** connected to the upper end of the lower portion **112**. The lower surface of the lower portion **112**, that is, the lower surface of the second housing **111** is a mounting surface **111b** facing the surface of the second board.

The lower portion **112** is a substantially square cylindrical member and includes a pair of rectangular flat plate-shaped

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long wall portions **112a** that are mutually parallel and extend in the longitudinal direction; short wall portions **112b** that are a pair of mutually parallel rectangular thick plate shaped or square columnar shaped members extending in the width direction, and that are connected to both ends of the long wall portions **112a**; and a lower cavity **134** demarcated by the four surfaces of the long wall portion **112a** and the short wall portions **112b** that is a square columnar shaped cavity that opens downward (positive direction of Z-axis). Note, as illustrated in FIG. **15A**, the upper end of the lower cavity **134** is closed by a boundary wall **135** that is a flat plate-shaped plate material extending in the X-Y direction and is a boundary of the lower portion **112** and the upper portion **131**. In addition, as illustrated in FIG. **12C**, the inside of the lower cavity **134** is divided into a plurality of demarcations by partition plates **134a** that are a plurality of plate materials extending in the Y-Z direction. Furthermore, outward protruding portions **112c** are formed at the bottom of the short wall portions **112b**. Also, metal fitting retention recesses **112d** are formed on the outward protruding portions **112c** and housing fixation metal fittings **171** are stowed and retained in the metal fitting retention recesses **112d**.

Note, lower end notches **112g** are formed in locations adjacent to the long wall portions **112a** at the lower end of the respective short wall portions **112b**. Furthermore, as illustrated in FIG. **12C**, the lower ends of power supply terminal retention recesses **112e** are open at locations corresponding to the lower end notches **112g** on the bottom surfaces of the short wall portions **112b**. The power supply terminal retention recesses **112e** are long narrow grooves or holes that penetrate the inside of the short wall portions **112b** in the vertical direction, and of the second terminals provided in the second connector **101**, the power terminals, in other words, retained portions **152** of the second power supply terminals **151** that are power supply terminals, are retained in each of the power supply terminal retention recesses **112e**. Furthermore, a tail portion **153** of the second power supply terminal **151** that extends outward in the width direction (Y-axis direction) from the lower end of the retained portion **152** is stowed in the lower end notch **112g**. Note, in the example illustrated in the drawings, there are two power supply terminal retention recesses **112e** formed on each of the short wall portions **112b** at both ends of the second housing **111** in the longitudinal direction but the number thereof is changed according to the number of second power supply terminals **151**.

In addition, as illustrated in FIG. **12C** and FIG. **15A**, a plurality of long narrow groove shaped signal terminal retention recesses **112f** extending in the vertical direction are formed on the inner surfaces of the long wall portions **112a**. Retained portions **162** of the second signal terminals **161** that are a signal terminal of the second terminals provided in the second connector **101** are stowed in each of the signal terminal retention recesses **112f**. Note, in the example illustrated in the drawings, there are thirty of the signal terminal retention recesses **112f** formed in each of the left and right long wall portions **112a** of the second housing **111** but the number thereof changes according to the number of second signal terminals **161**.

The upper portion **131** is a substantially square cylindrical member similar to the lower portion **112**, but has a smaller external dimension than the lower portion **112**. Furthermore, the upper portion **131** includes a pair of rectangular flat plate-shaped long wall portions **131a** that are mutually parallel and extend in the longitudinal direction; short wall portions **131b** that are a pair of mutually parallel rectangular thick plate-shaped or square columnar shaped members

extending in the width direction, and that are connected to both ends of the long wall portions **131a**; and an upper cavity **133** demarcated by the four surfaces of the long wall portions **131a** and the short wall portions **131b** that is a square columnar shaped cavity that opens upward (negative direction of Z-axis). Note, the lower end of the long wall portions **131a** and the short wall portions **131b** are integrally connected to the upper end of the long wall portions **112a** and short wall portions **112b** of the lower portion **112**. In addition, the upper cavity **133** has nearly the same cross-sectional shape and area as the lower cavity **134** but as illustrated in FIG. 15A, the lower end thereof is closed by the boundary wall **135** and is divided from the lower cavity **134**. Note, there is no member corresponding to the partition plates **134a** inside the upper cavity **133**.

In addition, an upward protruding end portion **131c** with a substantially quadrangular trapezoid shape is formed on the upper ends of the respective short wall portions **131b**. Furthermore, on the side surfaces of each of the short wall portions **131b** extending in the longitudinal direction of the second connector **101**, there are groove shaped power supply terminal stowing recesses **131e** extending in the vertical direction. The number of these power supply terminal stowing recesses **131e** formed is equal to the number of power supply terminal retention recesses **112e** and the lower ends thereof communicate with the upper ends of the corresponding power supply terminal retention recesses **112e**. Furthermore, a contact portion **155** of the second power supply terminals **151** is stowed in each power supply terminal stowing recess **131e** and at least a portion of a contact protruding portion **155a** protrudes outside of the side surface of the short wall portion **131b**.

In addition, as illustrated in FIG. 11 and FIG. 15A, a plurality of long narrow groove shaped signal terminal stowing recesses **131f** extending in the vertical direction are formed on the inner surfaces of the long wall portions **131a**. The number of signal terminal stowing recesses **131f** is equal to the number of signal terminal retention recesses **112f** and the lower ends thereof communicate with the upper ends of the corresponding signal terminal retention recesses **112f**. A contact portion **165** of the second signal terminal **161** is stowed in each signal terminal stowing recess **131f**. The surface of the contact portion **165** is substantially flush with the inner surface of the long wall portion **131a** and protrudes a slight amount more into the upper cavity **133** than this inner surface.

Each of the second signal terminals **161** are long narrow rod-shaped members integrally formed and provided elasticity by performing processing such as punching and bending of a conductive metal plate and as illustrated in FIGS. 14A and 14B, include a retained portion **162**, a tail portion **163** as a board connecting portion connected to the lower end of the retained portion **162**, and a contact portion **165** connected to the upper end of the retained portion **162**. In the example illustrated in the drawings, the second signal terminals **161** are configured to form a pair of left and right rows extending in the longitudinal direction of the second connector **101** and are arranged with the same pitch as the first signal terminals **61** (for example roughly 0.3 to 0.5 [mm]) and the same number (for example, thirty). The number and pitch of the second signal terminals **161** can be appropriately changed in the same manner as the first signal terminals **61**. In addition, the second signal terminals **161** in each of the left and right rows are arranged so as to be symmetrical as viewed in the longitudinal direction of the second connector **101**. Furthermore, each second signal terminal **161** is formed of a plate material extending in the

longitudinal direction of the second connector **101** in a parallel plane and bent within a cross-sectional surface of the second connector **101** as illustrated in FIG. 15A.

The tail portion **163** is bent at approximately 90 degrees relative to the retained portion **162** that extends in the vertical direction and is connected thereto and extends outward in the width direction of the second housing **111**. Furthermore, the lower surface thereof is substantially flush with the mounting surface **111b**, which is the lower surface of the second housing **111**, and is connected and fixed to a connection pad formed on the surface of the second board (not shown) by means of soldering or the like. This connection pad is typically connected to a conductive trace for transmitting signals on the second board. In addition, as illustrated in FIG. 12A, the tail portions **163** are disposed so that at least the tips thereof are positioned to the outside of the long wall portions **112a** of the lower portion **112** of the second housing **111** in the width direction and so as to form a row extending along the long wall portions **112a** extending in the longitudinal direction of the second connector **101**. Accordingly, in the work of mounting the second connector **101** to the surface of the second board, the operator can readily visually confirm the connection state of the tail portions **163** to connection pads formed on the surface of the second board. Furthermore, the length of the tips of the tail portions **163** that protrude outside the long wall portions **112a** of the lower portion **112** of the second housing **111** is set small to enable shortening the length thereof. This enables preventing an increase in the impedance of the second signal terminal **161**.

Furthermore, the retained portion **162** includes a straight rod-shaped or strip-shaped main body portion **162a** having a rectangular cross-sectional shape and uniform dimensions in the width direction (X-axis direction) and the thickness direction (Y-axis direction), and a plurality of protrusions **162b** protruding outward in the width direction from both side surfaces in the width direction (X-axis direction) of the main body portion **162a**. Furthermore, the main body portion **162a** is configured such that the width dimension thereof is smaller than the width dimension of the signal terminal retention recess **112f**, and the distance between the tips of the left and right protrusions **162b** is set to be greater than the width dimension of the signal terminal retention recess **112f**. Thus, the retained portions **162** of the second signal terminals **161** inserted from the mounting surface **111b** of the second housing **111** into the signal terminal retention recesses **112f** are stably retained based on the left and right protrusions **162b** penetrating into the left and right wall surfaces of the signal terminal retention recess **112f**.

In addition, the contact portion **165** connected to the upper end of the retained portion **162** has dimensions in the width direction (X-axis direction) and thickness direction (Y-axis direction) that are the same straight rod shape or strip shape as the main body portion **162a** of the retained portion **162** and extends straight upward (Z-axis negative direction). Furthermore, most of the contact portion **165** is stowed inside the signal terminal stowing recess **131f** and the center side surface of the contact portion **165** in the width direction of the second connector **101** is exposed in the upper cavity **133**.

Similar to the first signal terminal **61**, the second signal terminal **161** is used to transmit the signal, but in recent years, a high-frequency signal can be transmitted at, for example, 8 (GHz) (16 (Gbps)) as the speed of various types of signals has increased. If this manner of radio-frequency signals are transmitted, the impedance needs to be matched. Therefore, with the present embodiment, a signal terminal

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stowing recess first thickness reduction **131d1** and a signal terminal stowing recess second thickness reduction **131d2** that are spaces made up of air having a lower dielectric constant than the second housing **111** are established at a position behind the contact portion **165** in the long wall portion **131a** of the upper portion **131** of the second housing **111**, that is a dielectric, enabling adjusting the dielectric constant of the dielectric surrounding the periphery of the contact portion **165** and thus enabling adjusting the impedance of the second signal terminal **161**.

Each of the second power supply terminals **151** are members integrally formed and provided elasticity by performing processing such as punching and bending of a conductive metal plate and as illustrated in FIG. **14B**, include a retained portion **152**, a tail portion **153** as a board connecting portion connected to the lower end of the retained portion **152**, a center portion **154** connected to the upper end of the retained portion **152**, and a contact portion **155** connected to the upper end of the center portion **154**. In the example illustrated in the drawings, the second power supply terminals **151** are arranged on the pair of side surfaces extending in the longitudinal direction of the second connector **101** near both ends in the longitudinal direction, respectively. Thus, the second connector **101** is provided with power supply lines respectively near both ends in the longitudinal direction in which positive and negative power supply current flows. In addition, the left and right second power supply terminals **151** near each of both ends in the longitudinal direction of the second connector **101** are configured so as to be symmetrical in the longitudinal direction of the second connector **101**. Each second power supply terminal **151** is formed, in general, with plate material extending in the longitudinal direction of the first connector **101** [sic] bent within a plane parallel to the second connector **101** in the longitudinal direction.

The retained portion **152** includes a flat plate-shaped main body portion **152a** extending in the longitudinal direction and the vertical direction of the second connector **101** and a plurality of protrusions **152b** protruding outward from both side surfaces of the main body portion **152a** in the width direction (X-axis direction). Furthermore, the main body portion **152a** is configured such that the width dimension thereof is smaller than the width dimension of the power supply terminal retention recess **112e**, and the distance between the tips of the left and right protrusions **152b** is set to be greater than the width dimension of the power supply terminal retention recess **112e**. Thus, the second power supply terminal **151** retained portion **152** inserted from the mounting surface **111b** side of the second housing **111** is stably retained through penetrating of the left and right protrusions **152b** into the left and right wall surfaces of the power supply terminal retention recess **112e**.

In addition, the center portion **154** is a long narrow strip-shaped member facing upward (negative direction of Z-axis) from the upper end of the main body portion **152a** of the retained portion **152** and the dimension in the width direction (X-axis direction) thereof is set smaller than the dimension of the main body portion **152a** in the width direction.

Furthermore, the contact portion **155** is a portion that contacts the first power supply terminal **51** of the first connector **10** having dimensions in the width direction (X-axis direction) and thickness direction (Y-axis direction) set as a strip-shaped portion the same as the center portion **154**, and is a portion that functions as a cantilever with the lower end connected to the upper end of the center portion **154** as a base end (fixed end) and the upper end as a free end.

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Specifically, the contact portion **155** has a contact arm portion **155b**, the lower end of which is connected to the center portion **154**, has an overall gradual slope in an initial state with no external force applied, and has an upper end which is positioned more to the outside of the second housing **111** than the center portion **154**, a contact protruding portion **155a** connected to the upper end of the contact arm portion **155b** that bulges toward the outside of the second housing **111** in the width direction, and a contact tip portion **155c** at the upper end of the contact protruding portion **155a** facing the inside of the second housing **111** in the width direction. Furthermore, as illustrated in FIG. **11**, the contact portion **155** stowed in the power supply terminal stowing recess **131e** of the second housing **111** upper portion **131** is such that at least a portion of the contact protruding portion **155a** protrudes to the outside of the side surface of the short wall portion **112b**.

Furthermore, the tail portion **153** is bent at approximately 90 degrees relative to the main body portion **152a** of the retained portion **152** that extends in the vertical direction and is connected thereto and extends outward in the width direction of the second housing **111**. Note, the tail portion **153** includes a relay portion **153b** connected to the main body portion **152a** of the retained portion **152** and a main body portion **153a** that is connected to the relay portion **153b** via a stepped portion **153c** and extends outwardly in the width direction of the second housing **111**. The lower surface of the main body portion **153a** is substantially flush with the mounting surface **111b**, which is the lower surface of the second housing **111**, and is connected and fixed to a connection pad formed on the surface of the second board (not shown) by means of soldering or the like. This connection pad is connected to a conductive trace for transmitting power supply current on the second board. The relay portion **153b** connected to the main body portion **153a** via the stepped portion **153c** is positioned higher than the main body portion **153a** and the lower surface thereof is separated from the surface of the second board. In addition, as illustrated in FIG. **12A**, as viewed from above, the tail portion **153** has at least the tip of the main body portion **153a** positioned to the outside in the width direction of side walls of an outward protruding portion **112c** of the second housing **111**. Therefore, similar to the tail portions **163** of the second signal terminals **161**, the operator can readily visually confirm the connection state of the tail portions **153** of the second power supply terminals **151** on the connection pads formed on the surface of the second board.

The housing fixation metal fitting **171** is a member that is integrally formed and provided elasticity by performing punching and bending of a metal plate and as illustrated in FIG. **13**, includes a retained portion **172**, and a tail portion **173** as a board connecting portion connected to the lower end of the retained portion **172**. In the example illustrated in the figure, the housing fixation metal fittings **171** are arranged one by one at both ends of the second connector **101** in the longitudinal direction, and attached so as to be parallel to the end surface of the outward protruding portion **112c** of the lower portion **112** facing outward in the longitudinal direction of the second connector **101**.

The retained portion **172** includes a flat plate-shaped main body portion **172a** extending in the width direction and the vertical direction of the second connector **101** and a plurality of protrusions **172b** protruding outward in the width direction from both side surfaces of the main body portion **172a** in the width direction (Y-axis direction). Further, the main body portion **172a** has a width dimension that is smaller than the width dimension of the metal fitting retention recess

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112d of the outward protruding portion 112c and the distance between the tips of the left and right protrusions 172b is set to be greater than the width dimension of the metal fitting retention recess 112d. Thus, the retained portion 172 of the housing fixation metal fitting 171 inserted into the metal fitting retention recess 112d from the side opposite the mounting surface 111b of the second housing 111 is stably retained by the left and right protrusions 172b penetrating into the left and right wall surfaces of the metal fitting retention recess 112d.

Two tail portions 173 are formed for one housing fixation metal fitting 171. Specifically, the tail portion 173 is bent at approximately 90 degrees relative to the retained portion 172 that extends in the vertical direction and is connected thereto and extends outward in the longitudinal direction of the second housing 111. Furthermore, the lower surface thereof is substantially flush with the mounting surface 111b, which is the lower surface of the second housing 111, and is connected and fixed to a connection pad formed on the surface of the second board (not shown) by means of soldering or the like. The connection pad is typically not connected to the conductive trace of the second board and functions only to fix the housing fixation metal fitting 171 to the surface of the second board.

In this manner, the housing fixation metal fitting 171 is fixed on the surface of the second board. Therefore, the second housing 111 is strongly attached to the surface of the second board and as a result, the mounted state of the second connector 101 on the second board is more reliable.

Here, for convenience of explanation, the second connector 101 is described as a so-called straight type connector in a state of being erected on the surface of the second board, in other words, mounted with the upper cavity 133 open upwards (negative direction Z-axis), and the upper cavity 133 is described as a connector extending in a direction perpendicular to the second board but the second connector 101 is not limited to a straight type and a so called right angle type connector can also be used. For a right angle type, the long wall portions 112a of the lower portion 112 and the long wall portions 131a of the upper portion 131 are mounted substantially parallel with respect to the surface of the second board, and the upper cavity 133 also becomes substantially parallel with respect to the surface of the second board.

Next, the operation of mating the first connector 10 and the second connector 101 with the above configuration is described.

FIGS. 16A-16C are three views illustrating a mated state of the first connector and the second connector according to Embodiment 1. FIG. 17 is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 1 in a mated state and is a sectional view taken along the line K-K in FIG. 16A. Note, FIG. 16A is a plan view as viewed from the second connector side, FIG. 16B is a sectional view taken along the line I-I in FIG. 16A, and FIG. 16C is a sectional view taken along the line J-J in FIG. 16A.

Here, the first connector 10 is connected by means of soldering the tail portions 63 of the first signal terminals 61 and the tail portions 53 of the first power supply terminals 51 to the connection pad on the surface of the first board (not shown) and is thus surface mounted to the first board. In addition, connection pads connected to the tail portions 63 of the first signal terminals 61 are connected to a conductive trace for transmitting signals on the first board. The connection pads that the tail portions 53 of the first power

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supply terminals 51 are connected to are connected to a conductive trace for transmitting power supply on the first board.

In a similar manner, the second connector 101 is connected by soldering the tail portions 163 of the second signal terminals 161, the tail portions 153 of the second power supply terminals 151, and the tail portions 173 of the housing fixation metal fitting 171 to connection pads on the surface of the second board and is thus surface mounted to the second board. In addition, connection pads connected to the tail portions 163 of the second signal terminals 161 are connected to a conductive trace for transmitting signals on the second board. The connection pads that the tail portions 153 of the second power supply terminals 151 are connected to are connected to a conductive trace for transmitting power supply on the second board.

First, the operator sets the mating surface of the first connector 10 (upper surface in FIG. 2) and the mating surface of the second connector 101 (upper surface in FIG. 11) facing each other and makes adjustments aligning the position of the upper portion 131 of the second housing 111 of the second connector 101 with the position of the counterpart housing cavity for stowing 35 of the first movable housing 31 of the first connector 10. Therefore, as illustrated in FIG. 1B, positioning of the first connector 10 and the second connector 101 is complete.

In this state, as the first connector 10 and/or the second connector 101 are moved in a direction closer to the counterpart side, in other words, in a mating direction, the upper portion 131 of the second housing 111 is inserted into the counterpart housing cavity for stowing 35 of the main body portion 32 of the first movable housing 31 and the central wall portion 34 of the counterpart housing cavity for stowing 35 is inserted into the upper cavity 133 of the upper portion 131. Note, the substantially quadrangular trapezoid shape upward protruding end portion 131c having a sloped surface is formed at the upper end of the upper portion 131 at both ends in the longitudinal direction so when mating, the upper portion 131 is smoothly inserted into the counterpart housing cavity for stowing 35. In addition, the upper end parts of the short wall portions 32b and the upper end parts of the long wall portions 32a near the corner portions 32h of the first movable housing 31 are covered by the first upper end cover portion 57a and second upper end cover portion 57b that contain metal so during mating, even if the second housing 111 comes into contact with the upward protruding end portion 131c, no damage or injury will occur.

Thus, as illustrated in FIGS. 1A, 16, and 17, when the first connector 10 and second connector 101 are in a mated state, through mating of the first movable housing 31 and the second housing 111, the first signal terminals 61 and second signal terminals 161, that are mutually compatible, become conductive and the first power supply terminals 51 and the second power supply terminals 151, that are mutually compatible, become conductive.

Specifically, at least a portion near the upper end of the upper portion 131 of the second housing 111 enters in between the long wall portion 32a in the counterpart housing cavity for stowing 35 and the central wall portion 34 and the surface of the contact portion 165 of the second signal terminal 161 stowed in the signal terminal stowing recess 131f of the upper portion 131 in the width direction on the center side of the second connector 101 comes into contact with the portion of the contact protrusion portion 65a of the contact portion 65 of the first signal terminal 61 stowed in the signal terminal contact portion groove for stowing 34a of the central wall portion 34 protruding outside the central

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wall portion 34 side surface, and thus the first signal terminal 61 and the second signal terminal 161 become conductive. Here, the contact portion 65 of the first signal terminal 61 functions as a cantilever and the contact protrusion portion 65a is pushed against the surface of the second signal terminal 161 contact portion 165 by the elasticity exhibited by this cantilever; therefore, the conducting state of the first signal terminal 61 and second signal terminal 161 is reliably maintained.

In addition, at least the portion of the upper portion 131 of the second housing 111 close to the upper end of the short wall portion 131b enters into the counterpart housing cavity for stowing 35 close to the short wall portion 32b and the portion of the contact protruding portion 155a of the contact portion 155 of the second power supply terminal 151 stowed in the power supply terminal stowing recess 131e formed on the side surface of the short wall portion 131b extending in the longitudinal direction and protruding outside the side surface of the short wall portion 131b comes into contact with the contact surface that is the flat plate-shaped contact portion 55 connected to the lower end of the second upper end cover portion 57b that covers the upper end of the long wall portion 32a near the corner portion 32h of the first movable housing 31. Thus, the first power supply terminal 51 and the second power supply terminal 151 are electrically connected together. Here, the contact portion 155 of the second power supply terminal 151 functions as a cantilever and the contact protruding portion 155a is pushed against the contact surface of the first power supply terminal 51 contact portion 55 based on the elasticity exhibited by this cantilever; therefore, the conducting state of the first power supply terminal 51 and the second power supply terminal 151 is reliably maintained.

In this manner, with the first connector 10 and second connector 101 mated, if there is a fluctuation in the positional relationship of the first board and the second board and the second connector 101 is displaced relative to the first connector 10 or there is an external force such as an impact or vibration or the like, the bent portion 54 of the first power supply terminal 51 and the bent portion 64 of the first signal terminal 61 flexibly deform, appropriately absorbing displacement relative to the first fixed housing 11 of the first movable housing 31 mated to the second housing 111. Therefore, the mated state of the first connector 10 and the second connector 101 is maintained without being released.

In this manner, with the present embodiment, the first connector 10 includes the first fixed housing 11, the first movable housing 31 that is movable relative to the first fixed housing 11, and the first power supply terminals 51 retained in the first fixed housing 11 and first movable housing 31. Furthermore, the first fixed housing 11 includes the long wall portions 12a that extend in the longitudinal direction of the first connector 10 and short wall portions 12b that extend in the width direction of the first connector 10 and connect the end portions of the long wall portions 12a. The first movable housing 31 includes the long wall portions 32a that extend in the longitudinal direction of the first connector 10 and short wall portions 32b that extend in the width direction of the first connector 10 and connect the end portions of the long wall portions 32a. The first power supply terminals 51 include the fixed-side retained portion 52 retained in the short wall portions 12b of the first fixed housing 11, the movable-side retained portion 56 retained in the first movable housing 31, and the bent portion 54 that connects the fixed-side retained portion 52 and the movable-side retained portion 56. The movable-side retained portion 56 includes a first main body portion 56a at least a part of which extends

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along the short wall portions 32b and a second main body portion 56b at least a part of which extends along the long wall portions 32a.

Therefore, the degree of freedom of displacement of the first movable housing 31 is high, increasing the robustness of the first movable housing 31, enabling reliably maintaining the mated state of the second connector 101. In addition, this enables size reduction of the first connector 10 along with simplifying the structure, reducing component count, simplification of manufacturing, reducing cost, and improving reliability.

In addition, the first main body portion 56a includes the first upper end cover portion 57a that covers the upper end portion of the short wall portions 32b and the second main body portion 56b includes the second upper end cover portion 57b that covers the upper end portion of the long wall portions 32a. Therefore, even if the second housing 111 makes contact when mating with the second connector 101, there is no damage or injury to the long wall portions 32a or end wall portions 32b of the first movable housing 31.

Furthermore, the contact portion 55 for contacting the second power supply terminal 151 is arranged inside the long wall portion 32a and extends along the inner surface 32a1 of the long wall portion 32a and preferably, the contact portion 55 extends in the vertical direction parallel to the inner surface 32a1 of the long wall portion 32a and is connected to the second upper end cover portion 57b. Therefore, the conduction state between the first power supply terminal 51 and the second power supply terminal 151 is reliably maintained.

Furthermore, the bent portion 54 is bent in a substantially S shape as viewed from the width direction of the first connector 10; a first end thereof is connected to the upper end of the fixed-side retained portion 52 and a second end is connected to the lower end of the first main body portion 56a of the movable-side retained portion 56. Therefore, the first power supply terminal 51 mainly has flexibility in the longitudinal direction of the first connector 10 and the first movable housing 31 can stably be displaced relative to the first fixed housing 11.

Further, the second main body portion 56b of the movable-side retained portion 56 includes the engagement opening 58 that engages with the first movable housing 31. Therefore, the movable-side retained portion 56 is reliably retained in the first movable housing 31.

Further, the first power supply terminal 51 includes a tail portion 53 connected to the fixed-side retained portion 52 and the tail portion 53 includes a relay portion 53b extending from the lower end of the fixed-side retained portion 52 toward the center of the first connector 10 in the longitudinal direction and a main body portion 53a that is connected to the relay portion 53b and extends to the outside of the first connector 10 in the width direction. Therefore, since the tail portion 53 is not positioned outside the first fixed housing 11 in the longitudinal direction, the mounting surface of the first connector 10 on the surface of the first board does not increase in the longitudinal direction.

The first power supply terminals 51 are arranged in pairs at both ends in the longitudinal direction of the first connector 10, and arranged so as to be lined up in the width direction of the first connector 10 respectively at both ends in the longitudinal direction. Therefore, the first movable housing 31 is stably and flexibly displaceably retained with respect to the first fixed housing 11. In addition, the first connector 10 can be provided with power supply lines respectively at both ends in the longitudinal direction in which positive and negative power supply current flows.

Next, embodiment 2 will be described below. Note that, for portions having the same structure as that of embodiment 1, descriptions thereof are omitted by giving the same reference numerals thereto. Moreover, descriptions of the same operations and effects as those of embodiment 1 will be omitted.

FIG. 18 is an exploded view of the first connector according to Embodiment 2. FIGS. 19A and 19B are perspective views diagonally from below of the relationship between the movable housing and the signal terminal of the first connector according to Embodiment 2. FIGS. 20A-20C illustrate a mounting portion of the power supply terminal of the first connector to the movable housing according to Embodiment 2.

FIG. 19A is a perspective view including a cross section as seen diagonally from below the first connector, and FIG. 19B is a perspective view including a side cross section as seen diagonally from below the first connector. FIG. 20A is a top view of the first connector, FIG. 20B is a sectional view taken along the line L-L of FIG. 20A, and FIG. 20C is a sectional view taken along the line M-M in FIG. 20A.

In Embodiment 1, the movable-side retained portion 56 of the first power supply terminal 51 includes the first upper end cover portion 57a connected to the upper end of the first main body portion 56a, the second upper end cover portion 57b connected to the upper end of the second main body portion 56b, and the engagement opening 58 formed on the second upper end cover portion 57b and second main body portion 56b. The movable-side retained portion 56 moves from the top to bottom relative to the first movable housing 31 and is mounted close to the corner portions 32h of the first movable housing 31. The first upper end cover portion 57a and second upper end cover portion 57b cover the upper end portion of the short wall portions 32b and the upper end portion of the long wall portions 32a near the corner portions 32h of the first movable housing 31.

On the other hand, with the present embodiment, the movable-side retained portion 56 of the first power supply terminal 51 does not include the first upper end cover portion 57a, the second upper end cover portion 57b, or the engagement opening 58. The movable-side retained portion 56 moves relatively from bottom to top with respect to the first movable housing 31 and is attached close to the corner portions 32h of the first movable housing 31.

In addition, with the present embodiment, the shape of the tail portion 53 of the first power supply terminal 51, the structure of a part of the first movable housing 31, and the structure of a part of the first fixed housing 11 are different from those of Embodiment 1.

With the present embodiment, the first main body portion 56a of the movable-side retained portion 56 of the first power supply terminal 51 is a long narrow strip plate shape extending in the width direction of the first connector 10 and the dimension in the vertical direction is smaller than that of the first main body portion 56a in Embodiment 1. In addition, the second main body portion 56b of the present embodiment is bent at approximately 90 degrees in the width direction of the first connector 10 and connected to the outer edge of the first main body portion 56a, is a long narrow strip plate shape extending towards the center of the first connector 10 in the longitudinal direction, and the dimension thereof in the vertical direction is smaller than that of the second main body portion 56b of Embodiment 1. Note, the second main body portion 56b in the present embodiment does not include the engagement opening 58.

Furthermore, the contact portion 55 is connected to the second main body portion 56b via a contact portion con-

necting piece 55d. Specifically, a contact portion connecting piece 55d is bent at approximately 90 degrees towards the center edge of the first connector 10 in the longitudinal direction and connected to the lower edge of the second main body portion 56b and is a narrow long strip plate shape extending toward the inside of the first connector 10 in the width direction. The contact portion 55 is a narrow long strip plate shaped member extending in the vertical direction, is bent at approximately 90 degrees, and connected to the tip of the contact portion connecting piece 55d. In addition, an engaging protruding piece 58a extends upwards from a portion at the upper edge of the second main body portion 56b near the first main body portion 56a as an engaged portion that engages with the first movable housing 31.

In addition, power supply terminal movable-side recesses for stowing 32g for stowing portions of the movable-side retained portion 56 of the first power supply terminal 51 are formed at the four corner portions 32h of the first movable housing 31 main body portion 32. However, the power supply terminal movable-side recesses for stowing 32g of the present embodiment include the second recesses for stowing 32g2 formed on the inner surfaces 32a1 of the long wall portions 32a but not the first recesses for stowing 32g1 formed on the inner surfaces 32b1 of the short wall portions 32b. Further, engaging protrusions 32d are formed on the outer surfaces of the long wall portions 32a.

However, a thickened portion 32j is formed on the outer surfaces of the long wall portions 32a at a portion near both ends in the longitudinal direction. Furthermore, the power supply terminal movable-side recesses for stowing 32g of the present embodiment include a third stowing recess 32g3 as an engagement recess formed in the thickened portion 32j. The third stowing recess 32g3 is a slit shaped cavity extending in the vertical direction, the lower end opening at a downward facing surface of the first movable housing 31, specifically, the bottom surface of the thickened portion 32j, and is a cavity into which the engaging protruding piece 58a of the first power supply terminal 51 movable-side retained portion 56 is relatively inserted and retained from below.

In addition, the second recess for stowing 32g2 penetrates the main body portion 32 bottom portion 32c of the first movable housing 31 in the vertical direction and the lower end thereof is open to the lower surface of the bottom portion 32c. Furthermore, the contact portion 55 of the first power supply terminal 51 movable-side retained portion 56 is inserted into the second recess for stowing 32g2 relatively from below and stowed.

The foot portion 33a is not connected to the leg portion 33 of the first movable housing 31 in the present embodiment. Accordingly, in the first fixed housing 11, the lower end notch 12c is omitted.

In addition, the tail portion 53 of the first power supply terminal 51 in the present embodiment has a relay portion 53b bent approximately 90 degrees with respect to the first connector 10 in the width direction and is connected to the outer edge of the main body portion 52a of the fixed-side retained portion 52 extending in the vertical direction and is formed so as to extend toward the center of the first fixed housing 11 in the longitudinal direction. Furthermore, the main body portion 53a of the tail portion 53 is bent approximately 90 degrees and connected to the relay portion 53b and extends outwardly in the width direction of the first fixed housing 11.

Note that the basic configuration of another point of the first connector 10 in the present embodiment is the same as that of Embodiment 1 described above; therefore, a description thereof is omitted.

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Next, the configuration of the second connector **101** will be described.

FIG. **21** is an exploded view illustrating the second connector according to Embodiment 2.

Unlike the second connector **101** in Embodiment 1, the second connector **101** in the present embodiment does not include the housing fixation metal fitting **171**. Thus, regarding the second housing **111**, the outward protruding portion **112c** formed on the metal fitting retention recess **112d** is omitted.

Note that the basic configuration of another point of the second connector **101** in the present embodiment is the same as that of Embodiment 1 described above; therefore, a description thereof is omitted.

Next, the operation of mating the first connector **10** and the second connector **101** with the above configuration is described.

FIGS. **22A-22C** are three views illustrating a mated state of the first connector and the second connector according to Embodiment 2. FIG. **23** is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 2 in a mated state and is a sectional view taken along the line P-P in FIG. **22A**. Note, in FIG. **22A** is a plan view as viewed from the second connector side, FIG. **22B** is a sectional view taken along the line N-N in FIG. **22A**, and FIG. **22C** is a sectional view taken along the line O—O in FIG. **22A**.

As illustrated in FIGS. **22A-22C** and FIG. **23**, when the first connector **10** and second connector **101** are in a mated state, through mating of the first movable housing **31** and the second housing **111**, the first signal terminals **61** and second signal terminals **161**, that are mutually compatible, become conductive and the first power supply terminals **51** and the second power supply terminals **151**, that are mutually compatible, become conductive.

Specifically, at least a portion near the upper end of the upper portion **131** of the second housing **111** enters in between the long wall portion **32a** in the counterpart housing cavity for stowing **35** and the central wall portion **34** and the surface of the contact portion **165** of the second signal terminal **161** stowed in the signal terminal stowing recess **131f** of the upper portion **131** in the width direction on the center side of the second connector **101** comes into contact with the portion of the contact protrusion portion **65a** of the contact portion **65** of the first signal terminal **61** stowed in the signal terminal contact portion groove for stowing **34a** of the central wall portion **34** protruding outside the central wall portion **34** side surface, and thus the first signal terminal **61** and the second signal terminal **161** become conductive. Here, the contact portion **65** of the first signal terminal **61** functions as a cantilever and the contact protrusion portion **65a** is pushed against the surface of the second signal terminal **161** contact portion **165** by the elasticity exhibited by this cantilever; therefore, the conducting state of the first signal terminal **61** and second signal terminal **161** is reliably maintained.

In addition, at least the portion of the upper portion **131** of the second housing **111** close to the upper end of the short wall portion **131b** enters into the counterpart housing cavity for stowing **35** close to the short wall portion **32b** and the portion of the contact protruding portion **155a** of the contact portion **155** of the second power supply terminal **151** stowed in the power supply terminal stowing recess **131e** formed on the side surface of the short wall portion **131b** extending in the longitudinal direction and protruding outside the side surface of the short wall portion **131b** comes into contact with the contact surface that is the flat plate-shaped contact

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portion **55** stowed in the second recess for stowing **32g2** formed in the inner surface **32a1** of the long wall portion **32a** near the corner portion **32h** of the first movable housing **31**. Thus, the first power supply terminal **51** and the second power supply terminal **151** are electrically connected together. Here, the contact portion **155** of the second power supply terminal **151** functions as a cantilever and the contact protruding portion **155a** is pushed against the contact surface of the first power supply terminal **51** contact portion **55** based on the elasticity exhibited by this cantilever; therefore, the conducting state of the first power supply terminal **51** and the second power supply terminal **151** is reliably maintained.

In this manner, with the first connector **10** and second connector **101** mated, if there is a fluctuation in the positional relationship of the first board and the second board and the second connector **101** is displaced relative to the first connector **10** or there is an external force such as an impact or vibration or the like, the bent portion **54** of the first power supply terminal **51** and the bent portion **64** of the first signal terminal **61** flexibly deform, appropriately absorbing displacement relative to the first fixed housing **11** of the first movable housing **31** mated to the second housing **111**. Therefore, the mated state of the first connector **10** and the second connector **101** is maintained without being released.

In this manner, with the present embodiment, the engaging protruding piece **58a** is inserted into the third stowing recess **32g3** with a lower end open to the downward facing surface of the first movable housing **31** from below and engaged. Accordingly, similar to the movable-side retained portion **66** of the first signal terminal **61**, the movable-side retained portion **56** of the first power supply terminal **51** can be relatively inserted into and fixed in the first movable housing **31**. Therefore, the assembly process of the first connector **10** is simplified, enabling reducing manufacturing costs of the first connector **10**.

Note that an operation of mating the first connector **10** and the second connector **101** in the present embodiment and basic configurations and effects of the state of being mated and other points of the first connector **10** and the second connector **101** are the same as those of Embodiment 1 described above; therefore, descriptions thereof are omitted.

Next, Embodiment 3 will be described. Note that, for those having the same structure as those of Embodiments 1 and 2, descriptions thereof are omitted by giving the same reference numerals thereto. Moreover, descriptions of the same operations and effects as those of Embodiments 1 and 2 will be omitted.

FIGS. **24A** and **24B** are perspective views illustrating the first connector and the second connector according to Embodiment 3. FIG. **25** is an exploded view of the first connector according to Embodiment 3. FIGS. **26A** and **26B** are perspective views diagonally from below of the relationship between the movable housing and the signal terminal of the first connector according to Embodiment 3. FIGS. **27A-27C** illustrate a mounting portion of the power supply terminal of the first connector to the movable housing according to Embodiment 3.

Note, FIG. **24A** is a diagram illustrating a mated state and FIG. **24B** is a diagram illustrating a state prior to mating. FIG. **26A** is a perspective view including a cross section as seen diagonally from below the first connector, and FIG. **26B** is a perspective view including a side cross section as seen diagonally from below the first connector. FIG. **27A** is a top view of the first connector, FIG. **27B** is a sectional view taken along the line Q-Q in FIG. **27A**, and FIG. **27C** is a sectional view taken along the line R-R in FIG. **27A**.

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As illustrated in FIGS. 24A and 24B, the first connector 10 and second connector 101 of the present embodiment have dramatically reduced dimensions in the vertical direction (Z-axis direction) compared to the first connector 10 and the second connector 101 of Embodiments 1 and 2.

Specifically, with the first movable housing 31 of the first connector 10, the dimension in the vertical direction of the main body portion 32 is nearly the same as that in Embodiments 1 and 2, while the dimension in the vertical direction of the leg portion 33 is dramatically reduced compared to that of Embodiments 1 and 2. In accordance with this, the dimensions in the vertical direction of the first fixed housing 11, the first signal terminal 61, and the first power supply terminal 51 are dramatically reduced compared to those of Embodiments 1 and 2.

The first signal terminal 61 of the present embodiment is changed in that the first straight portion 64a1 and the first bent portion 64b1 of the bent portion 64 are omitted and in addition, the second bent portion 64b2 is provided with a bottom portion straight portion 64b23 that extends straight in the width direction of the first connector 10, as well as a bottom portion first bent portion 64b21 and bottom portion second bent portion 64b22 connected to both ends of the bottom portion straight portion 64b23. The bottom portion first bent portion 64b21 is a bent portion bent at approximately 180 degrees with a first end connected to the lower end of the main body portion 66a of the movable-side retained portion 66 and a second end connected to the bottom portion straight portion 64b23. In addition, the bottom portion second bent portion 64b22 is a bent portion bent at approximately 180 degrees with a first end connected to the lower end of the second straight portion 64a2 and a second end connected to the bottom portion straight portion 64b23. Thus, the dimensions in the vertical direction of the first signal terminal 61 are dramatically reduced compared to Embodiments 1 and 2 such that the contact tip portion 65c of the contact portion 65 is positioned only slightly above the upper end of the third bent portion 64b3. Note, the height of the upper end of the third bent portion 64b3 (position in vertical direction) is nearly the same as that of Embodiments 1 and 2.

In addition, the first power supply terminal 51 of the present embodiment has a first straight portion 54a1 with a dramatically reduced length (dimension in vertical direction) compared to that of Embodiment 2, and the movable-side retained portion 56 that was positioned above the second bent portion 54b2 in Embodiment 2 is changed and is positioned below the second bent portion 54b2.

Furthermore, the first fixed housing 11 of the present embodiment has the dimension in the vertical direction dramatically reduced compared to Embodiments 1 and 2 such that the brim portion 13c that demarcates the upper end of the signal terminal fixed-side retention recess 13a formed on the inner surface of the long wall portion 12a was positioned in the center or the lower half of the long wall portion 12a in the vertical direction in Embodiments 1 and 2 is changed to be positioned near the upper end of the long wall portion 12a.

Note that the basic configuration of another point of the first connector 10 in the present embodiment is the same as that of Embodiment 2 described above; therefore, a description thereof is omitted.

Next, the configuration of the second connector 101 will be described.

FIG. 28 is an exploded view illustrating the second connector according to Embodiment 3.

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The second connector 101 of the present embodiment has dimensions of the second housing 111 in the vertical direction of the upper portion 131 that are nearly the same as those of Embodiments 1 and 2 while the dimension in the vertical direction of the lower portion 112 is dramatically reduced compared to that of Embodiments 1 and 2. In accordance with this, the dimensions in the vertical direction of the retained portion 162 of the second signal terminal 161 and the retained portion 152 of the second power supply terminal 151 are dramatically reduced compared to those of Embodiments 1 and 2.

Note that the basic configuration of another point of the second connector 101 in the present embodiment is the same as that of Embodiment 2 described above; therefore, a description thereof is omitted.

Next, the operation of mating the first connector 10 and the second connector 101 with the above configuration is described.

FIGS. 29A-29C are three views illustrating a mated state of the first connector and the second connector according to Embodiment 3. FIG. 30 is a vertical cross-sectional view illustrating the first connector and the second connector according to Embodiment 3 in a mated state and is a sectional view taken along the line U-U in FIG. 29A. Note, FIG. 29A is a plan view as viewed from the second connector side, FIG. 29B is a sectional view taken along the line S-S in FIG. 29A, and FIG. 29C is a sectional view taken along the line T-T in FIG. 29A.

As illustrated in FIGS. 24A, 29A-29C, and 30, when the first connector 10 and second connector 101 are in a mated state, through mating of the first movable housing 31 and the second housing 111, the first signal terminals 61 and second signal terminals 161, that are mutually compatible, become conductive and the first power supply terminals 51 and the second power supply terminals 151, that are mutually compatible, become conductive.

In this manner, with the first connector 10 and second connector 101 mated, if there is a fluctuation in the positional relationship of the first board and the second board and the second connector 101 is displaced relative to the first connector 10 or there is an external force such as an impact or vibration or the like, the bent portion 54 of the first power supply terminal 51 and the bent portion 64 of the first signal terminal 61 flexibly deform, appropriately absorbing displacement relative to the first fixed housing 11 of the first movable housing 31 mated to the second housing 111. Therefore, the mated state of the first connector 10 and the second connector 101 is maintained without being released.

Note that an operation of mating the first connector 10 and the second connector 101 in the present embodiment and basic configurations and effects of the state of being mated and other points of the first connector 10 and the second connector 101 are the same as those of Embodiment 2 described above; therefore, descriptions thereof are omitted.

Next, Embodiment 4 will be described. Note that, for those having the same structure as that of Embodiments 1 to 3, descriptions thereof are omitted by giving the same reference numerals thereto. Moreover, descriptions of the same operations and effects as those of Embodiments 1 to 3 will be omitted.

FIG. 31 is an exploded view of the first connector according to Embodiment 4. FIGS. 32A and 32B are perspective views diagonally from below of the relationship between the movable housing and the signal terminal of the first connector according to Embodiment 4. FIGS. 33A-33C

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illustrate a mounting portion of the power supply terminal of the first connector to the movable housing according to Embodiment 4.

FIG. 32A is a perspective view including a cross section as seen diagonally from below the first connector, and FIG. 32B is a perspective view including a side cross section as seen diagonally from below the first connector. FIG. 33A is a top view of the first connector, FIG. 33B is a sectional view taken along the line V-V in FIG. 33A, and FIG. 33C is a sectional view taken along the line W-W in FIG. 33A.

In Embodiments 1 to 3, the contact portion 55 of the first power supply terminal 51 is a long rectangular plate material in the vertical direction and the surface thereof functions as a contact surface for contacting with the second power supply terminal 151 of the second connector 101. Furthermore, the contact portion 55 is stowed in the second recess for stowing 32g2 formed on the inner surface 32a1 of the first movable housing 31 long wall portion 32a. In addition, the contact surface is parallel with the inner surface 32a1.

On the other hand, with the present embodiment, the contact portion 55 of the first power supply terminal 51 includes a contact arm portion 55b stowed in the second recess for stowing 32g2, a contact protruding portion 55a that is connected to the upper end of the contact arm portion 55b and bulges toward the inside of the first movable housing 31 in the width direction, and a contact tip portion 55c at the upper end of the contact protruding portion 55a that faces outwardly in the width direction of the first movable housing 31. Furthermore, in a state prior to mating of the first connector 10 and the second connector 101, in other words, an initial state without application of an external force, the contact arm portion 55b is at an overall gradual incline and the upper end faces inward in the width direction of the first connector 10, in other words, inward in the width direction of the first movable housing 31. At least a portion of the contact protruding portion 55a protrudes from the inner surface 32a1 to the inside of the first movable housing 31 in the width direction.

Specifically, similar to Embodiments 2 and 3, the first main body portion 56a of the first power supply terminal 51 movable-side retained portion 56 is a narrow long strip plate shape extending in the width direction of the first connector 10 and the second main body portion 56b is bent approximately 90 degrees and connected to the outer edge of the first main body portion 56a in the width direction of the first connector 10 and extends toward the center of the first connector 10 in the longitudinal direction. However, the second main body portion 56b of the present embodiment is a plate material with a substantially L shape in side view (as viewed from the width direction of the first connector 10) and includes a strip-shaped vertical portion 56b1 that extends downward from the contact portion with the first main body portion 56a and a strip-shaped horizontal portion 56b2 that extends from the lower end of the vertical portion 56b1 toward the center of the first connector 10 in the longitudinal direction.

Furthermore, the engaging protruding piece 58a extends upwards from the upper edge near the tip of the horizontal portion 56b2 and the upper end of the engaging protruding piece 58a is connected to the lower end of the contact portion 55, in other words, the lower end of the contact arm portion 55b. The engaging protruding piece 58a extends in the vertical direction (Z-axis direction), whereas the contact arm portion 55b is inclined and connects to the engaging protruding piece 58a and extends diagonally upward while facing the inside of the first movable housing 31 in the width

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direction. Note, for the present embodiment, the contact portion connecting piece 55d is omitted.

In addition, similar to Embodiments 2 and 3, the thickened portion 32j is formed on the long wall portion 32a of the first movable housing 31 main body portion 32 and the power supply terminal movable-side recess for stowing 32g includes the third stowing recess 32g3 formed inside the thickened portion 32j. However, with the present embodiment, the thickened portion 32j is formed over a broad area of the long wall portion 32a and the third stowing recess 32g3 is formed with the lower end thereof open to the lower surface of the thickened portion 32j and the upper end thereof connected to the lower end of the second recess for stowing 32g2 formed on the inner surface 32a1 of the long wall portion 32a. Furthermore, the contact portion 55 of the first power supply terminal 51 movable-side retained portion 56 is relatively inserted through the third stowing recess 32g3 into the second recess for stowing 32g2 from below and stowed. Note that at least a part of the contact protruding portion 55a projects inward in the width direction of the first movable housing 31 from the inner surface 32a1. In addition, the engaging protruding piece 58a connected to the lower end of the contact portion 55 is stowed and retained in the third stowing recess 32g3.

Note that the basic configuration of another point of the first connector 10 in the present embodiment is the same as that of Embodiment 3 described above; therefore, a description thereof is omitted.

Next, the configuration of the second connector 101 will be described.

FIG. 34 is an exploded view illustrating the second connector according to Embodiment 4.

With Embodiments 1 to 3, the contact portion 155 of the second power supply terminal 151 in an initial state without an external force applied includes the contact arm portion 155b with an overall gradual incline, the contact protruding portion 155a connected to the upper end of the contact arm portion 155b that bulges outwardly in the width direction of the second housing 111, and the contact tip portion 155c that is the upper end of the contact protruding portion 155a and faces inward in the width direction of the second housing 111. Furthermore, the contact portion 155 stowed in the power supply terminal stowing recess 131e of the second housing 111 upper portion 131 is such that at least a portion of the contact protruding portion 155a protrudes to the outside of the side surface of the short wall portion 112b.

In contrast, with the present embodiment, the contact portion 155 of the second power supply terminal 151 is a long rectangular plate material in the vertical direction and the surface thereof functions as the contact surface for contacting the first power supply terminal 51 of the first connector 10. Further, the contact portion 155 is stowed in the power supply terminal stowing recess 131e and is stationary. In addition, the contact surface is parallel to the side surface of the short wall portion 112b but is preferably indented into the short wall portion 112b beyond the side surface.

Note that the basic configuration of another point of the second connector 101 in the present embodiment is the same as that of Embodiment 3 described above; therefore, a description thereof is omitted.

Next, the operation of mating the first connector 10 and the second connector 101 with the above configuration is described.

FIGS. 35A-35C are three views illustrating a mated state of the first connector and the second connector according to Embodiment 4. FIG. 36 is a vertical cross-sectional view

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illustrating the first connector and the second connector according to Embodiment 4 in a mated state and is a sectional view taken along the line Z-Z in FIG. 35A. Note, in FIG. 35A is a plan view as viewed from the second connector side, FIG. 35B is a sectional view taken along the line X-X in FIG. 35A, and FIG. 35C is a sectional view taken along the line Y-Y in FIG. 35A.

As illustrated in FIGS. 35A-35C and FIG. 36, when the first connector 10 and second connector 101 are in a mated state, through mating of the first movable housing 31 and the second housing 111, the first signal terminals 61 and second signal terminals 161, that are mutually compatible, become conductive and the first power supply terminals 51 and the second power supply terminals 151, that are mutually compatible, become conductive.

Specifically, at least a portion of the short wall portion 131b of upper portion 131 of the second housing 111 near the upper end enters into the counterpart housing cavity for stowing 35 near the short wall portion 32b and the portion of the contact protruding portion 55a of the contact portion 55 stowed in the second recess for stowing 32g2 formed on the inner surface 32a1 of the long wall portion 32a near the corner portion 32h of the first movable housing 31 protruding from the inner surface 32a1 in the width direction of the first movable housing 31 contacts the contact surface that is the surface of the contact portion 155 of the second power supply terminal 151 stowed in the power supply terminal stowing recess 131e formed on the side surface extending in the longitudinal direction of the short wall portion 131b. Thus, the first power supply terminal 51 and the second power supply terminal 151 are electrically connected together. Here, the contact arm portion 55b of the first power supply terminal 51 functions as a cantilever and the contact protruding portion 55a is pushed against the contact surface of the second power supply terminal 151 contact portion 155 based on the elasticity exhibited by this cantilever; therefore, the conducting state of the first power supply terminal 51 and the second power supply terminal 151 is reliably maintained.

Note that an operation of mating the first connector 10 and the second connector 101 in the present embodiment and basic configurations and effects of the state of being mated and other points of the first connector 10 and the second connector 101 are the same as those of Embodiment 3 described above; therefore, descriptions thereof are omitted.

Moreover, the disclosure herein describes features relating to suitable typical embodiments. Various other embodiments, modifications, and variations within the scope and spirit of the claims appended hereto will naturally be conceived of by those skilled in the art upon review of the disclosure herein.

The present disclosure can be applied to a connector.

The invention claimed is:

1. A connector comprising:

a fixed housing including a long wall portion extending in a longitudinal direction of the fixed housing and a short wall portion extending in a width direction of the fixed housing, the short wall portion being connected to an end portion of the long wall portion;

a movable housing movable relative to the fixed housing, the movable housing including a long wall portion extending in a longitudinal direction of the movable housing and a short wall portion extending in a width direction of the movable housing and connected to an end portion of the long wall portion of the movable housing; and

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power supply terminals retained in the fixed housing and in the movable housing, each power supply terminals comprising a fixed-side retained portion retained in the short wall portion of the fixed housing, a movable-side retained portion retained in the movable housing, and a bent portion connected to the fixed-side retained portion and connected to the movable-side retained portion, and

wherein in each power supply terminal, the movable-side retained portion includes a first retained portion with at least a portion extending along the short wall portion of the movable housing and a second retained portion with at least a portion extending along the long wall portion of the movable housing.

2. The connector according to claim 1, wherein in each power supply terminal, the first retained portion includes a first upper end cover portion covering an upper end portion of the short wall portion of the movable housing, and the second retained portion includes a second upper end cover portion covering an upper end portion of the long wall portion of the movable housing.

3. The connector according to claim 1, wherein each power supply terminal further comprises a contact portion connected to the second retained portion, wherein the second retained portion is arranged inside the long wall portion of the movable housing, and the contact portion is configured to come into contact with a counterpart power supply terminal.

4. The connector according to claim 3, wherein in each power supply terminal, the contact portion of each power supply terminal extends in a vertical direction along an inner surface of the long wall portion of the movable housing.

5. The connector according to claim 3, wherein in each power supply terminal, the contact portion includes a contact arm portion that extends in a vertical direction at an incline with respect to an inner surface of the long wall portion of the movable housing, and each power supply terminal further comprises a contact protruding portion connected to an upper end of the contact arm portion and which bulges inward in the width direction of the movable housing.

6. The connector according to claim 1, wherein the movable-side retained portion of each power supply terminal further comprises a second retained portion which includes an engaged portion that is engaged with the movable housing.

7. The connector according to claim 6, wherein the engaged portion of each power supply terminal is inserted and engaged from below in an engagement recess in which a lower end thereof is open to a downward facing surface of the movable housing.

8. The connector according to claim 1, wherein in each power supply terminal, the bent portion is bent in a substantially S-shape when viewed from the width direction of the movable housing, wherein a first end of the bent portion is connected to an upper end of the fixed-side retained portion and a second end of the bent portion is connected to a lower end of the first retained portion of the movable-side retained portion.

9. The connector according to claim 1, wherein each power supply terminal comprises a board connecting portion connected to the fixed-side retained portion, and wherein each board connecting portion includes a relay portion extending from the fixed-side retained portion toward a center of the fixed housing in the longitudinal direction of the fixed housing and a main body portion connected to the

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relay portion that extends towards an outside of the fixed housing in the width direction.

10. The connector according to claim 1, wherein the power supply terminals are arranged in pairs at both ends of the fixed housing in the longitudinal direction of the fixed housing and lined up in the width direction of the fixed housing respectively at both ends of the fixed housing in the longitudinal direction of the fixed housing.

11. The connector according to claim 2, wherein each power supply terminal further comprises a contact portion connected to the second retained portion, wherein the second retained portion is arranged inside the long wall portion of the movable housing, and the contact portion is configured to come into contact with a counterpart power supply terminal.

12. The connector according to claim 11, wherein in each power supply terminal, the contact portion of each power supply terminal extends in a vertical direction along an inner surface of the long wall portion of the movable housing.

13. The connector according to claim 11, wherein in each power supply terminal, the contact portion includes a contact arm portion that extends in a vertical direction at an incline with respect to an inner surface of the long wall portion of the movable housing, and each power supply terminal further comprises a contact protruding portion connected to an upper end of the contact arm portion and which bulges inward in the width direction of the movable housing.

14. The connector according to claim 11, wherein each power supply terminal comprises a board connecting portion connected to the fixed-side retained portion, and wherein each board connecting portion includes a relay portion extending from the fixed-side retained portion toward a center of the fixed housing in the longitudinal direction of the fixed housing and a main body portion connected to the relay portion that extends towards an outside of the fixed housing in the width direction.

15. The connector according to claim 2, wherein the movable-side retained portion of each power supply terminal

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further comprises a second retained portion which includes an engaged portion that is engaged with the movable housing.

16. The connector according to claim 15, wherein the engaged portion of each power supply terminal is inserted and engaged from below in an engagement recess in which a lower end thereof is open to a downward facing surface of the movable housing.

17. The connector according to claim 15, wherein each power supply terminal comprises a board connecting portion connected to the fixed-side retained portion, and wherein each board connecting portion includes a relay portion extending from the fixed-side retained portion toward a center of the fixed housing in the longitudinal direction of the fixed housing and a main body portion connected to the relay portion that extends towards an outside of the fixed housing in the width direction.

18. The connector according to claim 2, wherein in each power supply terminal, the bent portion is bent in a substantially S-shape when viewed from the width direction of the movable housing, wherein a first end of the bent portion is connected to an upper end of the fixed-side retained portion and a second end of the bent portion is connected to a lower end of the first retained portion of the movable-side retained portion.

19. The connector according to claim 2, wherein each power supply terminal comprises a board connecting portion connected to the fixed-side retained portion, and wherein each board connecting portion includes a relay portion extending from the fixed-side retained portion toward a center of the fixed housing in the longitudinal direction of the fixed housing and a main body portion connected to the relay portion that extends towards an outside of the fixed housing in the width direction.

20. The connector according to claim 2, wherein the power supply terminals are arranged in pairs at both ends of the fixed housing in the longitudinal direction of the fixed housing and lined up in the width direction of the fixed housing respectively at both ends of the fixed housing in the longitudinal direction of the fixed housing.

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